9

Gravitational radiation

KIP S. THORNE†

9.1 Introduction

9.1.1 The motivations for gravitational-wave research

The discovery of cosmic radio waves in the 1930s and their detailed study in the 40s, 50s, and 60s created a revolution in our view of the universe (Kellerman and Sheets, 1983; Sullivan, 1982, 1984). Previously the universe, as viewed by light, was regarded as serene and quiescent – dominated by stars and planets that wheel smoothly in their orbits, shining steadily and evolving (with few exceptions) on timescales of millions or billions of years. By contrast, the universe as viewed by radio waves was violent: galaxies in collision, jets ejected from galactic nuclei, quasars with luminosities far greater than our galaxy varying on timescales of hours, pulsars with gigantic radio beams rotating at several or many rotations per second; these were the typical strong radio emitters.

The radio revolution was so spectacular because the information carried by radio waves is so different from that carried by light. A factor 10^7 difference in wavelength meant the difference between photons predominantly thermal in origin (light) and photons predominantly nonthermal (radio), the difference between the bremsstrahlung and atomic transitions of stellar and planetary atmospheres on one hand, and the synchrotron radiation of intergalactic magnetized plasmas on the other.

As different as cosmic radio and optical radiations may be, their differences pale by comparison with those of electromagnetic waves and gravitational waves: cosmic gravitational waves should be emitted by, and carry detailed information about, coherent bulk motions of matter (e.g., collapsing stellar cores) or coherent vibrations of spacetime curvature (e.g.,

† Supported in part by the National Science Foundation (AST-8514911).

black holes). By contrast, cosmic electromagnetic waves are usually incoherent superpositions of emission from individual atoms, molecules, and charged particles. Gravitational waves are emitted most strongly in regions of spacetime where gravity is relativistic and where the velocities of bulk motion are near the speed of light. But electromagnetic waves come almost entirely from weak-gravity, low-velocity regions, since strong-gravity regions tend to be obscured by surrounding matter. Gravitational waves pass through surrounding matter with impunity, by contrast with electromagnetic waves which are easily absorbed and scattered, and even by contrast with neutrinos which, although they easily penetrate normal matter, should scatter thousands of times while leaving the core of a supernova.

These differences make it likely that, if cosmic gravitational waves can be detected and studied, they will create a revolution in our view of the universe comparable to or greater than that which resulted from the discovery of radio waves.

It might be argued that we are now so sophisticated and complete in our electromagnetically based (radio, millimeter, infrared, optical, ultraviolet, X-ray, gamma-ray, cosmic ray) understanding of the universe, compared to the optically based astronomers of the 1930s and 1940s, that a gravitational-wave revolution will be far less spectacular than was the radio revolution. It seems unlikely to me that we are so sophisticated. I am painfully aware of our lack of sophistication when I contemplate the sorry state of present estimates of the gravity waves bathing the earth (Section 9.4 below): for each type of gravity-wave source that has been studied, with the exception of binary stars and their coalescences, either (i) the strength of the source's waves for a given distance from earth is uncertain by several orders of magnitude; or (ii) the rate of occurrence of that type of source, and thus the distance to the nearest one, is uncertain by several orders of magnitude; or (iii) the very existence of the source is uncertain.

Although these uncertainties make us unhappy when we try to plan for the design and construction of gravitational-wave detectors, we will be rewarded with great surprises when gravity waves are ultimately detected and studied: the waves will give us extensive information about the universe that we are unlikely ever to obtain in any other way.

Detailed studies of cosmic gravitational waves are also likely to yield experimental tests of fundamental laws of physics which cannot be tested in any other way.

The first discovery of gravitational waves would directly verify the

predictions of general relativity, and other relativistic theories of gravity, that such waves should exist. (There has already been an indirect verification, in the form of the observed inspiral of the binary pulsar due to gravitational-radiation reaction; Weisberg and Taylor, 1984; Taylor, 1987.)

By comparing the arrival times of the first bursts of light and gravitational waves from a distant supernova, one could verify general relativity's prediction that electromagnetic and gravitational waves propagate with the same speed – i.e., that they couple to the static gravity (spacetime curvature) of our Galaxy and other galaxies in the same way. For a supernova in the Virgo cluster of galaxies (15 Mpc distant), first detected optically one day after the light curve starts to rise, the electromagnetic and gravitational speeds could be checked to be the same to within a fractional accuracy (1 light day)/(15 Mpc) $= 5 \times 10^{-11}$.

By measuring the polarization properties of the gravitational waves, one could verify general relativity's prediction that the waves are transverse and traceless – and thus are the classical consequences of spin-two gravitons (Eardley, Lee and Lightman, 1973; Eardley *et al.*, 1973).

By comparing the detailed wave forms of observed gravitational wave bursts with those predicted for the coalescence of black-hole binaries (which will be computed by numerical relativity in the next few years, see Section 9.3.3(e) below), one could verify that certain bursts are indeed produced by black-hole coalescences – and, as a consequence, verify unequivocally the existence of black holes and general relativity's predictions of their behavior in highly dynamical circumstances. Such verifications would constitute by far the strongest test ever of Einstein's laws of gravity.

9.1.2 A brief history of gravitational-wave research

Einstein (1916) laid the foundations of gravitational-wave theory within months after his final formulation of general relativity – restricting himself to weak (linearized) waves emitted by bodies with negligible self-gravity and propagating through flat, empty spacetime. During the next few years Einstein (1918), Weyl (1922), and Eddington (1924) elaborated on Einstein's initial work so that by the mid-1920s the linearized theory of gravitational waves was fully understood. However, it was clear – at least to Eddington (1924) – that for sources with significant self-gravity (e.g. binary systems), the linearized analysis was invalid.

Landau and Lifshitz (1941) gave the first fairly satisfactory treatment of the emission of waves by self-gravitating systems; but a series of failed

attempts to analyze radiation reaction in such systems in the late 1940s and 1950s (pp. 73 and 74 of Damour, 1983) shook physicists' faith in the ability of the waves to carry off energy, and even in the correctness of the Landau–Lifshitz formula for the emitted wave field. It required a clever thought experiment by Bondi (1957) to restore faith in the energy of the waves, and a series of beautiful and rigorous studies of the asymptotic properties of the waves at infinity by Bondi and collaborators (Bondi, 1960; Bondi, van der Burg, and Metzner, 1962; Sachs, 1962, 1963; Penrose, 1963*a*,*b*) and of the propagation of short-wavelength waves through a curved background spacetime by Isaacson (1968*a*,*b*) to restore faith that the fundamental theory of gravitational waves is soundly based.

The experimental search for cosmic gravitational waves was initiated by Joseph Weber (1960) at a time when almost nothing was known about possible cosmic sources and when nobody else had the vision to see that there were technological possibilities of ultimate success. After a decade of effort, Weber (1969) announced to the world tentative evidence that his resonant-bar gravity-wave detectors – one near Washington, DC, the other near Chicago – were being excited simultaneously by gravitational waves. There followed a six-year period of excitement and feverish effort as 15 other research groups around the world tried to construct and operate similar bar detectors (Tyson and Giffard, 1978; Amaldi and Pizella, 1979; de Sabbata and Weber, 1977; Weber, 1986, and references therein). Sadly, even with markedly improved sensitivities, these efforts gave no convincing evidence that gravity waves were actually being seen.

In parallel with this experimental effort, astrophysicists worldwide struggled through the early 1970s to milk, from electromagnetic observations of the universe and from fundamental theory, as much information as possible about the characteristics of the gravity waves that might be bathing the earth. By the mid-1970s a fuzzy but helpful picture had begun to emerge: While the sensitivities of the detectors to kilohertz-frequency bursts arriving, say, three times per year had improved during the early 70s from dimensionless amplitude $h_{3/\mathrm{yr}} \sim 1 \times 10^{-15}$ to $h_{3/\mathrm{yr}} \sim 3 \times 10^{-16}$ (a factor 10 improvement in energy flux), it seemed highly unlikely that such bursts bathing the earth would exceed $h_{3/\mathrm{yr}} \sim 1 \times 10^{-16}$; a reasonable probability of success would require $h_{3/\mathrm{yr}} \sim 10^{-20}$ or better; and a high probability would require $h_{3/\mathrm{yr}} \sim 10^{-21}$ to $h_{3/\mathrm{yr}} \sim 10^{-22}$ (Smarr, ed., 1979; Fig. 9.4 below). Although these estimates were discouraging, the theoretical efforts that produced them were making clear the enormous potential payoff that could follow the successful detection of gravity waves.

Fortunately, the experimental efforts of the early 1970s had pointed the way toward major possible detector improvements; and, consequently, although most of the first-generation experimental groups became discouraged and dropped out, a handful of highly talented groups continued onward into the 1980s with a second-generation effort involving major technological changes such as cooling the bars to liquid-helium temperatures, changing bar materials, switching from passive to active transducers, and even developing completely new types of detectors, most notably laser-interferometer gravity-wave detectors (called 'beam' detectors in this chapter). These second-generation efforts have reached fruition in the last few years: bars with kilohertz burst sensitivities $h_{3/\mathrm{yr}} \sim 10^{-17}$ (30 times higher in amplitude than the first-generation and 1000 higher in energy) are now collecting data in coordinated searches (Section 9.5.2(d) below); and small-scale beam detectors with $h_{3/\mathrm{yr}} \sim 5 \times 10^{-17}$ are now operating (Section 9.5.3(d) below) as prototypes for full-scale detectors with projected ultimate sensitivities in the 10^{-22} region (Section 9.5.3(g) below). The regime of possible success has been reached, and the regimes of reasonably probable success and highly probable success look reachable – though only with vigorous continuing efforts and the expenditure of non-trivial sums of money.

In parallel with these 1980s' second-generation efforts, theorists have redoubled their struggle to firm up our understanding of the waves bathing the earth, but with only modest results: the problem of knowing what kinds of sources actually occur, and how frequently, is hampered by the paucity of electromagnetic information; and, as a result, the apparent recent improvements in our knowledge (Section 9.4 below) might be little more than changes of fashion. On the other hand, given a specific scenario for how a postulated source behaves, theorists have become far more adept than before – thanks not least to supercomputers – at computing the details of the gravitational waves it should emit (Section 9.3.3 below). As a consequence, when waves are ultimately detected, the prospects have become reasonable for deciphering from them the details of their sources.

While the present, 1987, gravity-wave searches might bring success, it is not likely they will. Thus, we must anticipate a continued vigorous effort at technology development during the coming years, with the prospects of success improving significantly at each step along the way. In parallel, we must anticipate a continuing major effort by relativity theorists to refine their ability to decipher the source behaviors corresponding to postulated gravitational-wave forms, and a continuing effort by astrophysicists to give

better guidance as to what kinds of sources actually exist and in what numbers. The efforts must be great; but jointly they are likely to give an extremely valuable payoff.

9.1.3 Overview of this chapter

This chapter reviews all aspects of gravitational-wave research – experimental, theoretical relativity, and theoretical astrophysics. This review is intended to be readable by physicists (including advanced students) who are not specialists in general relativity, in experimental gravity, or in astrophysics.

A word of warning: in preparing this review I have *not* done a thorough literature search (I lacked the necessary energy!); nor have I cited all major original references of which I am aware (that would have made the reference list even longer than it is!). However, I have attempted to present all significant ideas and issues with which I am familiar, citing wherever possible the earliest occurrence of the idea or issue and one or more recent, thorough discussions of it.

This review is divided into four major parts: the physical and mathematical description of gravitational waves (Section 9.2), the generation and propagation of gravitational waves (Section 9.3), astrophysical sources of gravitational waves (Section 9.4), and the detection of gravitational waves (Section 9.5).

The physical and mathematical description of gravitational waves (Section 9.2) is presented in a form that compactifies and updates the corresponding material in the textbook that I coauthored fourteen years ago (Misner, Thorne, and Wheeler, 1973; cited henceforth as MTW). Emphasis focuses on the 'shortwave approximation' as a tool for defining waves mathematically (Section 9.2.1), and on measurements in the proper reference frame of an observer as a tool for defining waves physically (Section 9.2.2). A special 'TT coordinate system' is then introduced (Section 9.2.3) for use in analyzing systems large compared to a wavelength of the waves; and the energy, momentum, and quantization of gravitational waves are discussed (Section 9.2.4).

The theory of the generation and propagation of gravitational waves (Section 9.3) is far more sophisticated today than when MTW was written: we now understand how, in realistic situations, to split the problem of wave generation off from that of wave propagation, and how to mesh generation and propagation together using the technique of 'matched asymptotic expansions' (Section 9.3.1). We also understand more deeply the most

elementary of all ways to compute wave generation, the 'quadrupole formalism', and its relationship to radiation reaction in the emitting system (Section 9.3.2). Unfortunately, the most strongly emitting systems should violate the mathematical approximations that underlie the quadrupole formalism and thus can be analyzed only in rough order of magnitude using it; for more accurate analyses one must use a more sophisticated wave-generation formalism. A catalog of more sophisticated formalisms is given in Section 9.3.3 – which is an updated version of a review I wrote ten years ago (Thorne, 1977). The theory of the propagation of the waves from source to earth is sketched in Section 9.3.4, and various wave-propagation effects (absorption, scattering, dispersion, tails, gravitational focusing, diffraction. parametric amplification by background curvature, non-linear coupling of waves to themselves, and generation of background curvature by the waves' energy and momentum) are described in Section 9.3.5 – which with 9.3.4 is a shortened version of my recent (Thorne, 1983), long review of wave propagation. Section 9.3 concludes with very brief descriptions of elegant mathematical work on idealized wave-propagation situations: the asymptotic structure of waves propagating toward 'future null infinity' in an asymptotically flat spacetime (Section 9.3.6), and exact, analytic solutions to the Einstein equations for wave generation and propagation (Section 9.3.7).

It is nearly a decade since a detailed and thorough review has been written of astrophysical sources and detectors for gravitational waves (Smarr, ed., 1979; Douglass and Braginsky, 1979); and in the intervening time these topics have changed enormously. (For recent reviews of a number of subtopics see the chapters in Deruelle and Piran, 1983.) Sections 9.4 and 9.5 attempt a comprehensive review in a manner that closely ties the sources to the detection efforts. In these sections the sources and the detection strategies are split up into three categories: those for gravitational-wave 'bursts' (Section 9.4.1 and Fig. 9.4); those for periodic gravitational waves (Section 9.4.2 and Fig. 9.6); and those for a stochastic gravitational-wave background (Section 9.4.3 and Fig. 9.7). All previous reviews have been cavalier about factors of 2 in the definitions of wave strengths and detector sensitivities. This review tries to standardize the definitions, including factors of 2. The standardization is based on signal-to-noise-ratio analyses that are given at the beginnings of Sections 9.4.1, 9.4.2, and 9.4.3. For each source that looks favorable for wave detection, Section 9.4 gives a description of our current state of knowledge of the source and gives current estimates of the wave strengths and other wave characteristics.

The burst sources treated in Section 9.4.1 include supernovae (collapse of

a normal stellar core to form a neutron star, subsection c), the collapse of a star or star cluster to form a black hole (subsection d), the inspiral and coalescence of compact binaries (neutron stars and black holes, subsection e), and the fall of stars and small holes into supermassive holes (subsection f). Because our knowledge of sources is so poor, it is useful to estimate how strong the strongest wave bursts bathing the earth could be without violating our cherished beliefs about the laws of physics and the nature of the universe; this is done in subsection g. The periodic sources treated in Section 9.4.2 include rotating neutron stars (rigidly rotating pulsars, and neutron stars spun up by accretion until they encounter a radiation-reaction-driven instability, subsection b), and binary stars (including unevolved binaries, WUMa stars, white-dwarf binaries, and neutron-star binaries, subsection c). The stochastic sources in Section 9.4.3 include large numbers of binary stars whose waves superpose stochastically (subsection b), pre-galactic, Population III stars (subsection c), the big-bang singularity in which the universe began – with subsequent parametric amplification of its waves by background curvature in inflationary and other scenarios (subsection d), phase transitions in the subsequent but still early universe (subsection e), and cosmic strings produced by phase transitions (subsection f). Present estimates of the strengths of the waves from all these sources are shown in Figs. 9.4 (burst), 9.6 (periodic), and 9.7 (stochastic) along with the sensitivities of present and proposed detectors.

The detectors in Section 9.5 are divided into those that operate in the high-frequency regime, $f \gtrsim 10$ Hz (Sections 9.5.2, 9.5.3, and 9.5.4), those for low frequencies, $10\ \text{Hz} \gtrsim f \gtrsim 10^{-5}$ Hz (Section 9.5.5), and those for very low frequencies, $f \lesssim 10^{-5}$ Hz (Section 9.5.6). The high-frequency detectors are all earth-based; but because of seismic and gravity-gradient noise, the low- and very-low-frequency detectors must be space-based. Sections 9.5.2 and 9.5.3 describe in great detail the earth-based, high-frequency bar and beam detectors which have been under development for many years and show great promise for the future. Section 9.5.4 describes briefly other types of earth-based, high-frequency detectors. Section 9.5.5 describes low-frequency detectors including doppler tracking of spacecraft (subsection a), beam detectors in space which hold great promise for the turn of the century (subsection b), the normal modes of the earth and sun (subsections c and d), the vibrations of blocks of the earth's crust (subsection e), and the earth-orbiting skyhook (subsection f). Section 9.5.6 describes very-low-frequency detectors including the timing of pulsars (neutron-star rotations), which recently has placed interesting observational limits on a stochastic

background (subsection a), the timing of orbital motions of binary and planetary systems (subsection b), and astronomical observations of anisotropies in the temperature of the cosmic microwave radiation (subsection c), deviations from the Hubble flow (subsection d), and products of primordial nucleosynthesis (subsection d).

9.1.4 Notation and conventions

Throughout this chapter, unless otherwise stated, we shall assume that general relativity correctly describes classical gravitational waves. Our notation will be that of Misner, Thorne, and Wheeler (1973) – cited as MTW throughout – including, e.g., the use of Greek indices for spacetime (running from 0 to 3) and Latin for space (running from 1 to 3), the use of commas for partial derivatives and semicolons for covariant derivatives, the use of the Einstein summation convention, and the use of geometrized units in which Newton's gravitation constant G and the speed of light c are set equal to unity. Sometimes, particularly when discussing gravitational-wave detectors, we shall restore the Gs and cs to the equations and use cgs units.

9.2 The physical and mathematical description of a gravitational wave

9.2.1 Shortwave approximation

General relativistic gravitational waves are ripples in the curvature of spacetime that propagate with the speed of light. Because gravity is nonlinear, it is not possible in a fully precise manner to separate the contributions of gravitational waves to the curvature from the contributions of the earth, the sun, the galaxy, or anything else; and since such a separation underlies the very concept of a gravitational wave, this means that gravitational waves are not precisely defined entities.

On the other hand, in realistic astrophysical situations the lengthscale $\lambda\!\!\!^{-}$ on which the waves vary (their reduced wavelength, $\lambda\!\!\!^{-}=\lambda/2\pi$) is very short compared to the lengthscales $\mathscr{L}$ on which all other important curvatures vary; and this difference in lengthscale makes possible a high-accuracy, but approximate, split of the Riemann curvature tensor $R_{\alpha\beta\gamma\delta}$ into a 'background curvature' $R^{\rm B}_{\alpha\beta\gamma\delta}$ plus a contribution $R^{\rm GW}_{\alpha\beta\gamma\delta}$ due to gravitational waves: the background $R^{\rm B}_{\alpha\beta\gamma\delta}$ is the average of $R_{\alpha\beta\gamma\delta}$ over several wavelengths

$$R^{\rm B}_{\alpha\beta\gamma\delta}\equiv\langle R_{\alpha\beta\gamma\delta}\rangle; \tag{1a}$$

and the waves' curvature $R^{\rm GW}_{\alpha\beta\gamma\delta}$ is the rapidly varying difference

$$R^{\rm GW}_{\alpha\beta\gamma\delta}\equiv R_{\alpha\beta\gamma\delta}-R^{\rm B}_{\alpha\beta\gamma\delta}. \tag{1b}$$

Heuristically, $R^{B}_{\alpha\beta\gamma\delta}$ is like the large-scale (10 cm) curvature of an orange, while $R^{GW}_{\alpha\beta\gamma\delta}$ is like the fine-scale granulation of the skin of the orange (lengthscale a few millimeters).

This method of defining a gravitational wave, introduced into general relativity by Wheeler (1955) and Power and Wheeler (1957), is a special case of a standard technique in mathematical physics variously called 'shortwave approximation' or 'two-timing' or 'two-lengthscale expansion' or 'two-variable expansion' (see e.g. Chapter 3 of Cole, 1968); and it is intimately connected to the 'WKB approximation'. There is an elegant shortwave-approximation formalism for gravitational-wave theory due largely to Brill and Hartle (1964) and to Isaacson (1968*a*,*b*). Not surprisingly the formalism reveals that general relativistic gravitational waves propagate through vacuum in essentially the same manner as light – with the same speed, with the same changes of amplitude due to curvature of the wavefronts, with the same diffraction effects when focussed by a gravitational lens, etc. Because that formalism has been reviewed extensively elsewhere (e.g. Thorne, 1983), I shall not delve into it here, except for a brief description in Section 9.3.4 and an enumeration and brief discussion of some of its predictions in Section 9.3.5 below.

9.2.2 Measurements in the proper reference frame of an observer

In general relativity the Riemann curvature tensor is defined, operationally, by the relative accelerations it produces between adjacent particles (e.g. the 'equation of geodesic deviation', Sections 8.7 and 11.3 of MTW). Correspondingly, a gravitational wave can be defined operationally in the following way:

Consider an observer (freely falling or accelerated, it doesn't matter so long as the acceleration is slowly varying). Let the observer carry with herself a small Cartesian latticework of measuring rods and synchronized clocks (a 'proper reference frame' in the sense of Section 13.6 of MTW, with spatial coordinates x^j that measure proper distance along orthogonal axes). She is to measure the 'force of gravity' F_j that acts on a particle of mass m, momentarily at rest at location x^j. For example, she might let the particle fall freely, measure its acceleration g_j in her proper reference frame, and multiply it by the particle's mass m to get the force $F_j = mg_j$. Alternatively, she might measure the force required to hold the particle fixed in the coordinate grid of her proper reference frame and equate F_j to minus that force. Of course, this is not different in any way from what she would do were she a Newtonian physicist rather than a relativistic physicist. The difference

lies in how she analyzes and thinks about the force of gravity F_j. As a relativist, she recognizes (cf. Box 37.1 of MTW) that F_j is made up of a nearly steady, nearly position-independent component caused by her own failure to fall freely, plus a component proportional to the particle's Cartesian coordinate position x^j (relative acceleration of particle and origin of coordinates) which is caused by spacetime curvature $R_{\alpha\beta\gamma\delta}$. This latter contribution,

$$F_j = -mR_{j0k0}x^k \tag{2}$$

(where the index 0 denotes a component along her time basis vector), she splits up into a piece that changes slowly in time (background curvature contribution) plus a piece that is rapidly varying. She makes certain that there are no rapidly moving or rapidly changing nearby sources of gravity to account for the rapid variations; if there are none, then she can attribute the rapidly varying component of the force to gravitational waves

$$F_j^{\mathrm{GW}} = -mR^{\mathrm{GW}}_{j0k0}x^k. \tag{3}$$

It is conventional to use, as the primary entity for describing a gravitational wave, not the Riemann curvature tensor $R^{\mathrm{GW}}_{\alpha\beta\gamma\delta}$ which has dimensions 1/time^2 (or 1/length^2), but rather a dimensionless 'gravitational-wave field' h^{TT}_{jk}. In terms of the force-producing components of the Riemann tensor $R^{\mathrm{GW}}{}_{j0k0}$ and proper time t as measured by our observer, h^{TT}_{jk} is defined by (cf. Section 2.3 of Thorne 1983)

$$\frac{\partial^2 h^{\mathrm{TT}}_{jk}}{\partial t^2} \equiv -2R^{\mathrm{GW}}_{j0k0}. \tag{4}$$

The convenience of this gravitational-wave field lies in its simple relationship to displacements produced by the waves: if, for simplicity, the observer is freely falling and keeps the axes of her coordinate grid tied to gyroscopes and is in a region of spacetime where gravitational waves are the only source of spacetime curvature, then the waves will produce tiny oscillatory changes δx^j in the position of a test particle relative to the origin of her coordinate grid; and these changes will satisfy the equation of motion ('equation of geodesic deviation')

$$m\frac{\mathrm{d}^2\,\delta x^j}{\mathrm{d}t^2} = F_j^{\mathrm{GW}} = -mR^{\mathrm{GW}}_{j0k0}x^k = \frac{1}{2}m\frac{\partial^2 h^{\mathrm{TT}}_{jk}}{\partial t^2}x^k. \tag{5}$$

Because any realistic wave is so weak that the oscillatory changes δx^j are miniscule compared to the distance of the particle from the origin, x^k can be regarded as essentially constant on the right-hand side, and equation (5) can

then be integrated easily to give

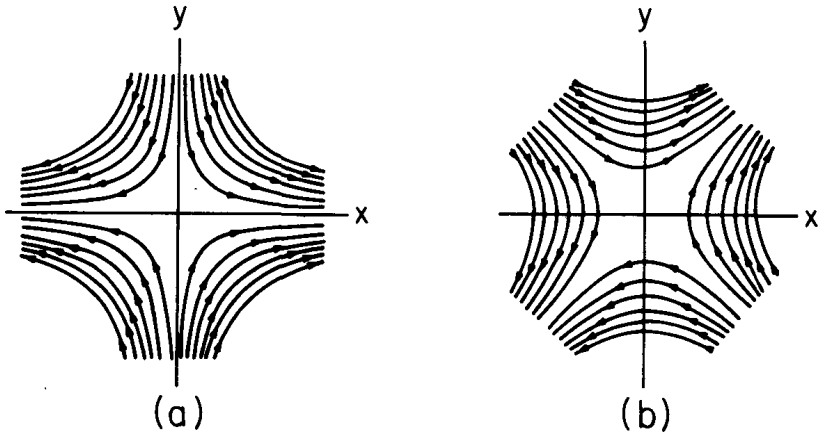
$$\delta x^j = \frac{1}{2} h^{\mathrm{TT}}_{jk} x^k. \tag{6}$$

Thus, aside from a factor $\frac{1}{2}$, h^{TT}_{jk} plays the role of the 'dimensionless strain of space', or the 'time-integrated shear of space': it is the ratio of the wave-induced displacement of a free particle relative to the origin, to its orginal displacement from the origin. (In the above equations and below it does not matter whether a spatial index is up or down, since the spatial coordinates are Cartesian.)

The superscript TT on the gravitational-wave field is to remind us that, according to general relativity, the field is 'transverse and traceless'. More specifically: the Einstein field equations guarantee that $R^{\mathrm{GW}}_{\alpha\beta\gamma\delta}$ and hence also h^{TT}_{jk} propagate with the speed of light. Since the sources of cosmic gravity waves are very far away, the waves look very nearly planar as they pass through the observer's proper reference frame. If we orient the x, y, z spatial axes so the waves propagate in the z direction, then the 'transversality' of the waves means that the only non-zero components of the wave field are h^{TT}_{xx}, $h^{\mathrm{TT}}_{xy} = h^{\mathrm{TT}}_{yx}$, and h^{TT}_{yy}; and the 'trace-free' property means that $h^{\mathrm{TT}}_{xx} = -h^{\mathrm{TT}}_{yy}$. Thus, the gravitational waves, like electromagnetic waves, have only two independent components – two polarization states.

Because of their TT nature, gravitational waves produce a quadrupolar, divergence-free force field (equation (5) and Fig. 9.1). This force field has two components corresponding to the two polarization states of the waves: the

Fig. 9.1. Lines of force for gravitational waves (equations (5) and (7)): (*a*) with '+' polarization, $m\,\delta\ddot{x} = \frac{1}{2}\ddot{h}_+ x$ and $m\,\delta\ddot{y} = -\frac{1}{2}\ddot{h}_+ y$; and (b) with '×' polarization, $m\,\delta\ddot{x} = \frac{1}{2}\ddot{h}_\times y$ and $m\,\delta\ddot{y} = \frac{1}{2}\ddot{h}_\times x$ – where dots denote time derivatives.



quantity

$$h_+ \equiv h^{TT}_{xx} = -h^{TT}_{yy} \tag{7a}$$

produces a force field with the orientation of a '+' sign, while

$$h_\times \equiv h^{TT}_{xy} = h^{TT}_{yx} \tag{7b}$$

produces one with the orientation of a '×' sign. Thus, h_+ and $h_\times$ are called the 'plus' and 'cross' (or + and ×) gravity-wave amplitudes. From these amplitudes and the polarization tensors $e^+_{xx} = -e^+_{yy} = 1$, $e^\times_{xy} = e^\times_{yx} = 1$ (all other components zero), one can reconstruct the full wave field

$$h^{TT}_{jk} = h_+ e^+_{jk} + h_\times e^\times_{jk}. \tag{7c}$$

It is straightforward to show that, if one rotates the x and y axes in the transverse plane through an angle $\Delta\psi$, the gravity-wave amplitudes are changed to

$$\begin{aligned} h^{new}_+ &= h^{old}_+ \cos 2\Delta\psi + h^{old}_\times \sin 2\Delta\psi, \\ h^{new}_\times &= -h^{old}_+ \sin 2\Delta\psi + h^{old}_\times \cos 2\Delta\psi. \end{aligned} \tag{7d}$$

The quadrupolar symmetry of the force field, together with the tenets of canonical field theory, tells us that general relativistic gravitational waves must be associated with quanta of spin two ('gravitons'). The spin is always the ratio of 360 degrees to the angle, about the propagation direction, through which one must rotate an instantaneous field to make it return to its original state – the 'return angle'. For electromagnetic waves the return angle is 360 degrees and the spin is one; for general relativistic gravitational waves the return angle is 180 degrees and the spin is two. In other relativistic theories of gravity the wave field has other symmetries and therefore other spins for its quanta – and in most relativistic theories, by contrast with general relativity, the symmetries are not even frame-invariant (local Lorentz-invariant); and, as a result, the waves of those theories cannot be incorporated into canonical field theory and cannot be quantized by canonical techniques. For details see Eardley, Lee and Lightman (1973), and Eardley *et al.* (1973).

The above definition of the gravitational-wave field h^{TT}_{jk} relies on a specific choice of reference frame. It turns out that, if one pursues this definition in two different reference frames related by a boost in some arbitrary direction, and if one chooses the spatial axes of the two frames so they are unrotated relative to each other, then the instantaneous wave fields h^{TT}_{jk} (and correspondingly h_+ and $h_\times$) will be the same. Stated more precisely – but in a language I shall not explain – the general-relativistic gravitational-wave field h^{TT}_{jk} has 'boost-weight zero' (it is a scalar under boosts); but, as

discussed above, it has 'spin-weight two' (it behaves like a spin-two field under rotations). For further details see Section 2.3.2 of Thorne (1983), and for the mathematics that goes along with the concepts of 'spin-weight' and 'boost-weight', see Geroch, Held and Penrose (1973).

9.2.3 TT coordinate system

The above description of the force law and force fields produced by a gravitational wave is just the leading order in a power series expansion in distance $r=(\delta_{jk}x^j x^k)^{\frac{1}{2}}$ from the origin of the observer's proper reference frame. The higher-order fractional corrections to the forces are of order $(r/\lambda\!\!\!^{-})^2$ and, correspondingly, the proper reference frame's coordinates x^j fail to measure proper distance by fractional amounts of order $h^{\rm TT}_{jk}(r/\lambda\!\!\!^{-})^2$ (see, e.g. Zhang, 1986, for discussion in the case of a freely falling observer). Thus, the above description of gravity-wave forces is accurate only if the region of interest is spatially small compared to a reduced wavelength. When the region is large, an alternative description is needed.

The nicest alternative description makes use of a 'TT coordinate system' (Section 35.4 of MTW), i.e. a coordinate system which is nearly Minkowski throughout the spacetime region of interest, and in which the contribution of the waves to the deviations from the Minkowski metric is embodied in the same $h^{\rm TT}_{jk}$ as we introduced above.

Because the TT coordinates must be nearly Minkowski, they cannot cover a spacetime region that is too large: their extent in both time and space must be far smaller than the background radius of curvature $\mathscr{R}_{\rm B}\sim|R^{\rm B}_{\alpha\beta\gamma\delta}|^{-\frac{1}{2}}$. In typical situations, this limitation is very mild compared to the limitation on the size of an observer's proper reference frame: one can stretch TT coordinates over any region small compared to the Hubble distance (cutting out holes in the vicinities of black holes and neutron stars), but if the waves have a frequency of a kilohertz, then any proper reference frame used to study their forces must be small compared to $\lambda\!\!\!^{-}\sim 50$ km.

In a TT coordinate system the spacetime metric coefficients take the form

$$g_{\alpha\beta}=\eta_{\alpha\beta}+h^{\rm B}_{\alpha\beta}+h^{\rm TT}_{\alpha\beta}, \tag{8}$$

where $\eta_{\alpha\beta}$ are the Minkowski metric coefficients (diagonal -1, $+1$, $+1$, $+1$), $h^{\rm B}_{\alpha\beta}$ is the background metric perturbation which varies on a long lengthscale $\mathscr{L}$, and $h^{\rm TT}_{\alpha\beta}$ is the gravity-wave metric perturbation which varies on the short lengthscale $\lambda\!\!\!^{-}$. The time-time and space-time components of the gravity-wave perturbation $h^{\rm TT}_{00}$ and $h^{\rm TT}_{0j}=h^{\rm TT}_{j0}$ vanish, and the space-space components $h^{\rm TT}_{jk}$ are the same as the gravity-wave field that would be

computed from the Riemann tensor (equation (4)) in the proper reference frame of any observer who is nearly at rest in the TT coordinate system. For proofs and discussions, see e.g. Sections 35.4, 37.1, and 37.2 of MTW.

Whereas physics in a proper reference frame can be formulated in the Newtonian language of three-dimensional forces, including gravitational forces, physics in a TT coordinate system must be formulated in the relativistic language of geodesic motion and vanishing divergence of the stress-energy tensor; cf. Section 9.5.1 below.

9.2.4 *Energy, momentum, and quantization of gravitational waves*

In the 1940s and early 1950s a controversy raged over whether or not gravitational waves can carry energy. The controversy was ultimately resolved by Herman Bondi (1957) using a simple thought experiment: place several beads on a rough stick and let a gravitational wave pass. The above description of the forces produced by the waves (which was first understood fully by Bondi and by Felix Pirani 1956 and their colleagues) guarantees that – if the stick is not *too* rough – the wave will push the beads back and forth on the stick, heating it. Surely, if the wave can heat a stick, it must carry energy.

It was not until the late 1960s that a fully satisfactory mathematical description of the energy in a gravity wave was devised: Richard Isaacson (1968*a*,*b*), using the shortwave approximation which he had developed in detail on the basis of cruder earlier work of Wheeler (1955), Power and Wheeler (1957) and Brill and Hartle (1964), introduced a stress-energy tensor $T^{GW}_{\alpha\beta}$ for gravitational waves. This stress-energy tensor, like the background curvature $R^{B}_{\alpha\beta\gamma\delta}$, is smooth on the lengthscale $\lambda\!\!\!^{-}$; it is obtained, in fact, by averaging the squared gradient of the wave field over several wavelengths:

$$T^{GW}_{\alpha\beta} = \frac{1}{32\pi} \sum_{i,j} \langle h^{TT}_{ij,\alpha} h^{TT}_{ij,\beta} \rangle, \tag{9}$$

where $\langle \cdots \rangle$ means 'average over several wavelengths'. (For a pedagogical derivation and discussion see Sections 35.7–35.15 of MTW; for a beautiful rederivation by the method of averaged Lagrangians see MacCallum and Taub, 1973.) If the waves are propagating in the z direction, this stress-energy tensor takes the standard form for a bundle of zero-rest-mass particles (gravitons) moving at the speed of light in the z direction:

$$T^{GW}_{00} = -T^{GW}_{0z} = -T^{GW}_{z0} = T^{GW}_{zz} = \frac{1}{16\pi} \langle (\partial h_+/\partial t)^2 + (\partial h_\times/\partial t)^2 \rangle. \tag{10}$$

In order of magnitude, restoring the factors of G and c, the energy flux in the waves if they have frequency $f = c/2\pi\lambda$ is

$$-T_{0z}^{GW} \simeq \frac{\pi}{4}\frac{c^3}{G} f^2 \langle h_+^2 + h_\times^2 \rangle = 320 \frac{\text{erg}}{\text{cm}^2\,\text{s}} \left(\frac{f}{1\,\text{kHz}}\right)^2 \left\langle \frac{h_+^2 + h_\times^2}{(10^{-21})^2} \right\rangle. \tag{11}$$

The numbers in this equation correspond to a strongly emitting supernova in the Virgo cluster of galaxies, where there are several supernovae per year. Contrast this huge gravity-wave energy flux with the peak electromagnetic flux at the height of the supernova, $\sim 10^{-9}$ erg cm^{-2} s^{-1}; but note that the gravity waves should last for only a few milliseconds, while the strong electromagnetic output lasts for days.

Corresponding to the huge energy flux (11) in an astrophysically interesting gravitational wave is a huge occupation number for the quantum states of the gravitational-wave field: it is not hard to show that for the above supernova burst only a handful of quantum states are occupied; and they each contain $n \sim 10^{75}$ gravitons (equations (6)–(8) of Thorne *et al.*, 1979). This means that the waves behave exceedingly classically; quantum-mechanical corrections to the classical theory have fractional magnitude $1/\sqrt{n} \sim 10^{-37}$. (Although the full quantization of the gravitational field is exceedingly difficult and not yet fully under control, the quantization of weak gravitational waves propagating through a smooth background spacetime – equivalent to weak waves in flat spacetime – has been well understood for decades; see, e.g., the most elementary aspects of Feynman, 1963; Dewitt, 1967*a*,*b*.)

Isaacson's stress-energy tensor (9) for gravitational waves has the same properties and plays the same role as the stress-energy tensor for any other field or form of matter in the background spacetime. For example, $T_{\alpha\beta}^{GW}$ generates background curvature through the Einstein field equations (averaged over several wavelengths of the waves); also $T_{\alpha\beta}^{GW}$ has vanishing divergence (conservation of gravity-wave energy and momentum) in spacetime regions where the waves are not being generated, absorbed, or scattered. For full details see Isaacson (1968*b*) or Section 35.15 of MTW.

9.3 The generation and propagation of gravitational waves

9.3.1 Wave propagation split off from wave generation

Turn, now, to the generation of gravitational waves and their propagation from their source to the earth. Mathematically, the wave-generation problem and the wave-propagation problem are each difficult – though for very different reasons, so that to handle the difficulties requires two very

different sets of mathematical tools and approximations. For ease of analysis, then, it is important to split the propagation problem off from the generation problem and treat them separately. This is accomplished by dividing the space around the source into three regions (Section III of Thorne, 1980*b*): a 'wave-generation region' at distances from the source $r \lesssim r_{\rm I}$ = ('Inner radius'); a 'local wave zone' at distances $r_{\rm I} \lesssim r \lesssim r_{\rm O}$ = ('Outer radius'); and a 'distant wave zone' at distances $r \gtrsim r_{\rm O}$. The theory of wave generation is developed with one set of mathematical tools in the wave-generation region and the local wave zone, i.e. at distances $r \lesssim r_{\rm O}$; the theory of propagation to earth is developed with the other set of tools in the local wave zone and the distant wave zone, i.e. at distances $r \gtrsim r_{\rm I}$; and the two theories are matched together in their domain of overlap, the local wave zone $r_{\rm I} \lesssim r \lesssim r_{\rm O}$.

The inner radius $r_{\rm I}$ is far enough out to be in the wave zone, $r_{\rm I} \gg \lambda\!\!\!^{-}$; far enough to be in a region where the source's gravity is weak, $r_{\rm I} \gg 2M \equiv$ (Schwarzschild radius of source) = 2 × (mass of source); and far enough to be outside the source, $r_{\rm I} \gg L \equiv$ (size of source). The outer radius $r_{\rm O}$ is far enough beyond the inner radius to leave many wavelengths in the local wave zone, $r_{\rm O} - r_{\rm I} \gg \lambda\!\!\!^{-}$; but not so far that the gravitational redshift can produce a significant net phase shift during propagation through the local wave zone, $\delta\phi = (M/\lambda\!\!\!^{-})\ln(r_{\rm O}/r_{\rm I}) \ll 1$; and not so far that the background curvature of the external universe can significantly affect the propagation, $r_{\rm O} - r_{\rm I} \ll \mathscr{R}_{\rm B} \equiv |R^{\rm B}_{\alpha\beta\gamma\delta}|^{-\frac{1}{2}} \equiv$ (radius of curvature of background spacetime). These choices of $r_{\rm I}$ and $r_{\rm O}$ permit one to ignore, in the local wave zone $r_{\rm I} \lesssim r \lesssim r_{\rm O}$, the background curvature both of the source and of the external universe; i.e. they permit one to regard the waves in the local wave zone as propagating through flat spacetime. This greatly simplifies calculations.

For a given, astrophysically interesting source the wave-generation task consists of computing with reasonable accuracy the dynamical behavior of the source's gravitational field in the wave-generation region $r \lesssim r_{\rm I}$, and further computing how that dynamical curvature develops into outward propagating waves in the local wave zone, $r_{\rm I} \lesssim r \lesssim r_{\rm O}$. Once this is done, the theorist can switch tasks and mathematical formalisms from wave generation to wave propagation. The wave-propagation task takes as input the propagating waves in the local wave zone and carries them outward (typically using the shortwave approximation and formalism) through the universe from source to earth.

As an addendum to the wave-generation task, one often computes the

back-reaction effects of the wave emission on the source, i.e. the 'radiation reaction'.

9.3.2 The quadrupole formalism for wave generation and radiation reaction

Of all the techniques for computing wave generation, one, the 'quadrupole formalism', is especially important because it is highly accurate for many sources and is accurate in order of magnitude for most. The quadrupole formalism was derived originally by Einstein (1916, 1918) for sources with negligible self-gravity and slow internal motions. Later, in a series of steps by Landau and Lifshitz (1941), Fock (1959), Ipser (1971) and Thorne (1980*b*, Sections VII and XII), it became clear that the quadrupole formalism requires for high accuracy no constraints whatsoever on the strength of the source's internal gravity; all that is required is slow motion – more specifically, that the source's size L be small compared to the reduced wavelength $\lambda\!\!\!^{-}$ of the waves it emits. In other words, the quadrupole formalism is to gravity-wave generation what the 'poor-antenna approximation' (dipole formalism) is to radio-wave generation. Moreover, as in the radio-wave problem, the quadrupole formalism typically is accurate to within factors of order 2 even for sources with sizes of order a reduced wavelength $\lambda\!\!\!^{-}$ (cf. equation (4.8) of Thorne, 1980*b*); and since very few astrophysical systems are larger than a reduced wavelength of the waves they emit, this justifies the use of the quadrupole formalism for order-of-magnitude astrophysical estimates in almost all situations.

The quadrupole formalism writes the gravitational-wave field in the source's local wave zone (where the background curvature can be ignored) in the following simple form:

$$h_{jk}^{\mathrm{TT}} = \frac{2}{r}\frac{\partial^2}{\partial t^2}\left[\mathscr{I}_{jk}(t-r)\right]^{\mathrm{TT}}. \tag{12}$$

Here r is the distance to the source's center, t is proper time as measured by an observer at rest with respect to the source, $t-r$ is retarded time, the superscript TT means 'algebraically project out and keep only the part that is transverse to the (radial) direction of propagation and is traceless' (Box 35.1 of MTW), and $\mathscr{I}_{jk}(t-r)$ is the source's mass quadrupole moment evaluated at the retarded time $t-r$.

The meaning of 'mass quadrupole moment' is well known when the source has weak internal gravity and small internal stresses, so Newtonian

gravity is a good approximation to general relativity inside and near the source. Then $\mathscr{I}_{jk}$ is the symmetric, trace-free (STF) part of the second moment of the source's mass density ρ, as computed in a Cartesian coordinate system centered on the source:

$$\mathscr{I}_{jk}(t)=\left[\int \rho(t)x^j x^k \,\mathrm{d}^3x\right]^{\mathrm{STF}}=\int \rho(t)\left[x^j x^k-\frac{1}{3}r^2\delta_{jk}\right]\mathrm{d}^3x; \tag{13a}$$

equivalently, it is the coefficient of the $1/r^3$ part of the source's Newtonian gravitational potential

$$\Phi=-\frac{M}{r}-\frac{3}{2}\frac{\mathscr{I}_{jk}(t)x^j x^k}{r^5}-\frac{5}{2}\frac{\mathscr{I}_{ijk}x^i x^j x^k}{r^7}+\cdots. \tag{13b}$$

If the source has strong internal gravity, one can no longer express its mass quadrupole moment as the simple integral (13a). However, so long as the source has slow internal motions ($L\ll\lambda\!\!\!^{-}$), there will be a region of space far enough from the source to be in vacuum ($r>L$) and far enough for gravity to be weak ($r\gg 2M$), yet near enough for retardation and wave behavior to be unimportant ($r\ll\lambda\!\!\!^{-}$). In this 'weak-field, vacuum, near-zone' gravity can be described with high accuracy as Newtonian; and the Newtonian potential is $\Phi\cong-\frac{1}{2}(g_{00}+1)$, where g_{00} is the time-time part of the metric in a coordinate system that is as Minkowski as possible throughout the weak-field, vacuum near zone. The mass quadrupole moment can then be read off this Newtonian potential using the standard formula (13b). For further discussion see the review in Section 3 of Thorne (1983) or the original treatment in Part Two of Thorne (1980*b*).

For order-of-magnitude calculations of h^{TT}_{jk} (equation (12)), one can approximate the TT part of the second time derivative of the mass quadrupole moment by that portion of the source's internal kinetic energy which is associated with non-spherical motions, $E^{\mathrm{ns}}_{\mathrm{kin}}$; and, unless the source is at very large cosmological redshifts $z\gg 1$, one can propagate the waves to earth as though the intervening spacetime were flat – with the result that the local wave-zone formula for the waves, equation (12), is valid also at earth. The result is the simple order-of-magnitude formula

$$h\sim\frac{E^{\mathrm{ns}}_{\mathrm{kin}}}{r} \tag{14}$$

for the magnitude h of the gravity-wave field h^{TT}_{jk} at earth.

From the 'exact' quadrupole formula (12) for the wave field in the local wave zone and Isaacson's formula (9) for the stress-energy tensor of the waves, one can compute the fluxes of energy and angular momentum carried

by the waves. By integrating those fluxes over a sphere surrounding the source in the local wave zone, one obtains for the rates of emission of energy (Einstein, 1916, 1918) and angular momentum (Peters, 1964)

$$\frac{dE^{GW}}{dt}=\frac{1}{5}\sum_{j,k}\left\langle\left(\frac{d^3\mathcal{I}_{jk}}{dt^3}\right)^2\right\rangle, \tag{15}$$

$$\frac{dJ_i^{GW}}{dt}=\frac{2}{5}\sum_{j,k,l}\varepsilon_{ijk}\left\langle\frac{\partial^2\mathcal{I}_{jl}}{\partial t^2}\frac{\partial^3\mathcal{I}_{lk}}{\partial t^3}\right\rangle. \tag{16}$$

The linear momentum carried off by the waves vanishes when one computes it by the quadrupole formalism; but when one includes higher-order corrections to the field emitted by slow-motion sources, one finds for the rate of emission of linear momentum (first derived by Papapetrou, 1962, 1971; for a more modern derivation in the notation of this chapter see Section IV.C of Thorne, 1980*b*)

$$\frac{dP_i^{GW}}{dt}=\frac{2}{63}\sum_{j,k}\left\langle\frac{\partial^3\mathcal{I}_{jk}}{\partial t^3}\frac{\partial^4\mathcal{I}_{jki}}{\partial t^4}\right\rangle+\frac{16}{45}\sum_{j,k,a}\varepsilon_{ijk}\left\langle\frac{\partial^3\mathcal{I}_{ja}}{\partial t^3}\frac{\partial^3\mathcal{S}_{ka}}{\partial t^3}\right\rangle. \tag{17}$$

Here $\mathcal{I}_{ijk}$ is the source's 'mass octupole moment' and $\mathcal{S}_{ij}$ is its 'current quadrupole moment' (gravitational analog of magnetic quadrupole moment). For sources with weak internal gravity and stresses (nearly Newtonian sources), these moments are computable from the simple volume integrals

$$\mathcal{I}_{ijk}=\left(\int\rho x^i x^j x^k\,d^3x\right)^{STF}, \tag{18a}$$

$$\mathcal{S}_{ij}=\left(\int\rho\varepsilon_{ipq}x^p v^q x^j\,d^3x\right)^{STF}, \tag{18b}$$

where, as in equation (15a), STF means 'make it symmetric and trace-free', i.e. 'symmetrize on all free indices and remove the traces on all pairs of free indices'. Independently of the strengths of the internal gravity and stresses, the moments can be read off the Newtonian potential $\Phi\cong-\frac{1}{2}(g_{00}+1)$ (equation (13b)) and off the 'gravitomagnetic potential' $\beta_i\equiv g_{0i}$ in the source's weak-field near zone:

$$\beta_i=-2\varepsilon_{ipq}\frac{J^p x^q}{r^3}-4\varepsilon_{ipq}\frac{\mathcal{S}_{pa}x^q x^a}{r^5}-\cdots. \tag{19}$$

In equation (19) J^p, the moment in the leading, dipolar term, is the source's angular momentum. For further details, discussions, and derivations see Thorne (1983) or Thorne (1980*b*). For a discussion of the gravitomagnetic potential see, e.g., Chapter 3 of Thorne, Price and Macdonald (1986).

The laws of conservation of energy, angular momentum, and linear momentum imply that radiation reaction should deplete the source's

energy, angular momentum, and linear momentum at just the right rates to compensate for the losses (15), (16), and (17); and a detailed analysis of the radiation reaction forces reveals that this is so (Peres, 1960). Particularly convenient in analyzing the radiation reaction in a source with weak self-gravity is a Newtonian-type radiation-reaction potential (Burke, 1969; Thorne, 1969; Chandrasekhar and Esposito, 1970; Section 36.8 of MTW).

Different physicists feel comfortable with different levels of rigor. In recent years these differences have shown up strongly and publicly in a controversy over derivations of the quadrupole wave-generation formula (12) and the formula for the energy sapped from a source by radiation reaction (negative of equation (15)). Many physicists – myself among them – were quite satisfied with derivations at the level of rigor, e.g., of Landau and Lifshitz (1941) and Peres (1960). Others (e.g. Ehlers *et al.*, 1976) felt that those early derivations were inadequately rigorous and, correspondingly, that the quadrupole formulae were suspect for sources with non-negligible self-gravity. The controversy was heightened by the fact that there were mathematical errors in some (but not all) of the early derivations (see Walker and Will, 1980 and Section 3.4.2 of Thorne, 1983 for discussions).

The controversy was still raging in the early 1980s; see, e.g. Ashtekar (1983). However, during the mid 1980s it has largely subsided. There are now many new derivations of the quadrupole formulae, with much improved rigor, and they all produce the same, standard results; see, e.g., Anderson *et al.* (1982), Blanchet and Damour (1984); Christodoulou and Schmidt (1979); Isaacson, Welling and Winicour (1984); and for reviews see Will (1986), Schutz (1986*a*) and Damour (1987).

Of particular interest is radiation reaction in the binary pulsar PSR1913+16, which should cause the binary system's two neutron stars to spiral together slowly with a consequent gradual decrease in their orbital period. Because the duration of observations of the pulsar is short (12 years), the cumulative effects of radiation reaction on the orbit during those observations are 100 times smaller than post-Newtonian effects; and, consequently, the detailed effects of the radiation reaction could not be fully understood until the orbital equations were fully under control up through post-post-Newtonian order. Damour and Deruelle (1986) have now brought the orbital equations fully under control; and there is now a beautiful agreement between those equations – including the quadrupolar radiation reaction – and the observational data. For a detailed discussion see Chapter 6 of this book.

9.3.3 A catalog of formalisms for computing wave generation

The order-of-magnitude formula $h \sim E^{ns}_{kin}/r$ shows that of all sources at fixed distance r, the strongest emitters will be those with the largest non-spherical kinetic energies, i.e. those with the largest internal masses and with internal velocities approaching the speed of light. Thus, the strongest emitters are likely to violate the slow-motion assumption which underlies the quadrupole formalism, and will require for accurate analysis either higher-order corrections to the quadrupole formalism, or wave-generation formalisms that do not entail any slow-motion assumption.

There are a number of other wave-generation formalisms which can be applied to such sources. This section is a catalog of them, with references to detailed presentations and applications. For an out-of-date but unified presentation of most of these formalisms see Thorne (1977).

To be tractable with a minimum of numerical computation, a wave-generation formalism must break the extreme non-linearity of the Einstein field equations by imposing a power-series expansion in some small quantity and keeping only the lowest, linear order or the lowest few orders. Wave-generation formalisms can be classified according to their choice of the small expansion parameter. *Slow-motion formalisms* (of which the quadrupole formalism is an example) expand in $L/\lambda\!\!\!^{-}$=(size of source)/(reduced wavelength of waves); subsection (a) below. *Post-Minkowski formalisms* expand in the strength of the gravitational field inside the source, i.e. in the magnitude of the deviations of the metric coefficients from their Minkowski values; subsection (b). *Post-Newtonian formalisms* expand simultaneously in $L/\lambda\!\!\!^{-}$ and the strength of the internal gravitational field; subsection (c). *Perturbation formalisms* expand in the deviations of the metric from its form for some non-radiative, astrophysical system – e.g. from the Kerr metric for a rotating black hole, or from the metric for an equilibrium, rotating, relativistic stellar model, or from the Friedman–Robertson–Walker metric for a homogeneous, isotropic 'big-bang'; subsection (d).

Of all astrophysical sources, the very strongest emitters will entail gravitationally induced large-amplitude, high-velocity, non-spherical internal motions – e.g. the inspiral and coalescence of a binary black hole or binary neutron-star system. For such sources there is no small parameter in which one can expand. The only way to compute the full details of the wave field emitted by such sources is by the techniques of *numerical relativity*: the numerical solution of the full Einstein field equations on a supercomputer; subsection (e).

(a) *Slow-motion formalisms*

As for electromagnetic waves, so also for gravitational waves, slow-motion expansions give rise automatically to multipolar expansions: the electromagnetic vector potential is a sum of an electric dipolar term (typical magnitude $(Q/r)(L/\lambda\!\!\!^{-})$ where Q is the charge of the source), plus an electric quadrupole and a magnetic dipole [magnitude $(Q/r)(L/\lambda\!\!\!^{-})^2$], plus an electric octupole and a magnetic quadrupole [magnitude $(Q/r)(L/\lambda\!\!\!^{-})^3$], etc. Similarly, the gravitational-wave field h^{TT}_{jk} is the sum of a mass-quadrupole term $[\sim(\partial^2/\partial t^2)\mathscr{I}_{jk}/r\sim(M/r)(L/\lambda\!\!\!^{-})^2]$, plus a mass octupole $[\sim(\partial^3/\partial t^3)\mathscr{I}_{ijk}/r\sim(M/r)(L/\lambda\!\!\!^{-})^3]$ and a current quadrupole $[\sim(\partial^2/\partial t^2)\mathscr{S}_{ij}/r\sim(M/r)(L/\lambda\!\!\!^{-})^3]$, etc. In each case, electromagnetic and gravitational, there are two families of moments involved; and in each case as one goes to higher orders in $L/\lambda\!\!\!^{-}$ one is driven to include higher-order moments of the source.

Early foundations for these expansions in the gravitational case will be found in Bonnor (1959) and Pirani (1964); full mathematical details will be found in Thorne (1980*b*); and major improvements and elucidations will be found in Blanchet and Damour (1986), Blanchet (1987*a*) and Damour (1987).

A good example which shows how, as the source's internal motions are speeded up, the higher moments gradually become more and more important, is the gravitational bremsstrahlung radiation emitted when two stars fly past each other with some high initial velocity; see Kovacs and Thorne (1978) and Turner and Will (1978) for full details.

In some slow-motion systems the mass quadrupole contribution may be suppressed, leaving the current quadrupole or the mass octupole to dominate. A good example is the torsional oscillation of a neutron star, in which the motions are slow because the shear modulus is weak, and the mass moments vanish because of parity considerations leaving the current quadrupole to dominate the radiation; for details see Schumaker and Thorne (1983). Another example is the Chandrasekhar (1970)–Friedman–Schutz (1978) (CFS) instability in neutron stars, which preferentially excites mass octupole ($l=3$) or hexadecapole ($l=4$) or $l=5$ modes of pulsation causing them to radiate more strongly than quadrupole; see Section 4.3(b) below.

(b) *Post-Minkowski formalisms*

Post-Minkowski wave-generation formalisms are sometimes called 'post-linear' because they entail expanding in the strength of the gravitational field

beyond linear order; and they are sometimes called 'fast-motion' to contrast them with slow-motion formalisms.

There is a systematic way to take a post-Minkowski wave-generation formalism that is accurate to a given order in the strength of the source's internal gravity, and iterate it to obtain a formalism of higher accuracy (Thorne and Kovacs, 1975; Thorne, 1977).

The wave-generation formalism that is accurate to first post-Minkowski order (first order in the strength of internal gravity) is 'Linearized theory', i.e. the linear approximation to general relativity. Linearized theory is discussed in most textbooks, e.g. Chapter 18 and Sections 35.1–35.6 of MTW. Halpern and Desbrandes (1969) and, independently, Press (1977) have derived a particularly useful Linearized wave-generation formula for systems with sizes L large compared to a reduced wavelength $\lambda\!\!\!^{-}$. Examples of gravity-wave generation that have been analyzed by Linearized theory are: (i) the coherent (but painfully slow) transformation of electromagnetic waves into gravitational waves (first considered by Gertsenshtein, 1962, subsequent work reviewed in Section 4.1 of Grishchuk and Polnarev, 1980); and (ii) the waves emitted by the explosion of a non-spherical nuclear bomb (Wheeler, 1962; Wood *et al.*, 1970).

Linearized theory is completely ignorant of the source's internal gravity; it can correctly predict the emitted waves only if the source's motions are governed by non-gravitational forces – typically by electric or magnetic forces. For systems with significant but weak internal gravity (e.g. stellar pulsations, binary systems, and high-speed stellar encounters), one must use a wave-generation formalism accurate to the next, 'post-post-Minkowski' or 'post-Linear' order. For the details of such a formalism, see, e.g., Thorne and Kovacs (1975), Crowley and Thorne (1977); and for its application to high-speed stellar encounters (gravitational bremsstrahlung radiation) see Kovacs and Thorne (1977, 1978). For a recent review see Westpfahl (1985).

Thus far nobody has developed a post^2-Linear wave-generation formalism in detail – i.e. a formalism accurate to post^3-Minkowski order. There has been no great need for such a formalism, and the post-Linear formalism is sufficiently hard to work with in practice (cf. Kovacs and Thorne, 1977) that it is not clear whether post^2-Linear would be significantly easier than full-blown numerical relativity.

(c) *Post-Newtonian formalisms*

Post-Newtonian approximations to general relativity make the assumption – in accord with the virial theorem for gravitationally bound systems – that

inside the source the deviations of the metric from Minkowski (i.e. the dimensionless strength of the source's gravity) have a magnitude ε of order $(L/\lambda\!\!\!^{-})^2$; and accordingly they expand the Einstein field equations simultaneously in ε (post-Minkowski expansion) and $L/\lambda\!\!\!^{-}$ (slow-motion expansion). See, e.g., Burke (1979) for a review of the method.

The lowest-order wave generation formalism that results from this expansion is called 'Newtonian' because it computes the evolution of the source using Newton's laws of gravity and mechanics, then evaluates the source's time-evolving quadrupole moment using the standard Newtonian volume integral (13a), then inserts that quadrupole moment into the standard quadrupole wave-generation formula (12). It is this Newtonian version of the quadrupole formalism that has been especially controversial (see the end of Section 3.2 above) but is now almost universally agreed to be highly reliable.

There have been a large number of important wave-generation calculations with this formalism. Some examples are: (i) the waves emitted by binary systems in Newtonian, elliptical orbits (Peters and Mathews. 1963); (ii) the waves emitted by a variety of models of stars that collapse to form neutron stars (Saenz and Shapiro, 1978, 1981); and (iii) the waves emitted in the head-on collision of two compact stars (Gilden and Shapiro. 1984).

The Newtonian wave-generation formalism starts losing accuracy when the source's internal gravity becomes too strong ($\varepsilon \sim 0.05$) and its internal velocities too high ($v \sim 0.2$) (Turner and Will, 1978) – e.g. in the late stages of the spiraling together of a neutron-star binary system. In such a situation it is useful to include the next higher-order corrections (one order higher in the strength of gravity ε, two higher in the speed $L/\lambda\!\!\!^{-}$). The result is the post-Newtonian formalism (Epstein and Wagoner, 1975; Wagoner, 1977; Tsvetkov, 1984). Examples of calculations that have been performed with the post-Newtonian formalism are the radiation from a system of bodies whose sizes are all small compared to their separations (Wagoner and Will. 1976), gravitational bremsstrahlung at moderate velocities (Turner and Will, 1978), and the radiation emitted by a slowly rotating star that collapses to a neutron star (Turner and Wagoner, 1979).

The foundations for a post^2-Newtonian wave-generation formalism have also been worked out (Section V.E. of Thorne, 1980*b*); but it has never been developed in full detail or applied to any sources. Such a formalism – by contrast with the post^2-Linear – would likely be far more tractable than full-

blown numerical relativity; so it may one day prove useful in tying down the gravitational waves from large-amplitude processes involving neutron stars.

(d) *Perturbation formalisms*

Much can be learned about gravitational waves from neutron stars and black holes by studying weak, non-radial perturbations around their equilibrium structures. The theory of non-radial perturbations of non-rotating relativistic stars was developed by Campolattaro, Detweiler, Ipser, Price and Thorne (see pp. 195–201 of Thorne 1978 for a review and references); and extensions of the theory have been given by Schumaker and Thorne (1983), Detweiler and Lindblom (1985) and Finn (1986). Among its recent applications are computations by Lindblom and Detweiler (1983) of the normal modes of neutron stars with a variety of equations of state. For rotating stars the corresponding theory is due largely to Chandrasekhar, Friedman and Schutz (reviewed through 1977 on pp. 201–8 of Thorne 1978 and reviewed more recently by Schutz 1987). The most important recent applications are studies of the waves emitted by the 'Chandrasekhar–Friedman–Schutz instability' in rapidly rotating neutron stars; see Section 9.4.2(b) below.

For non-rotating (Schwarzschild) black holes the perturbation theory is due to Regge and Wheeler (1957) with major subsequent contributions by Bardeen, Chandrasekhar, Detweiler, Edelstein, Moncrief, Press, Vishveshwara and Zerilli; see pp. 180–8 of Thorne (1978) for a review. Recent applications include a definitive numerical evaluation of the eigenfrequencies and damping times of a Schwarzschild hole's quasinormal modes (Leaver, 1985, 1986*a*), and of the Green's function for arbitrary perturbations of a Schwarzschild hole (Leaver, 1986*b*). Finally, for rotating (Kerr) black holes the theory is due to Teukolsky (1972, 1973), Teukolsky and Press (1974), Wald (1973), Chrzanowski (1975), Cohen and Kegeles (1975), Chandrasekhar and Detweiler (1976), Detweiler (1977), Chandrasekhar (1983) and Sasaki and Nakamura (1982); and recent applications include the evaluation of the eigenfrequencies and damping times of a Kerr hole's quasinormal modes (Detweiler, 1980; Leaver, 1985, 1986*a*), and evaluations of the gravitational waves emitted when a compact body orbits a Kerr hole (Detweiler, 1978), scatters gravitationally off a Kerr hole (Kojima and Nakamura, 1984*b*), or plunges into a Kerr hole (Detweiler and Szedenits, 1979; Kojima and Nakamura, 1984*a*).

Primordial gravitational waves (waves created in or near the big-bang) cannot be analyzed by splitting space into a wave-generation region plus

wave zones (Section 9.3.1), because the transition from wave generation to wave propagation is a temporal rather than a spatial one: the transition occurs when the size of the cosmological horizon expands to become much larger than a wavelength, thereby unfreezing a set of frozen-in initial perturbations. The only way, today, to analyze primordial waves is by a perturbation formalism somewhat akin to that used for stars and black holes: the unperturbed configuration is typically a non-radiative. Friedman–Robertson–Walker cosmological model, and the perturbations are studied by linearizing the Einstein equations – or a quantized variant of them – around that model. The resulting theory is due to Lifshitz (1946); see also Section 7.3 of Zel'dovich and Novikov (1983), and references cited in Section 9.4.3(d) below.

(e) *Numerical relativity*

Numerical solution of the full Einstein equations is the only way, today, to study wave generation in the strongest and most interesting of all gravity-wave sources: those with high internal velocities, strong internal gravity, and large deviations from a non-radiating spacetime. During the past decade several dozen researchers have worked vigorously to develop the field of numerical relativity. The results of this effort are: an analytic formulation of the initial value problem and the dynamical evolution problem for the Einstein field equations, in a form that facilitates numerical solutions (York, 1983, and references cited therein); for axisymmetric systems, viable ways to slice spacetime and choose the spatial coordinates so as to avoid pathologies and to compute the emitted gravitational waves efficiently (Smarr and York, 1978; Piran, 1983; Bardeen and Piran, 1983; Sasaki, 1984; Stewart and Friedrich, 1983; Isaacson, Welling and Winicour, 1983; Gomez *et al.*, 1986; Anderson and Hobill, 1986; Evans and Abrahams, 1987); and several good computer codes for evolving axisymmetric systems and computing their waves (Nakamura, 1983; Stark and Piran, 1986; Piran and Stark, 1986; Evans, 1986).

Among the problems that have been studied successfully with the codes thus far are the gravitational radiation produced by a head-on collision of two Schwarzschild black holes (Smarr, 1977*a*), by the collapse of a rotating star to form a Kerr black hole (Stark and Piran, 1986), and by the vibrations of a neutron star (Evans, 1986).

The experts in numerical relativity are now beginning to move on from axisymmetric systems, which have two non-trivial spatial dimensions and one non-trivial time, to asymmetric, generic systems with three non-trivial

spatial dimensions and one time (e.g. Nakamura, 1987). This full '3 + 1' effort will require using the world's largest supercomputers, and will require new techniques for slicing spacetime into space plus time, for choosing the spatial coordinates, and for differencing the Einstein equations. The effort may absorb almost as many person-years as the development of gravitational-wave detectors; but it will be well worthwhile: the payoffs will include the ability to compute in detail the waveforms from the strongest gravity-wave sources in the universe, such as the spiraling together and coalescence of two black holes – waveforms that will be crucial to the interpretation of gravity-wave observations and to their use for strong-field, highly dynamical tests of general relativity.

One should not be misled into believing that numerical relativity will be the totally dominant tool for realistic gravitational-wave calculations in the coming years. On the contrary, we can expect a healthy interaction between numerical and analytical techniques; for discussion see Schutz (1986*c*).

9.3.4 Wave propagation in the real universe

Since I have recently written a detailed review of the theory of gravitational wave propagation (Section 2 of Thorne, 1983), I shall only sketch the main points briefly.

No matter how strong a source of gravity waves may be, once its waves are fully formed (once they have reached a location, e.g. in the local wave zone, where the inhomogeneity lengthscale $\mathscr{L}$ of the background curvature is large compared to their wavelength λ), they will have dimensionless amplitudes h small compared to unity. This can be seen as follows:

The stress-energy tensor of the gravitational waves, $T^{GW}_{\alpha\beta}$ (equation (9)), acts as a source for the background curvature through the Einstein equations $G^{B}_{\alpha\beta} = 8\pi(T^{GW}_{\alpha\beta} + T^{other}_{\alpha\beta})$. Since the background Einstein tensor $G^{B}_{\alpha\beta}$ has magnitude less than or of order $1/\mathscr{R}^2_B$ (where $\mathscr{R}_B$ is the background radius of curvature as defined from the Riemann tensor), and since the magnitude of the gravity-wave stress-energy tensor (9) is h^2/λ^2, the Einstein equations imply that

$$h \lesssim \lambda/\mathscr{R}_B. \tag{20}$$

Since the inhomogeneity lengthscale $\mathscr{L}$ of the background curvature is always less than or of order the radius of curvature $\mathscr{R}_B$, equation (20) implies the claimed result:

$$h \lesssim \lambda/\mathscr{R}_B \lesssim \lambda/\mathscr{L} \ll 1. \tag{21}$$

This permits one to study the subsequent propagation of the waves, once

they are fully formed, using a linearized approximation to the Einstein field equations (Sections 35.13 and 35.14 of MTW): (i) one introduces a field $\bar{h}_{\alpha\beta}$ (which is actually the trace-reversed contribution of the waves to the spacetime metric in a suitable gauge). This field is defined to be equal to $h^{\rm TT}_{\alpha\beta}$ in the region from which the waves are propagating (the local wave zone in the case of isolated sources; the very early universe in the case of primordial waves). One then evolves the field $\bar{h}_{\alpha\beta}$ out into the surrounding universe and to earth using the curved-spacetime wave equation (equation (35.64) of MTW)

$$\bar{h}_{\alpha\beta|\mu}{}^{|\mu} + g^{\rm B}_{\alpha\beta}\bar{h}^{\mu\nu}{}_{|\nu\mu} - 2\bar{h}_{\mu(\alpha}{}^{|\mu}{}_{|\beta)} + 2R^{\rm B}_{\mu\alpha\nu\beta}\bar{h}^{\mu\nu} - 2R^{\rm B}_{\mu(\alpha}\bar{h}_{\beta)}{}^{\mu} = -16\pi\delta T_{\alpha\beta}. \tag{22}$$

Here $g^{\rm B}_{\alpha\beta}$ is the background metric, the subscript and superscript $|$ denote covariant derivatives with respect to the background metric, $R^{\rm B}_{\alpha\beta}$ and $R^{\rm B}_{\alpha\beta;\delta}$ are the Ricci and Riemann curvature tensors of the background; and $\delta T_{\alpha\beta}$ is the perturbation in the non-gravitational stress-energy tensor produced by the trace-reversed metric perturbation $\bar{h}_{\alpha\beta}$ itself.

Although the field $\bar{h}_{\alpha\beta}$ initially is wavelike and thus has $\lambda\!\!\!^{-} \ll \mathscr{L} \lesssim \mathscr{R}_{\rm B}$, it might propagate into regions where the background has very short lengthscales, $\mathscr{L} \lesssim \lambda\!\!\!^{-}$. (For example, the waves produced by the Crab pulsar, with $\lambda\!\!\!^{-} \sim 1000$ km may propagate through a massive white dwarf with $\mathscr{L} \sim 1000$ km and $\mathscr{R}_{\rm B} \sim 30\,000$ km, or even through a neutron star with $\mathscr{L} \sim 10$ km and $\mathscr{R}_{\rm B} \sim 30$ km). If this happens, one need not worry. The wave equation (22), because it depends for its validity only on the weakness of the field $\bar{h}^{\alpha\beta}$ and not on the shortwave assumption, remains valid and carries the field through the region of short background lengthscale (where, strictly speaking, it is no longer a gravitational wave), and thence onward into regions of long lengthscale (where it once again is a gravitational wave).

In those regions where $\bar{h}_{\alpha\beta}$ is a wave, i.e. has $\lambda\!\!\!^{-} \ll \mathscr{L}$, one can compute from it the gravitational-wave field $h^{\rm TT}_{jk}$ by a very simple prescription: introduce the proper reference frame of a specific observer; and in that frame discard the time-time and time-space parts of $\bar{h}_{\alpha\beta}$, and algebraically project out from the space-space parts those pieces that are transverse to the propagation direction and are trace-free. The result will be $h^{\rm TT}_{jk}$. For further discussion and justifications see Box 35.1 of MTW and Section 2.4.2 of Thorne (1983).

As I shall discuss below, the effects of the wave-stimulated stress-energy perturbations $\delta T_{\alpha\beta}$ are never large enough to be astrophysically important, so one can ignore them in propagation calculations. Moreover, in regions

(nearly everywhere) where $\lambda\ll\mathscr{R}_B$, the contributions of the background curvature tensors $R^B_{\mu\alpha\nu\beta}$ and $R^B_{\mu\alpha}$ to the wave equation can be ignored, and one can specialize the gauge so as to make $\bar{h}_{\alpha\beta}$ divergence free. The wave equation (22) then assumes the simplest form imaginable:

$$\Box\bar{h}_{\alpha\beta}\equiv\bar{h}_{\alpha\beta|\mu}{}^{\mu}=0. \tag{23}$$

When, in addition, the radius of curvature of the wave fronts is large compared to λ (as is true everywhere except near the very rare focal points of gravitational lenses), the wave equation (23) can be solved easily by the techniques of geometric optics (Isaacson, 1968*a*; Exercise 35.15 of MTW; Section 2.5 of Thorne, 1983): the field $\bar{h}^{\alpha\beta}$ propagates along null rays; its polarization is parallel transported along the rays; and its amplitude, like the amplitude of light, varies along each ray as 1/(the radius of curvature of the wave front).

Because good gravitational lenses are so rare in the real universe, and because regions (black holes and neutron stars) with very strong curvature ($\mathscr{R}_B\lesssim\lambda$) are so rare and so small, the waves from almost every source will propagate to earth via pure geometric optics. Moreover, because the universe is almost globally Lorentz (flat) in its background geometry on lengthscales small compared to the Hubble distance, for sources at distances much less than Hubble (at cosmological redshifts $z\ll 1$), this geometric optics propagation will produce a simple $1/r$ falloff of amplitude and will preserve the wave form (the dependence on retarded time) and the polarization. More specifically, it will produce a gravitational-wave field in TT coordinates with the simple form

$$h^{TT}_{jk}=\frac{A^{TT}_{jk}(t-r,\,\theta,\,\phi)}{r}, \tag{24}$$

where r is distance to the source and θ, ϕ are spherical polar angles centered on the source. For a slow-motion source the function A^{TT}_{jk} is just twice the TT part of the second time derivative of the source's quadrupole moment, as one sees trivially by matching (24) onto (12) in the source's local wave zone.

For a source at a large cosmological redshift $z\gtrsim 1$, if one approximates the background spacetime geometry by that of a Friedmann–Robertson–Walker cosmological model, the geometric-optics propagation produces the same effects for gravity waves as for light: (i) the magnitude of h^{TT}_{jk} falls off with the same $1/R$ behavior as for light, where R is $(1/2\pi)\times$ (circumference of a sphere passing through the earth and centered on the source, at the time the waves reach earth) (equations (29.28)–(29.33) of MTW); (ii) the polarization, like that of light in vacuum, is parallel transported radially

from source to earth; and (iii) the time dependence of the wave form is unchanged by propagation, except for a frequency-independent redshift $f_{\text{received}}/f_{\text{emitted}} = 1/(1+z)$. For further details and derivations see Section 2.5.4 of Thorne (1983) or Section 7.2 of Thorne (1977).

9.3.5 A catalog of wave-propagation effects

In principle gravitational waves can experience almost all the familiar peculiarities of propagation that electromagnetic waves experience. Here I shall enumerate those that have been studied, mention briefly their importance or unimportance, and give references for further detail.

(a) *Absorption, scattering and dispersion by matter and electromagnetic fields*

Gravitational waves are so weakly absorbed by matter (Section 2.4.3 of Thorne, 1983) that absorption is astrophysically important only near the Planck era of the big-bang (Section 7.2 of Zel'dovich and Novikov, 1983). For experimenters, however, the tiny absorption that should occur in a gravity wave detector is very important; see Section 9.5.2(b) below.

The scattering of gravitational waves by matter is also so weak that it is never astrophysically important, except near the Planck era. In the analysis of resonant gravity wave detectors, however, scattering is conceptually important: the method of detailed balance (an idealized calculation in which a resonant detector is driven by monochromatic waves into vibrations of such unrealistically high amplitude that it reradiates at the same rate as it absorbs – i.e. it scatters the waves strongly) is a powerful way of computing the cross-section of a resonant detector; see Section 37.7 of MTW.

Coherent scattering by a medium produces dispersion – i.e. frequency-dependent propagation speeds. Although dispersion is very important for electromagnetic waves, it is never of any importance in the real universe for gravitational waves (Section 2.4.3 of Thorne, 1983). It is instructive, however, to imagine and study theoretically an unrealistic form of matter ('respondium') with such strong dispersion that it actually reflects gravitational waves (Press, 1979).

For detailed analyses of absorption, scattering, and/or dispersion by specific kinds of matter see the following references and the references cited therein: black holes, Matzner *et al.* (1985); De Logi and Kovacs (1977); neutron stars and other stars, Linet (1984); elementary particles, Section 7.2 of Zel'dovich and Novikov (1983); a magnetized plasma, Macedo and Nelson (1983); a uniform medium, Esposito (1971*a*,*b*), Papadopoulos and Esposito (1985), Szekeres (1971), Section 4.2 of Grishchuk and Polnarev

(1980), Section 2.4.3 of Thorne (1980); and a medium with boundaries, Dyson (1969), Carter and Quintana (1977).

Because gravitational and electromagnetic waves should propagate with the same speed, they can interact in a coherent way (Gertsenshtein, 1962). The interaction is so weak, however, that a substantial transformation of one into the other requires propagation over a distance of order the radius of curvature of the background spacetime which their own energy density produces. Thus, such coherent interaction is not likely ever to be important in the real universe – except possibly in gravity-wave detectors; see Section 9.5.4(a) below. For a review of the extensive literature on this subject see Grishchuk and Polnarev (1980). For a brief pedagogical discussion see Section 17.9 of Zel'dovich and Novikov (1983).

(b) *Scattering by background curvature, and tails of waves*

In regions where the background radius of curvature is comparable to or shorter than the reduced wavelength, $\mathscr{R}_B \lesssim \lambda\!\!\!^{-}$, the background strongly scatters the waves. This is very important in some sources of waves, as the waves are trying to form; for example, it is responsible for the normal-mode vibrations of black holes (Press, 1971), and it leads to the formation of 'tails' of the waves in a source's near zone (Price, 1972*a*,*b*; Thorne, 1972; Cunningham, Price and Moncrief, 1979) and to radiative tails in the wave zone (Leaver, 1986*b*; Blanchet, 1987) – which, however, are not likely ever to be observationally important.

(c) *Gravitational focusing*

Lumps of background curvature associated with black holes, stars, star clusters, and galaxies will focus gravitational waves in precisely the same manner as they focus electromagnetic waves; and just as this focusing is observationally important for the light and radio waves from a few very distant quasars, so it might also be important for very distant discrete sources of gravitational waves. Focusing by the sun, in the case of waves with sufficiently short wavelength can be significant, but not at earth; the focal point lies farther out in the solar system, near the orbit of Jupiter (Cyranski and Lubkin, 1974).

(d) *Diffraction*

Near the focal point of a gravitational lens the waves cease to propagate along null rays and begin to diffract, thereby lessening the strength of the focusing. The analysis of this is no different for gravitational waves than for

electromagnetic or scalar waves, since polarization plays no important role. Diffraction causes focusing to be significant only when the lens has a gravitational radius $2M$ that is larger than or of order the waves' reduced wavelength $\lambda\!\!\!^{-}$. For an order of magnitude discussion see, e.g. Section 2.6.1 of Thorne (1983); for full details see Bontz and Haugan (1981).

(e) *Parametric amplification by background curvature*

In regions of a dynamical spacetime (e.g. the expanding universe) in which the characteristic wavelength $\lambda\!\!\!^{-}$ of gravitational waves is larger than or comparable to the background radius of curvature $\mathscr{R}_{\rm B}$, $\lambda\!\!\!^{-} \gtrsim \mathscr{R}_{\rm B}$, the waves can be parametrically amplified by interaction with the dynamical background (Grishchuk, 1974, 1975*a*,*b*, 1977; Grishchuk and Polnarev, 1980). Viewed quantum mechanically, the interaction causes stimulated emission of new gravitons. This effect may well have enabled the expansion of the universe to amplify vacuum fluctuations from the big-bang singularity (from the Planck time) into a strong, stochastic background of gravitational waves today; see Section 9.4.3(d) below.

(f) *Non-linear coupling of the waves to themselves (frequency doubling, etc.)*

Because general relativistic gravity is non-linear, there is a non-linear coupling of gravitational waves to themselves; and in principle this leads to such non-linear conversion processes as frequency doubling. However, in practice these effects are not important in regions where the waves *are* waves (where $\lambda\!\!\!^{-} \ll \mathscr{L}$). This is because, in such regions, the dimensionless amplitude of the waves is very small compared to unity (equation (21) above). For a more detailed discussion see Section 2 of Thorne (1985).

(g) *Generation of background curvature by the waves*

The generation of background curvature by the stress-energy of the waves (Isaacson, 1968*b*, MTW Section 35.15) is important in cosmological models in any epoch when the waves are sufficiently strong that their energy density is comparable to that of matter; see, e.g. Hu (1978) and Chapter 17 of Zel'dovich and Novikov (1983). It is also important in a 'geon' – i.e. a bundle of gravitational waves that is held together by its own gravitational pull on itself (Wheeler, 1962; pp. 409–38 of Wheeler, 1964; Brill and Hartle, 1964). But geons surely do not exist in the real universe; they are only theoretical entities, useful for exploring issues in fundamental physics.

Wave-produced background curvature is also important in the idealized situation where one plane-fronted wave gets focused by passing through

another ('plane-wave collision'): the focusing itself is produced by the wave-generated background curvature. Moreover, if the waves are precisely planar, a spacetime singularity forms at the focal plane (Khan and Penrose, 1971; Szekeres, 1972; Nutku and Halil, 1977; Tipler, 1980; Matzner and Tipler, 1984; Chandrasekhar and Xanthopoulos, 1986; Yurtsever, 1987*a*), and the generation of background curvature plays a key role in the singularity. In the more realistic case (which, however, almost certainly does not occur in the real universe except conceivably near the big-bang), in which the waves are almost planar but die out slowly at large transverse distances, if the transverse size is sufficiently large compared to the initial wave amplitude, then the focusing probably still drives the amplitude up far enough – before diffraction can act – to make background curvature generation become strong and force a singularity to form (Yurtsever, 1987*b*).

9.3.6 Wave propagation in an idealized, asymptotically flat universe

The split of wave propagation and wave generation into two separate calculations is worrisome to those physicists who seek the highest levels of rigor. Unfortunately, in analyses of waves in the real universe, where the background spacetime is complex, the split is essential. Progress cannot be made without it. However, the split can be avoided to some extent in an idealized 'universe', where the source of interest resides alone in an otherwise empty and asymptotically flat spacetime.

In such an idealized universe, and only there, it has been possible to treat wave generation and wave propagation simultaneously, with a single, elegant formalism that spans the weak-gravity portions of the wave-generation region, and all of the local wave zone and distant wave zone. In my opinion, the nicest version of this unified formalism is a variant of the post-Minkowski formalism due to Blanchet and Damour (1986) and Blanchet (1987*a*). See Damour (1986) for a review of this and of earlier work by others. This formalism is especially nice because, although it entails an expansion in the strength of the gravitational field, the expansion has been carried out to all orders, and a number of beautiful theorems have been proved about it.

Also of great interest, elegance, and beauty, in an idealized asymptotically flat universe, are expansions of the spacetime curvature along the outgoing light cone in inverse powers of the distance to the source (Bondi, 1960; Bondi, van der Burg and Metzner, 1962; Sachs, 1962, 1963; Penrose, 1963*a*,*b*; Newman and Penrose, 1965). Such expansions, carefully

formulated and combined with conformal transformations that bring 'infinity' in to finite locations, reveal asymptotic structures of asymptotically flat spacetime that illuminate the properties of gravitational radiation. For recent reviews and references see Newman and Tod (1980), Walker (1983), Schmidt (1979, 1986), Ashtekar (1984).

9.3.7 Exact, analytic solutions for wave generation and propagation

Those who seek high rigor (and even less rigorous people like me) have also found pleasure in the few existing exact, analytic solutions of the Einstein field equations that describe sources which radiate into asymptotically flat spacetime (e.g. Bicak, 1968; Bicak, Hoenselaers and Schmidt, 1983; Bicak, 1985; and references therein); and in idealized exact solutions for cylindrical waves (Einstein and Rosen, 1936; Weber and Wheeler, 1957) and for planar waves (e.g. Rosen, 1937; Bondi, Pirani and Robinson, 1959; Ehlers and Kundt, 1962; Sections 35.9–35.12 of MTW). Also pleasing for its rigor is the extreme limit of geometric optics, where the wavelength becomes so short that the radiation is compacted into a *gravitational shock wave* (e.g. Pirani, 1957; Papapetrou, 1977 and references therein).

9.4 Astrophysical sources of gravitational waves

In the real universe it is useful to divide the anticipated gravitational waves into three classes, according to their temporal behaviors: *bursts*, which last for only a few cycles, or at most for times short compared to a typical observing run; *periodic waves*, which are superpositions of sinusoids with frequencies f_i that are more or less constant over times long compared to an observing run; and *stochastic waves*, which fluctuate stochastically and last for a time long compared to an observing run.

In this section I shall describe the present knowledge and speculations about various sources of gravitational waves. Attention will be restricted to those sources which are most promising for detection by present or planned detectors, with Section 9.4.1 focusing on burst sources, 9.4.2 on periodic sources, and 9.4.3 on stochastic sources.

As was emphasized in Section 9.1.1, for each source, with the single exception of binary stars and their coalescences (Section 9.4.1(e) below), either (i) the strength of the source's waves for a given distance from earth is uncertain by several orders of magnitude; or (ii) the rate of occurrence of that type of source, and thus the distance to the nearest one, is uncertain by several orders of magnitude; or (iii) the very existence of the source is uncertain. Despite these uncertainties, it is important in the wave-detection effort to estimate as best one can, for each source, the strengths of the waves

bathing the earth. Such estimates will be stated below, with references; and they are collected together below in Figs. 9.6 (burst sources), 9.7 (periodic sources) and 9.6 (stochastic sources).

9.4.1 Burst sources

(a) *Bursts with and without memory*

As Braginsky and Grishchuk (1985) have recently emphasized, gravitational-wave bursts can be subdivided into two classes: *normal bursts*, in which h_{jk}^{TT} begins zero before the burst and returns to zero afterward, and *bursts with memory*, in which h_{jk}^{TT} (by convention) begins zero before the burst and then settles down into a non-zero, constant value Δh_{jk}^{TT} (the burst's 'memory') after the burst is over.

Physically, a burst with memory arises whenever the source, before the burst or afterward or both, consists of several free bodies that are moving with uniform velocities relative to each other. For example, the explosion of a star into several pieces will produce a burst with memory, as will the collision of two freely moving (not binary) stars or black holes, or the gravitational scattering of a star by a black hole; but the birth of a black hole in non-spherical stellar collapse will produce a burst without memory. In all cases, the 'memory' is the change in the total '$1/r$' coulomb-type gravitational field of the source (Braginsky and Thorne, 1987). For example, in the quadrupole approximation, when one takes account of the fact that one can add a time-independent constant to h_{jk}^{TT} ('gauge change' that moves stuff between the background field and the wave field) so as to make h_{jk}^{TT} zero initially, equations (12) and (13a) give

$$\Delta h_{jk}^{TT} = \Delta \sum_A \left(\frac{4 m_A v_A^j v_A^k}{r} \right)^{STF} . \tag{25}$$

Here the summation is over the free bodies in the system, m_A and v_A^j are the mass and velocity of body A, and Δ denotes the change from before the burst is emitted to afterward.

As is discussed by Braginsky and Thorne (1987), the memory part of a burst, Δh_{jk}^{TT}, can be studied by any detector (with adequate sensitivity) that operates at a frequency lower than the burst's characteristic frequency, $f \lesssim f_c$. Put equivalently, one can think of a burst with memory as having a signal that extends down to all frequencies below f_c (cf. the 'zero-frequency limit' discussed by Smarr, 1977*b* and by Bontz and Price, 1979).

Current prejudice suggests that the strongest of burst sources (and thus

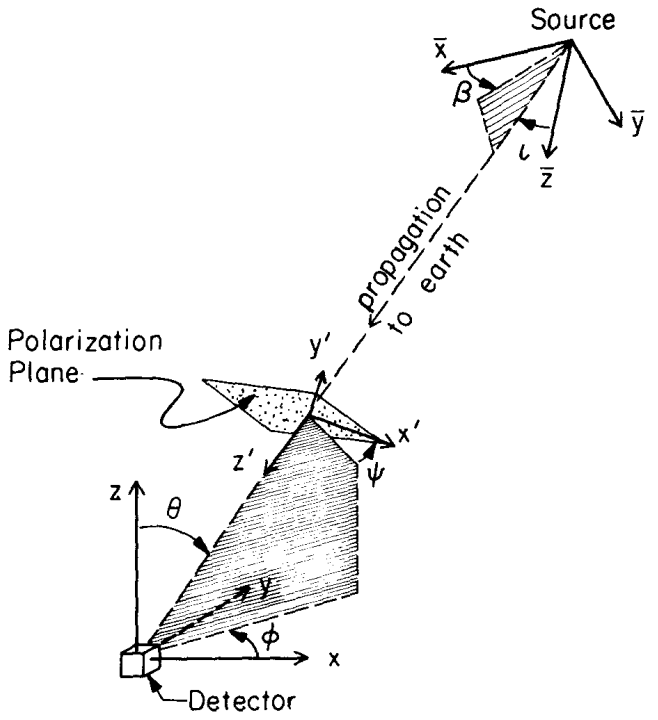
the most interesting) may produce normal bursts rather than bursts with memory; but this prejudice could perfectly well be wrong. In accord with this prejudice, the remainder of this section will focus on normal bursts.

(b) *Characterization of the waves from a normal burst source and the noise in a detector searching for them*

Burst sources are best characterized by their full wave forms $h_{jk}^{TT}(t)$. However, when comparing with detector sensitivities, it is helpful to have a more compact characterization. Past discussions have used a loosely defined 'characteristic amplitude' h_c and 'characteristic frequency' f_c. However, the factors of order 3 that are glossed over by the loose definitions are beginning to be important in the planning of gravity-wave searches, especially in the case of the inspiral and coalescence of binary neutron stars (subsection (e) below); and I therefore shall be quite careful in my definitions of h_c and f_c and subsequently in my corresponding discussions of detector sensitivities.

As an aid to defining h_c and f_c with care, I show in Fig. 9.2 a diagram of the emission, propagation, and absorption of the wave. The source has

Fig. 9.2. The angles $\iota, \beta, \psi, \theta, \phi$ which characterize the emission, propagation and detection of a gravitational wave.



preferred local Cartesian axes $(\bar{x}, \bar{y}, \bar{z})$ with respect to which its internal structure is especially simple. We shall denote by (ι, β) the direction toward earth (spherical polar angles) relative to those axes. The detector, similarly, has preferred local Cartesian axes (x, y, z) with respect to which its internal structure is especially simple; and we shall denote by (θ, ϕ) the direction toward the source (spherical polar angles) relative to those detector axes. The waves themselves, as they pass through the detector, are most simply described in a third set of Cartesian axes (x', y', z') with origin at the detector's center of mass, z' axis along the waves' propagation direction, and x' and y' axes so oriented in the polarization plane as to make the wave forms $h_+ = h^{TT}_{x'x'} = -h^{TT}_{y'y'}$, and $h_\times = h^{TT}_{x'y'} = h^{TT}_{y'x'}$, especially simple. We shall denote by ψ the angle between the x' polarization axis and the $\phi = 0$ plane.

From a model for the source one can compute the wave forms $h_+(t-z'; \iota, \beta)$ and $h_\times(t-z'; \iota, \beta)$ arriving at the detector. If the detector is small compared to a reduced wavelength, as we shall assume throughout Section 9.4 and in Sections 9.5.1–9.5.3 but not 9.5.4–9.5.6, then it will feel a simple linear combination of h_+ and $h_\times$; i.e. it will detect the wave form

$$h(t) = F_+(\theta, \phi, \psi)h_+(t; \iota, \beta) + F_\times(\theta, \phi, \psi)h_\times(t; \iota, \beta). \tag{26}$$

Here F_+ and $F_\times$ are detector beam-pattern functions (to be discussed in Sections 9.5.2 and 9.5.3 below), which depend on the direction of the source (θ, ϕ) and the orientation ψ of the polarization axes relative to the detector's orientation (Fig. 9.2) and which have values in the range $0 \leqslant |F_A| \leqslant 1$. There will be some special choice of θ, ϕ, ψ for which $F_+ = 1$ and $F_\times = 0$; we shall call this the 'optimum source direction and polarization' for the + mode of the waves.

In Section 9.5 we shall characterize the noise in a detector by a frequency-dependent spectral density $S_h(f)$ (with dimensions Hz^{-1}) defined as follows: if a precisely sinusoidal gravitational wave with known phase α, known frequency $f > 0$ and unknown rms amplitude h_0,

$$h = (2)^{\frac{1}{2}} h_0 \cos(2\pi f t + \alpha), \qquad \alpha = \text{const.}, \tag{27a}$$

impinges on the detector, and if the experimenters seek to detect the wave by Fourier analyzing the detector output with a bandwidth Δf (integration time $\hat{\tau} = 1/\Delta f$), then the amplitude signal-to-noise ratio will be

$$\frac{S}{N} = \frac{h_0}{[S_h(f)\Delta f]^{\frac{1}{2}}}. \tag{27b}$$

In much of the gravitational-wave literature this $S_h(f)$ is denoted $[h(f)]^2$; i.e. $h(f) \equiv [S_h(f)]^{\frac{1}{2}}$ (with dimensions $\mathrm{Hz}^{-\frac{1}{2}}$).

In this section we are interested not in periodic gravitational waves, but rather in bursts – i.e. in waves with complicated time dependences $h(t)$ and with durations short compared to the experimenters' observation time. We shall assume that the wave form $h(t)$ is known, and that our only question is whether the wave is really present or not – with some postulated starting time t_0. (Later we shall discuss the statistical consequences of an unknown starting time.) There is a well-known optimal strategy for searching for such a wave in the presence of noise with known spectral density:

(i) One constructs a *Wiener optimal filter* $K(t)$, that function of time whose Fourier transform $\tilde{K}(f)$ is the same as the transform $\tilde{h}(f)$ of the signal $h(t)$, weighted by $1/S_h(f)$ (so that noisy frequencies are suppressed):

$$\tilde{K}(f) \equiv \frac{\tilde{h}(f)}{S_h(f)}, \qquad \tilde{h}(f) \equiv \int_{-\infty}^{+\infty} h(t)\, e^{i2\pi f t}\, dt, \tag{28a}$$

$$K(t) = \int_{-\infty}^{+\infty} \tilde{K}(f)\, e^{-i2\pi f t}\, df. \tag{28b}$$

(ii) One then takes the output of the detector, which includes noise and possibly signal; from it one computes the gravitational wave form $h_{\text{output}}(t)$ that would have been required to produce the observed output if there were no noise present; and one then computes a quantity

$$W = \int_{-\infty}^{+\infty} K(t - t_0) h_{\text{output}}(t)\, dt. \tag{28c}$$

(iii) This quantity will have root-mean-square contribution N from noise; and if the signal was actually present with starting time t_0, it will have a signal contribution S (i.e. $W = N + S$), with squared signal-to-noise ratio

$$\frac{S^2}{N^2} = \int_0^{\infty} \frac{2|\tilde{h}(f)|^2}{S_h(f)}\, df. \tag{29}$$

See Wiener (1949), Sections 25–27 of Wainstein and Zubakov (1962), Kafka (1977), and Michelson and Taber (1984) for proofs and discussion. The integral in (29) is from 0 to ∞ rather than $-\infty$ to $+\infty$ and there is a factor 2 present because $S_h(f)$ is a 'one-sided spectral density' (defined only for $f \geqslant 0$; negative frequencies are folded into positive). The squared signal-to-noise ratio (29) will be the basis for our definitions of h_c and f_c.

Expression (29) is the squared signal-to-noise ratio for a specific ('fiducial') source at some specific distance r_0 from earth. Suppose that space is uniformly filled with sources, all identical to this fiducial source but with random directions (θ, ϕ), orientations (ι, β), and polarization angles ψ. Suppose, further, that inside the source's distance r_0 there is, on average, one

burst each D_0 days. Then what, on average, will be the squared signal-to-noise ratio $(S^2/N^2)_{\text{strongest}}$ of the strongest burst that occurs each D_0 days? One might expect the answer to be the fiducial S^2/N^2 of (29), averaged over all the angles θ, ϕ, ψ, ι, β. Not so. There is a statistical preference for directions and polarizations that give larger values of S^2/N^2, because they can be seen out to greater distances where the event rate is greater. This effect gives, assuming the event rate goes up as r^3 and h_{jk}^{TT} goes down as $1/r$, $(S^2/N^2)_{\text{strongest}} = \langle S^3/N^3 \rangle^{\frac{2}{3}}$ where $\langle \cdots \rangle$ denotes an average over randomly distributed angles. Now, the $\frac{2}{3}$ power is a pain to deal with in subsequent calculations, so we shall switch to a straightforward angular average $\langle S^2/N^2 \rangle$ and to compensate we shall insert a multiplicative factor $\frac{3}{2}$, which is (approximately) the ratio of $\langle S^3/N^3 \rangle^{\frac{2}{3}}$ to $\langle S^2/N^2 \rangle$ if both source and detector have quadrupole beam patterns ('slow-motion, poor-antenna regime'). This, together with $\langle F_+^2 \rangle = \langle F_\times^2 \rangle$ and $\langle F_+ F_\times \rangle = 0$ (true for any quadrupole-beam-pattern gravity-wave detector), permits us to write $(S^2/N^2)_{\text{strongest}}$ in the following approximate form, where we omit the subscript 'strongest' for ease of notation and where $\langle |\tilde{h}_+|^2 + |\tilde{h}_\times|^2 \rangle$ is averaged over source angles (ι, β):

$$\frac{S^2}{N^2} \cong 3\langle F_+^2(\theta, \phi, \psi) \rangle \int_0^\infty \frac{\langle |\tilde{h}_+|^2 + |\tilde{h}_\times|^2 \rangle}{S_h(f)} \, df. \tag{30}$$

We shall now specialize, for the remainder of this subsection, to gravity-wave searches with broad-band detectors, i.e. detectors for which the noise $S_h(f)$ is small over a band of frequencies $f_{\min} \lesssim f \lesssim f_{\max}$, with $f_{\max} \gtrsim 2 f_{\min}$. Narrow-band detectors will be treated separately in Section 9.5.2(b) below. For broad-band detectors equation (30) motivates us to define the characteristic frequency and amplitude of our fiducial source, at distance r_0, by

$$f_c \equiv \left[\int_0^\infty \frac{\langle |\tilde{h}_+|^2 + |\tilde{h}_\times|^2 \rangle}{S_h(f)} \, df \right]^{-1} \left[\int_0^\infty \frac{\langle |\tilde{h}_+|^2 + |\tilde{h}_\times|^2 \rangle}{S_h(f)} \, f \, df \right], \tag{31a}$$

$$h_c \equiv \left[3 \int_0^\infty \frac{S_h(f_c)}{S_h(f)} \langle |\tilde{h}_+|^2 + |\tilde{h}_\times|^2 \rangle f \, df \right]^{\frac{1}{2}}, \tag{31b}$$

where the average is over randomly distributed source orientation angles ι, β. Similarly, we shall define a characteristic detector noise amplitude at frequency f by

$$h_n(f) \equiv \frac{[f S_h(f)]^{\frac{1}{2}}}{\langle F_+^2(\theta, \phi, \psi) \rangle^{\frac{1}{2}}}. \tag{32}$$

In terms of these quantities equation (30) reads

$$\frac{S}{N}=\frac{h_c}{h_n(f_c)}. \tag{33}$$

This is the S/N of the strongest burst that arrives at earth, on average, at the same rate as bursts occur inside the fiducial source's distance r_0.

In Fig. 9.4 below we characterize detector burst sensitivities not by $h_n(f)$, but rather by a quantity $h_{3/yr}$ that answers the following question: *What is the characteristic strength* $h_c=h_{3/yr}$ *of a source with sufficiently large S/N (equation* (33)*) that, if it is seen three times per year (once each* 10^7 *seconds) by two identical detectors operating in coincidence, we can be* 90% *confident the detectors are not just seeing their own noise.* The use of two detectors permits (we shall presume) the elimination of non-Gaussian noise. Then the Gaussian distribution of the amplitude noise, together with the use of two detectors and the fact that in $\hat{\tau}=10^7$ s the experimenters must try roughly $2\pi f_c\hat{\tau}\simeq f_c/10^{-8}$ Hz starting times t_0 for their Wiener optimal filter, implies that we require $S/N\simeq[\ln(f_c/10^{-8}\text{ Hz})]^{\frac{1}{2}}$; and, correspondingly,

$$h_{3/yr}\simeq\left[\ln\left(\frac{f_c}{10^{-8}\text{ Hz}}\right)\right]^{\frac{1}{2}}\frac{[f_cS_h(f_c)]^{\frac{1}{2}}}{\langle F_+^2(\theta,\phi,\psi)\rangle^{\frac{1}{2}}}\simeq(3\text{–}5)h_n(f_c). \tag{34}$$

Here the range 3–5 corresponds to the range of frequencies of interest for most burst searches, 10^{-4} Hz–10^{+4} Hz. Note that because the events being sought are so far out on the tail of the Gaussian probability distribution, changing by a factor 10 or 100 the number of trial starting times t_0, or asking for 99% or 99.9% confidence rather than 90%, would have a negligible effect on this $h_{3/yr}$.

It is often useful to rewrite the f_c and h_c of equations (31) in terms of the energy flux per unit frequency $dE_{GW}/dA\,df$ carried past the detector by the waves. From equation (10) for the waves' stress-energy tensor together with Parseval's theorem we infer that

$$\frac{dE_{GW}}{dA\,df}=\frac{\pi}{2}f^2(|\tilde{h}_+(f)|^2+|\tilde{h}_\times(f)|^2), \tag{35}$$

where the extra factor 2 comes from folding negative frequencies into positive (so $f>0$). When this quantity is averaged over all directions ι, β, it gives $(4\pi r_0^2)^{-1}\,dE_{GW}/df$, where r_0 is the distance to the source; and consequently equations (31) become

$$f_c=\left[\int_{-\infty}^{\infty}\frac{dE_{GW}/df}{fS_h(f)}\,d\ln f\right]^{-1}\left[\int_{-\infty}^{\infty}\frac{dE_{GW}/df}{S_h(f)}\,d\ln f\right], \tag{36a}$$

$$h_c = \left[\int_{-\infty}^{\infty} \frac{S_h(f_c)}{S_h(f)} \frac{3}{2\pi^2 r_o^2} \frac{dE_{GW}}{df} \, d\ln f \right]^{\frac{1}{2}}. \tag{36b}$$

For most burst sources (e.g. supernovae) the wave form is so uncertain that a careful calculation of h_c is unjustified. In such cases it is useful to reexpress h_c (equation (36b)), approximately, in terms of the total energy ΔE_{GW} radiated:

$$h_c \simeq \left(\frac{3}{2\pi^2} \frac{\Delta E_{GW}/f_c}{r_o^2} \right)^{\frac{1}{2}} = 2.7 \times 10^{-20} \left(\frac{\Delta E_{GW}}{M_\odot c^2} \right)^{\frac{1}{2}} \left(\frac{1\,\text{kHz}}{f_c} \right)^{\frac{1}{2}} \left(\frac{10\,\text{Mpc}}{r_o} \right), \tag{37}$$

where $M_\odot$ is the mass of the sun and 10 Mpc is the distance to the center of the Virgo cluster of galaxies (assuming a Hubble constant of 100 km s^{-1} Mpc^{-1}).

We turn now to a discussion of specific burst sources, and their characteristic frequencies f_c and wave strengths h_c.

(c) *Supernovae (collapse to neutron star)*

Supernovae of 'type II' are believed, with a high level of confidence, to be created by the gravitational collapse, to a neutron-star state, of the cores of massive, highly evolved stars. Supernovae of 'type I', by contrast, are thought to result from nuclear explosions of white dwarfs that are accreting mass from close companions – explosions in which the stellar core probably does not, but might, collapse to a neutron-star state (Woosley and Weaver, 1986; Evans, Iben and Smarr, 1987, and references therein). In addition to these two types of optically observed supernovae, there may well be stellar collapses to a neutron star that produce little optical display ('optically silent supernovae').

The rate of occurrence of supernovae of types I and II is fairly well determined observationally: in our Galaxy roughly one type I each 40 years and one type II each 40 years; out to the distance of the center of the Virgo Cluster of Galaxies (10 Mpc) several type I and several type II per year; rate increasing roughly as (distance)^3 near and beyond Virgo; see, e.g. Tammann (1981). Thus, to have an interesting event rate one must have adequate sensitivity to reach the center of Virgo, about 10 Mpc. Optically silent supernovae could be more frequent, since statistics on normal stars massive enough to form neutron stars when they die, permit a neutron-star birth rate that could be as high as one every four years in our galaxy if mass loss in the late stages of stellar evolution is smaller than normally thought. Alternatively, it is conceivable that optically silent supernovae never occur.

The strengths of the waves from a supernova depend crucially on the

degree of non-sphericity in the stellar collapse that triggers it, and somewhat on the speed of collapse – i.e. on whether the collapse is nearly free-fall ('cold collapse') or is more gentle due to the resistance of thermal pressure ('hot collapse'). Perfectly spherical collapse will produce no waves; highly non-spherical collapse will produce strong waves. Little is known about the degree of non-sphericity in type II (which are surely due to stellar collapse), but current prejudice suggests that the typical type II might be quite spherical and thus poorly radiating. If type I are due to explosion of an accreting white dwarf and, contrary to current thought, the explosion is accompanied by collapse of the stellar core to a neutron star, then the white dwarf might be rapidly rotating due to the accretion, and centrifugal forces might then cause it to collapse very non-spherically and radiate strongly.

In the mid-1970s there was a swing of fashion from believing that the collapse is cold and fast to believing it is hot and slow (e.g. Wilson, 1974; Schramm and Arnett, 1975). Some astrophysicists were aghast at the consequence of this swing: for example, in the highly non-spherical but axisymmetric collapse models of Saenz and Shapiro (1978, their tables 1–4) the total energy radiated as gravitational waves was reduced from $\Delta E_{\rm GW}/M_\odot c^2 \sim 6\times 10^{-3}$ for cold and fast to $\sim 1\times 10^{-5}$ for hot and slow. This boded ill for attempts to detect gravitational waves, the astrophysicists thought. However, closer scrutiny revealed relatively little change in the prospects for detection: the total energy radiated $\Delta E_{\rm GW}$ is a rather poor indicator of detectability. Much more relevant is the amplitude signal-to-noise ratio $S/N = h_{\rm c}/h_{\rm n}(f_{\rm c})$ (equation (33)). The planned LIGO beam detectors, when optimized for frequency $f_{\rm c}$, have $h_{\rm n}(f_{\rm c}) \propto f_{\rm c}$ (Fig. 9.4 and equation (125a)); and this together with equation (37) gives

$$S/N \propto (\Delta E_{\rm GW}/f_{\rm c}^3)^{\frac{1}{2}}. \tag{38}$$

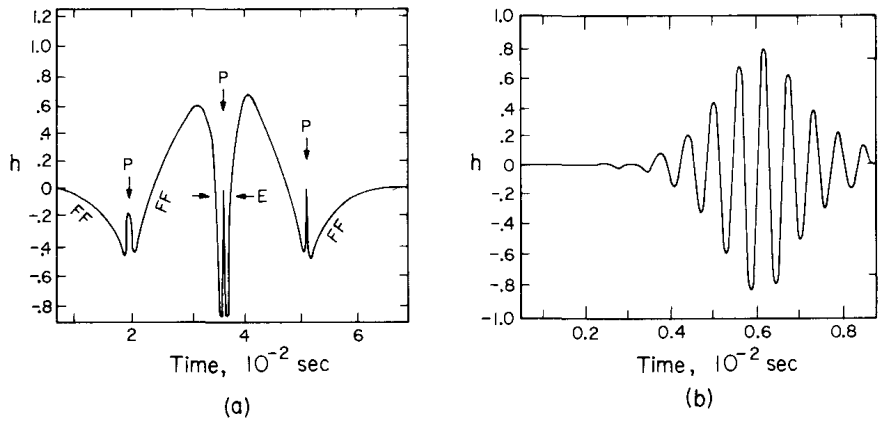
Although the new fashion (hot and slow) corresponded to a reduction in $\Delta E_{\rm GW}$ by a factor 600, the characteristic frequency of the waves also went down (from $\simeq 3000$ Hz to $\simeq 500$ Hz; see Saenz and Shapiro, 1978); and, correspondingly, S/N was reduced by less than a factor 2.

This illustrates the importance of thinking in terms of $h_{\rm c}$, $f_{\rm c}$, and $S/N = h_{\rm c}/h_{\rm n}(f_{\rm c})$ rather than in terms of energy radiated, when evaluating the strengths of gravitational waves.

Corresponding to our poor knowledge of the strengths of the waves from supernovae is a similar poor knowledge of the wave forms. It seems unlikely that theorists will be able to firm up their predictions of the waveforms before observers detect and study them. Thus, it is best to think of the computed wave forms as giving us a tool for translating future observations

into an understanding of what is happening in the stellar core. As an example consider Fig. 9.3, which shows the wave forms computed by Saenz and Shapiro (1978, 1981), using the quadrupole formalism, for two different stellar collapses. In each case – and in general, when one thinks the quadrupole formalism may be accurate – one can invert equation (12) to see how the source's quadrupole moment is behaving. The left curve (Saenz and Shapiro, 1978) shows several epochs labeled FF in which $h_+(t)$ varies approximately as $|t-t_0|^{-\frac{2}{3}}$, corresponding to free-fall motion; and these free-fall epochs are separated by three brief periods with sharply reversed peaks (labeled 'P' in the diagram) corresponding to a sharp acceleration in the direction opposite to the free fall. Considering the timescales of hundredths of a second for the free-fall epoch (characteristic of late stages of collapse to a neutron star) and ~ 0.5 milliseconds for the peak epochs (characteristic of neutron-star pulsation), the natural and correct interpretation is that these waves are from collapse to a neutron star in which the stellar core bounced sharply three times. The fact that the three sharp peaks are all in the same direction (up, not down) indicates that the sharp bounces were all along the same axis. Surely the other axis that projects on our sky should have bounced as well, or at least stopped its collapse; so there should be at least one sharp peak in the down direction. Indeed there is; it is superposed on the central up peak (region labeled E in the diagram). The natural and correct interpretation is that the star was centrifugally flattened by rotation; its pole collapsed fast and bounced three times (up peaks P) while its equator collapsed more slowly and bounced once (down peak E).

Fig. 9.3. Wave forms produced by two very different scenarios for the collapse of a normal star to form a neutron star. Wave form (*a*) is from Saenz and Shapiro (1978); (*b*) is from Saenz and Shapiro (1981).

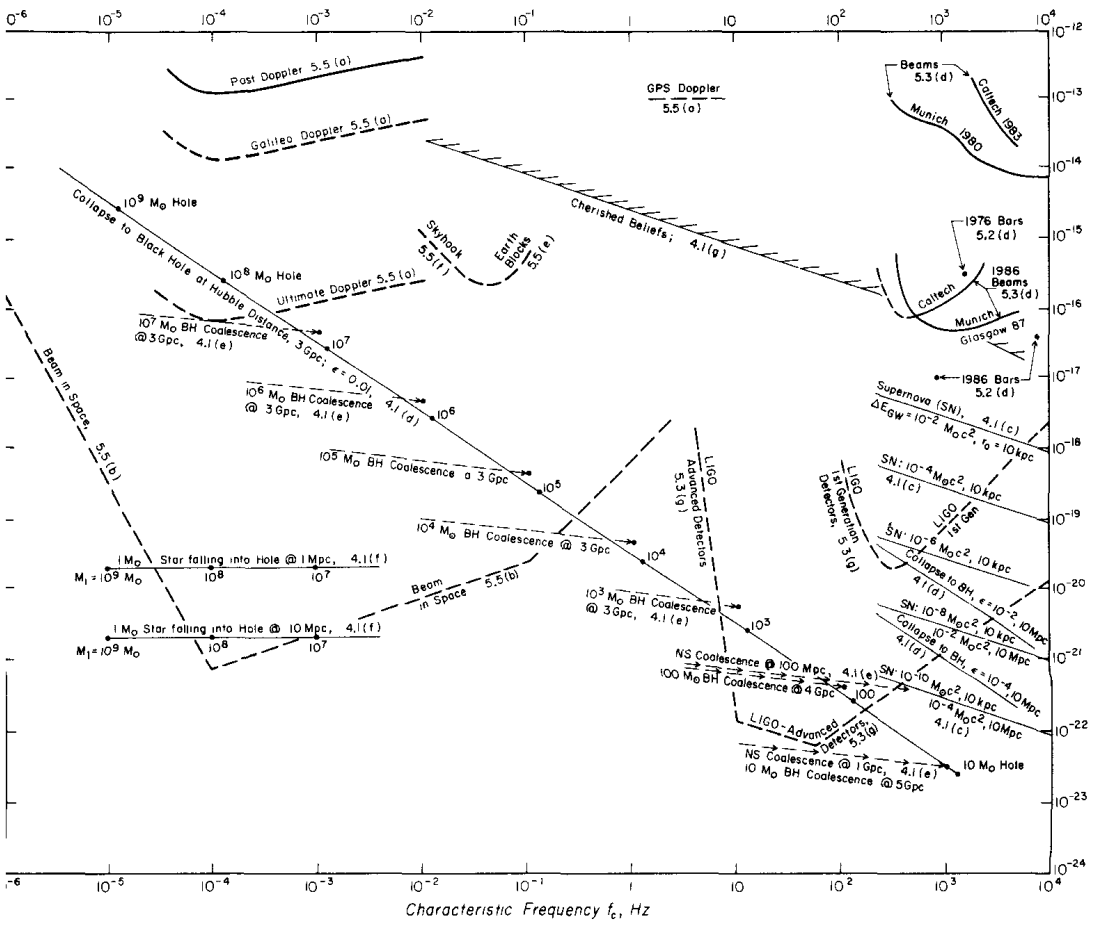


The very different right-hand curve in Fig. 9.3 (Saenz and Shapiro, 1981) implies a quadrupole moment that somehow is driven into sinusoidal oscillations which initially increase in amplitude and then die out. Again, the fact that the period (~ 0.6 ms) is that of a neutron-star pulsation suggests that something is triggering, then damping, such a pulsation. If this wave form was seen roughly one day before an optical supernova was found in the Virgo cluster, one would infer that the waves were from the supernova and one could deduce that the quadrupole-moment oscillations are so large in amplitude that they must have absorbed, say, 10^{-4} of the collapse energy. The natural explanation might be parametric amplification of quadrupole neutron star pulsations by a bouncing stellar collapse, followed by hydrodynamic damping – the process that gave rise to this computed wave form.

A large amount of effort has gone into model calculations of gravitational collapse to a neutron star and the waves it emits; but the effort has not produced a consensus by any means! For a detailed review of the literature up to 1982 see Eardley (1983); for an update on that review see Müller (1984). In the case of rapidly rotating collapse, where the emission should be strongest but the event rate is totally unknown (most collapses *could* be slowly rotating), there are three radically different scenarios and corresponding wave characteristics: (i) the star may remain axisymmetric throughout the collapse. In this case the best current 'wisdom' (but by no means a consensus) comes from calculations by Müller (1982) and is pessimistic. Those calculations predict the strongest emission to come in two different spectral regions: $f_c \sim 1000$ Hz where $\Delta E_{GW} \sim 1 \times 10^{-7} M_\odot c^2$ and $h_c \sim 1 \times 10^{-23}$ (10 Mpc/r_o) due to the initial collapse and bounce; and $f_c \sim 10^4$ Hz where $\Delta E_{GW} \sim 10^{-6} M_\odot c^2$ and $h_c \sim 1 \times 10^{-23}$ (10 Mpc/r_o) due to pulsations of the newly formed neutron star. (ii) The star may become unstable to an '$m=2$ bar-mode' deformation so it rotates end-over-end like an American football. In this case the best current 'wisdom' is more optimistic: the calculations of Ipser and Managan (1984) predict a highly monochromatic emission at $f \sim 1000$ Hz, lasting for ~ 30 cycles and producing $\Delta E_{GW} \sim 3 \times 10^{-4} M_\odot c^2$ and $h_c \sim 5 \times 10^{-22}$ (10 Mpc/r_o). (iii) The collapsing star may become so strongly unstable to non-axisymmetric perturbations that on the way down it breaks up into two or more discrete lumps. Very little is known about this possibility, and radiation reaction from the $m=2$ mode (case ii) might prevent it from occurring at all (Ipser, 1986). Eardley (1983) argues that if it does occur, it may produce quite strong waves: $\Delta E_{GW} \sim$ (a few) $\times 10^{-2} M_\odot c^2$ at $f_c \sim 1000$ Hz, corresponding to $h_c \sim 4 \times 10^{-21}$ (10 Mpc/r_o).

Because this best wisdom is so insecure, Fig. 9.4 shows wave strengths based not on these specific models, but rather on the general equation (37) for several possible values of ΔE_{GW} and r_o, and for the entire range of characteristic frequencies that have shown up in model calculations, $200\ \text{Hz} \lesssim f_c \lesssim 10\,000\ \text{Hz}$. Note that detectability depends strongly on whether the waves come off at low frequencies or high: a factor 8 reduction in ΔE_{GW} can be compensated by a factor 2 reduction in f_c.

Fig. 9.4. The characteristic amplitudes h_c (equation (31b)) and frequencies f_c (equation (31a)) of gravitational waves from several postulated *burst sources* (thin curves), and the sensitivities $h_{3/yr}$ of several existing and planned detectors (thick curves and circles) ($h_{3/yr}$ is the amplitude h_c of the weakest source that can be detected three times per year with 90% confidence by two identical detectors operating in coincidence). The abbreviations BH, NS and SN are used for black hole, neutron star and supernova. The sources are discussed in detail in the indicated subsections of Section 9.4.1, and the detectors in the indicated subsections of Section 9.5.



We note in passing that neutron-star pulsations might be excited not only as part of the star's birth throes, but also as a consequence of a sudden strain release or phase transition in an old neutron star (Thorne, 1978; Ramaty *et al.*, 1980; Haensel, Zdunik and Schaeffer, 1986). Since there are 10^8–10^9 old neutron stars in our galaxy, it is conceivable – though not highly likely – that our galaxy could produce an interesting event rate. To be detectable by the planned LIGOs, the waves would need to have $h_c \gtrsim 10^{-21}$ at $f_c \sim 3000$ Hz corresponding to

$$\Delta E_{GW} \gtrsim 7 \times 10^{45} \text{ erg} \times (10 \text{ kpc}/r_0)^2. \tag{39}$$

(d) *Collapse of a star or star cluster to form a black hole*

As with collapse to form a neutron star, so also for collapse to a black hole, the strengths of the waves produced are highly sensitive to the degree of non-sphericity, and the typical degree of non-sphericity is unknown. Equally unknown in the black-hole case, by contrast with the neutron star, is the frequency of occurrence of such collapses:

It is very likely that black holes exist in our universe with masses throughout the range $2\,M_\odot \lesssim M \lesssim 10^{10}\,M_\odot$ (see Chapter 8 of this book). The holes of lowest mass can only form by direct collapse of a star. Those of higher mass, however, can form by many routes (direct collapse; gradual growth from a small hole by accretion; collision and coalescence of smaller holes; . . .). For discussions of the routes that might occur in a dense galactic nucleus see Blandford (1979) and Rees (1983). Which routes actually occur and how often are almost totally unknown. However, roughly known upper limits on the birth rates of the smallest and the largest holes give – under the assumption that all births are by direct collapse – corresponding upper limits on the rate at which gravity-wave bursts from such collapses hit the earth. For holes of a few solar masses the birth rate under reasonable assumptions (Section 8.3.3 of Chapter 8) should not exceed $\sim \frac{1}{3}$ the birth rate of neutron stars; and correspondingly, at the distance of the Virgo cluster it should not exceed ~ 1/year. Bethe (1986) argues that the rate may actually be of this magnitude. At the other end of the spectrum, holes with masses $M \gtrsim 10^6\,M_\odot$ probably occur only in galactic nuclei; and over its lifetime each galactic nucleus might give birth, at maximum, to only a few such holes. Correspondingly (Thorne and Braginsky, 1976; Blandford, 1979), the maximum rate of collapse-births of supermassive holes, $M \gtrsim 10^6\,M_\odot$, is a few per year throughout the observable universe – i.e. out to the Hubble distance. It is fashionable to believe that the actual rate is much less than this upper limit (e.g. Rees, 1983).

In one respect collapse to a black hole is better understood than collapse to a neutron star: the final object is far simpler, and correspondingly the waves from its vibrations, if they are triggered by the collapse, are far better understood. Detailed calculations suggest, in fact, that black-hole vibrations are rather easy to trigger (e.g. Detweiler, 1977) and that when they are triggered, the most slowly damped one or two quadrupole modes will dominate. Thus, while the details of the initial burst of waves may depend on unknown details of the collapse, the late-time behavior will have a well-established damped oscillatory form from which one can read off the mass of the hole with excellent accuracy and its angular momentum with modest accuracy (Detweiler, 1980; Leaver, 1985, 1986*a*; Stark and Piran, 1986; Piran and Stark, 1986).

As a specific example, Fig. 9.5 shows the waves produced by a specific model of a collapsing, rotating star as computed using numerical relativity techniques by Stark and Piran (1986). The solid curve is the computed waveform, and the dashed curve is a fit to it using a mixture of the two most weakly dampled quadrupole modes of a non-rotating hole. The fact that the fit is so good shows (i) that the final ringdown is, indeed, due to the black-hole vibrations; and (ii) that the hole was not rotating extremely rapidly, i.e. it had $(1-a/M)\gtrsim 0.3$, where a is the specific angular momentum and M the mass. If the hole had had $1-a/M<0.3$, the 'Q' of the hole's oscillations would have been noticeably larger, and the ringdown of the waves would have been noticeably slower. (For details of the normal modes of black holes and the frequencies and damping times of the waves they should produce see Detweiler, 1980, and Leaver, 1985, 1986*a*.)

If collapse to a black hole radiates with an efficiency $\Delta E/Mc^2 \equiv \varepsilon$ and the hole is at a distance r_0 and has a mass M, then the characteristic frequency and amplitude of its waves will be (Stark and Piran, 1986; Piran and Stark, 1986; and equation (37) above)

$$f_c \cong \frac{1}{5\pi M} = (1.3\times 10^4 \text{ Hz})\left(\frac{M_\odot}{M}\right), \tag{40a}$$

$$h_c \cong \left(\frac{15}{2\pi}\varepsilon\right)^{\frac{1}{2}}\frac{M}{r_0} = 7\times 10^{-22}\left(\frac{\varepsilon}{0.01}\right)^{\frac{1}{2}}\left(\frac{M}{M_\odot}\right)\left(\frac{10\text{ Mpc}}{r_0}\right)$$
$$= 1.0\times 10^{-20}\left(\frac{\varepsilon}{0.01}\right)^{\frac{1}{2}}\left(\frac{10^3\text{ Hz}}{f_c}\right)\left(\frac{10\text{ Mpc}}{r_0}\right). \tag{40b}$$

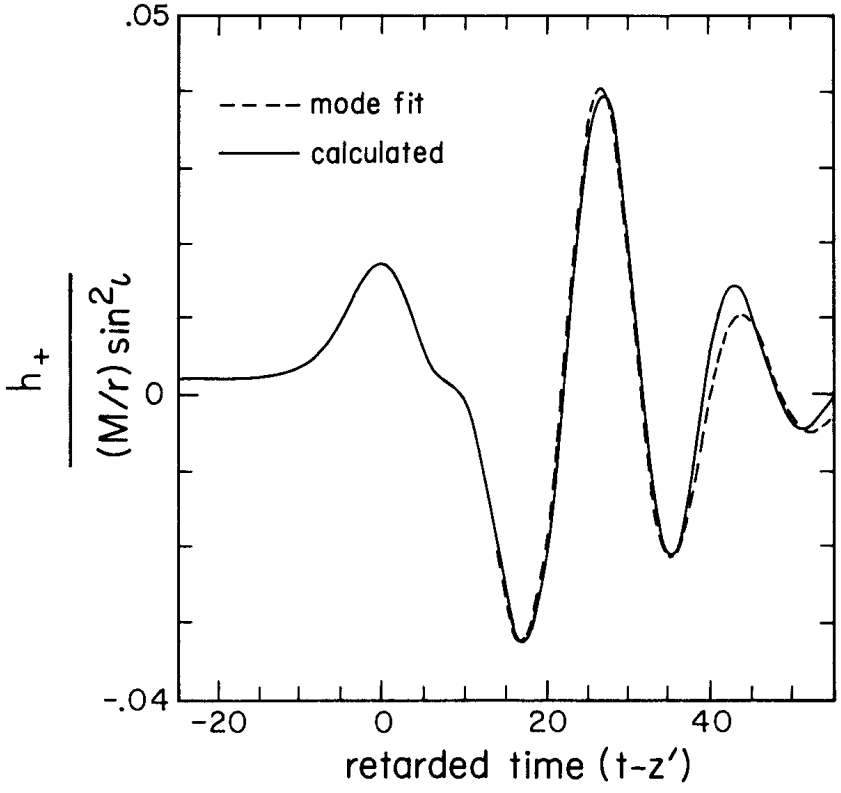
If the collapse is axisymmetric, then the efficiency ε probably does not exceed 7×10^{-4} (Stark and Piran, 1986). However, in the non-axisymmetric case (e.g. formation of an elongated configuration due to rapid rotation, or

bifurcation into one or more lumps during collapse (the 'collapse, pursuit. and plunge' scenario of Ruffini and Wheeler, 1971)), the efficiency might be in the range 0.01–0.1 (see, e.g., Eardley, 1983, and Rees, 1983). The source characteristics (40) are shown in Fig. 9.4 for black-hole births at the Hubble distance and at the distance of Virgo, with efficiencies of $\varepsilon = 10^{-2}$ and 10^{-4}.

(e) *Coalescence of compact binaries (neutron stars and black holes)*

Since a large fraction of all stars are in close binary systems, the dead remnants of stellar evolution may contain a significant number of binary

Fig. 9.5. The gravitational wave form produced by the gravitational collapse of an axisymmetric rotating star to produce a Kerr black hole, as computed by Piran and Stark (1986) using numerical relativity techniques. The star and the black hole it forms both have $J/M^2 \equiv a/M = 0.63$ (where J is angular momentum and M is mass). The dashed curve is a fit, to the wave form, of a superposition of the waves from the two most slowly damped quadrupolar normal modes of a non-rotating hole with mass M, $h_+ \sim \text{Real}\{A_1 e^{-i\omega_1 t} + A_2 e^{-i\omega_2 t}\}$ with $\omega_1 = (0.374 - 0.089i)/M$, $\omega_2 = (0.348 - 0.274i)/M$. The fitting amplitudes are $A_1 = -0.9 - 1.1i$, $A_2 = 0.9 + 1.4i$. Thus, these two modes are roughly equally excited by the collapse.



systems whose components are neutron stars or black holes, and are close enough together to be driven into coalescence by gravitational radiation reaction in a time less than the age of the universe. The binary pulsar PSR1913+16 is an example of such a system; it will coalesce 3.5×10^8 years from now.

As the two bodies in a compact binary spiral together, they emit periodic gravitational waves with a frequency that sweeps upward toward a maximum,

$$f_{\max} \simeq 1 \text{ kHz for neutron stars,} \tag{41a}$$

$$f_{\max} \simeq \frac{10 \text{ kHz}}{M_1/M_\odot} \text{ for holes with the larger having mass } M_1. \tag{41b}$$

The wave form during the nearly Newtonian part of the frequency sweep, $f \ll f_{\max}$, is easily computed from the quadrupole formalism (Section 9.3.2). The post-Newtonian corrections to this waveform will become more and more important as f rises toward $f_{\max}$; they are given in Wagoner and Will (1976); see also Gal'tsov, Matiukhin and Petukhov (1980). Ultimately, near $f_{\max}$, higher-order corrections or full non-linear relativity are needed to get the wave form reasonably accurately. The final, coalescence stage will be especially interesting and complex in the case of a neutron-star binary, and may be quite sensitive to the masses of the two stars; and as with supernovae, we might not understand reliably what to expect until gravitational-wave observations show us. For a first, preliminary theoretical effort at understanding, see Clark and Eardley (1977). For black holes, by contrast, numerical relativity is likely to give us, within the next five years or so, a detailed and highly reliable picture of the final coalescence and the wave forms it produces, including the dependence on the hole's masses and angular momenta. Comparison of the predicted wave forms with observed ones will constitute the strongest test ever of general relativity. (The wave forms for the astrophysically unlikely cases of head-on collisions of two identical non-rotating holes or neutron stars have already been evaluated by numerical relativity; see Smarr, 1977*a* for black holes, and Gilden and Shapiro, 1984 for neutron stars.)

Because the binary system spends far more time in the early, low-frequency part of the sweep than in the later, high-frequency part or in the final coalescence, and because planned gravity wave detectors have less amplitude noise at low frequencies, $f \sim 100$ Hz, than at high, $f \gg 100$ Hz (cf. Fig. 9.4), it will be easier for detectors to see the Newtonian regime of the sweep than the post-Newtonian regime or the final coalescence – except in

the case of black-hole binaries with $M_1 \sim 100$–$1000\,M_\odot$ or $M_1 \gtrsim 10^6\,M_\odot$ (Fig. 9.4). In the Newtonian regime, if we orient the polarization axes $\vec{e}_{x'}$ and $\vec{e}_{y'}$ along the major and minor axes of the projection of the orbital plane on the sky, then the wave form will be

$$h_+ = 2(1+\cos^2 \iota)(\mu/r)(\pi M f)^{\frac{2}{3}} \cos(2\pi f t), \tag{42a}$$

$$h_\times = \pm 4 \cos \iota (\mu/r)(\pi M f)^{\frac{2}{3}} \sin(2\pi f t). \tag{42b}$$

Here it is assumed that the orbit is circular because radiation reaction long ago will have forced circularization (Peters and Mathews, 1963); ι is the angle of inclination of the orbit to the line of sight; M and μ are the total and reduced masses

$$M = M_1 + M_2, \qquad \mu = M_1 M_2 / M; \tag{42c}$$

and f, the frequency of the waves (equal to twice the orbital frequency), is given as a function of time by (MTW equation (36.17))

$$f = \frac{1}{\pi}\left[\frac{5}{256}\frac{1}{\mu M^{\frac{2}{3}}}\frac{1}{(t_0 - t)}\right]^{\frac{3}{8}}. \tag{42d}$$

The most promising detectors for coalescing neutron-star binaries and low-mass black-hole binaries are beam detectors in the planned multi-kilometer LIGOs. As we shall see in Section 9.5.3(e), a beam detector can be operated in several different optical configurations. The optimum configuration for searching for coalescing binaries is likely to be one with *light recycling*, for which the spectral density of shot noise (the dominant noise above some 'seismic cutoff' frequency f_s) will have the form

$$S_h(f) = \text{const} \times f_k[1 + (f/f_k)^2] \quad \text{at } f > f_s. \tag{43a}$$

Here f_k is a 'knee frequency' which the experimenters can adjust by changing the reflectivities of certain mirrors in their detectors; see equation (117c) and Fig. 9.13 below, and associated discussion. The constant in equation (43a) is independent of the choice of f_k. At frequencies below the 'seismic cutoff' f_s seismic noise is likely to come on very strong; accordingly, we shall make the approximation

$$S_h(f) = \infty \quad \text{for } f < f_s. \tag{43b}$$

By Fourier-transforming the wave forms (42), squaring, and averaging over the source orientation angle ι, we obtain

$$\langle |\tilde{h}_+|^2 + |\tilde{h}_\times|^2 \rangle = \frac{\pi}{12}\left(\frac{\mu}{r}\right)^2 \frac{M^3}{\mu}\frac{1}{(\pi M f)^{\frac{7}{3}}}. \tag{44}$$

By inserting this source strength (44) and the detector noise (43) into equation (30) and maximizing the resulting signal-to-noise ratio with

respect to the knee frequency f_k, we find that the experimenter will do best to choose

$$f_k = 1.44 f_s. \tag{45}$$

For smaller choices of f_k there is not a wide enough frequency band between the seismic cutoff and the knee to take optimal advantage of the broad-band nature of the signal. For larger values of f_k the experimenter loses because the height $S_h(f)$ of the 'noise floor' at $f_s < f \lesssim f_k$ (equation (43a)) is proportional to f_k. With this choice of knee, equations (31a) and (43)–(45) give for the characteristic frequency $f_c = 0.909 f_k$. Below (equation (125a)) we shall characterize the sensitivites of beam detectors in full scale LIGOs, when searching for bursts, by the noise amplitude h_n *at the knee*; and, correspondingly, we here shall set f_c (which after all is somewhat arbitrary) to f_k rather than $0.909 f_k$:

$$f_c = f_k = 1.44 f_s. \tag{46a}$$

Equations (31b) and (43)–(45) then give for the characteristic amplitude of the waves from inspiraling binaries

$$h_c = 0.237 \frac{\mu^{\frac{1}{2}} M^{\frac{1}{3}}}{r_0 f_c^{\frac{1}{6}}} = 4.1 \times 10^{-22} \left(\frac{\mu}{M_\odot}\right)^{\frac{1}{2}} \left(\frac{M}{M_\odot}\right)^{\frac{1}{3}} \left(\frac{100\,\text{Mpc}}{r}\right) \left(\frac{100\,\text{Hz}}{f_c}\right)^{\frac{1}{6}}. \tag{46b}$$

The characteristic amplitude (46b) is enhanced over the actual rms wave strength $\langle h_+^2(t_c) + h_\times^2(t_c)\rangle^{\frac{1}{2}}$ the waves have at the time $t = t_c$ when they sweep through frequency f_c – enhanced by very nearly the square root of the number of periods, $n = (f^2/\dot{f})_{f=f_c} = (5/96\pi)(M/\mu)(\pi M f_c)^{-\frac{5}{3}}$, that the binary spends in the vicinity of the frequency f_c. This $\sqrt{n} \simeq 28(\mu/M_\odot)^{-\frac{1}{2}}(M/M_\odot)^{-\frac{1}{3}}(f_c/100\,\text{Hz})^{-\frac{5}{6}}$ enhancement corresponds to the enhancement in effective signal that the experimenters will achieve by optimal signal processing in their search for these frequency-sweeping bursts.

From a study of the waveform (42) using broad-band detectors at several widely spaced locations on the earth, one can deduce the following information: (i) the direction to the source (which comes from phase differences in the signal at different detectors in different locations); (ii) the inclination of the orbit to the line of sight (which comes from the amplitudes in the two different polarization modes); (iii) the direction the stars move in their orbit (which comes from the + or − sign in equation (42b)); (iv) the combination $(\mu^3 M^2)^{\frac{1}{5}}$ of the reduced and total masses; and (v) the distance r to the source. If the mass combination $(\mu^3 M^2)^{\frac{1}{5}}$ is $\lesssim 1.5 M_\odot$, one can be

fairly sure the binary was made of neutron stars; if it is much larger, one can be fairly sure that at least one of the bodies was a massive black hole.

Especially intriguing is the possibility (Schutz, 1986*b*) that, in the case of neutron stars the coalescence will produce electromagnetic emission (e.g. due to an explosion of the less massive star, Blinnikov *et al.*, 1984) that is strong enough to be detected at earth and thereby to pin down the source's location with far higher precision than can be obtained from the gravitational waves. In this case, a redshift will probably be obtainable from optical observations; and that redshift together with the gravitational-wave-determined distance r will give a value for the Hubble constant. Schutz (1986*b*), from a detailed study of the expected noise in future beam detectors, concludes that the prospects are good thereby to obtain a significantly better value for the Hubble constant than we now have. Even in the absence of electromagnetic signals from the coalescence, it may prove possible by statistical means to determine the Hubble constant using combined data for a number of coalescences; see Schutz (1986*b*) for details.

Clark, van den Heuvel and Sutantyo (1979) have estimated, from neutron-star observations in our own galaxy, that to see three coalescences of neutron-star binaries per year one must look out to a distance of 100^{+100}_{-40} Mpc, where the quoted uncertainties are at the 90% confidence level. Correspondingly, in Fig. 9.4 are shown the characteristic amplitude and frequency, for a range of values of the seismic cutoff f_s and corresponding values of $f_c = 1.44 f_s$, produced by the coalescence of two 1.4 solar mass neutron stars at 100 Mpc (estimated event rate about three per year) and at $\frac{1}{3}$ the Hubble distance (event rate about ten per day). Because much has been learned observationally about the statistics of neutron-star binaries since these estimates of event rates were made, a careful restudy of the estimates is much needed.

As Fig. 9.4 shows, future earth based beam detectors may be able to see black-hole coalescences throughout the universe, so long as the more massive of the two holes does not exceed $1000\,M_\odot/(1+z)$, where z is the hole's cosmological redshift. The coalescence rate for black-hole binaries of a few solar masses could be of order that for neutron-star binaries (a few per year at 100 Mpc), or might well be far lower. Particularly intriguing is a scenario, suggested as very plausible by Shapiro and Teukolsky (1985) and Quinlan and Shapiro (1987), in which a large fraction of galactic nuclei create, at some phase of their evolution, a dense cluster of neutron stars and small-mass black holes which – on a timescale of only a few years – form tight binaries that coalesce, with the coalesced holes then forming new tight

binaries that coalesce, ... until the cluster goes unstable and collapses to form a single large hole. This scenario suggests that in typical years the earth might be hit by a number of spiraling wave bursts from coalescing binaries of masses $3\,M_\odot$–$1000\,M_\odot$ at the Hubble distance, $z\sim 1$. Also intriguing but much less likely is a scenario discussed by Bond and Carr (1984) in which a sizable fraction of the mass of the universe is in black-hole binaries with masses ~ 100–$1000\,M_\odot$ for which the coalescence rate could be several per year in the local group of galaxies (distance ~ 1 Mpc).

One or more black hole binaries of any mass up to $\sim 10^8\,M_\odot$ *might* have formed in the nuclei of a reasonable fraction of all galaxies during the past life of the universe, leading to event rates $\gtrsim 1$/year out to the Hubble distance (cf. Fig. 1 of Rees, 1983) – or they might never form. There is actually observational evidence for supermassive black-hole binaries formed by the coalescence of galactic nuclei (Begelman, Blandford and Rees, 1980; Rees, 1983); but the rate of such events probably does not exceed $\frac{1}{100}$ years out to the Hubble distance (Rees, 1983).

(f) *The fall of stars and small holes into supermassive holes*

The supermassive ($M_1 \gtrsim 10^5\,M_\odot$) black holes thought to inhabit the nuclei of galaxies might typically grow by accretion on timescales as short as 10^8 years; see, e.g. Section 8.6 of Blandford and Thorne (1979). When such a hole grows larger than $10^9\,M_\odot$, normal stars can pass near or plunge through its horizon without being torn apart tidally, and the number of stars that so scatter or plunge could well be of order one per year or more (e.g. Dymnikova, Popov and Zentsova, 1982). For smaller supermassive holes, scattering or plunging normal stars will be tidally disrupted, reducing the strength of their waves; but the reduction will not be great, at least in the case of radial infall, unless the hole is below $10^6\,M_\odot$ (Nakamura and Sasaki, 1981; Haugan, Shapiro and Wasserman, 1982). For any hole, neutron stars and satellite holes can scatter or plunge through without enough disruption to strongly suppress their radiation; but the event rate (per supermassive hole) will typically be well below one per year.

The wave forms emitted when a star or small hole is scattered by or plunges into a supermassive hole have been evaluated with high precision using perturbation formalisms; see, e.g. Detweiler and Szedenits (1979), Kojima and Nakamura (1984*a*). The characteristic frequency and amplitude for typical (non-head-on) impact parameters are

$$f_c \cong \frac{1}{20M_1} = 10^{-4}\,\mathrm{Hz}\left(\frac{10^8\,M_\odot}{M_1}\right), \tag{47a}$$

$$h_c \cong \frac{M_2}{2r_o} = 2\times 10^{-21}\left(\frac{M_2}{M_\odot}\right)\left(\frac{10\ \text{Mpc}}{r_o}\right), \tag{47b}$$

where M_1 is the mass of the large hole, M_2 is that of the infalling body, and $r_o \sim 10$ Mpc might give a reasonable event rate since there are ~ 100 galaxies as massive as or more massive than our own inside this distance, including M87 for which observational data suggest a central black hole of mass $M_1 \sim 4\times 10^9\, M_\odot$ (Section 8.3.1.4 of Chapter 8). These h_c and f_c are plotted in Fig. 9.4 for several interesting sets of parameters. It is conceivable that such plunge bursts will be seen by beam detectors in space, if and when they are flown.

(g) *Cherished beliefs*

The above discussion makes clear how uncertain is our electromagnetically based knowledge of gravitational-wave sources. Correspondingly, it seems very likely that when gravitational waves are finally seen, they will come predominantly from sources we have not thought of or we have underestimated; and it seems quite possible that the waves will be stronger than the above estimates suggest.

In light of this, it is interesting to ask the following question: How strong could be the strongest bursts that strike the earth on average three times per year without violating our 'cherished beliefs' about the laws of physics and the universe? Zimmermann and Thorne (1980) have enumerated a set of cherished beliefs, including (i) that general relativity is correct, (ii) that we do not live in a special time or place in the universe, (iii) that there are no enormous primordial bursts, (iv) that no single, coherently radiating object in our galaxy has a mass exceeding $10^8\, M_\odot$, and (v) that the strongest bursts were not beamed by their sources into solid angles $\ll \pi$. From these cherished beliefs they have derived the upper limit on the 3/year burst strength which is shown in Fig. 9.4. That limit could actually be achieved at frequencies $f \gtrsim 30$ Hz by an unlikely but not implausible scenario in which a large fraction of the mass of the universe was cycled, long ago, through a pre-galactic population of massive stars ('Population III' stars), leaving much of the universe's mass in compact binary systems that inhabit the halos of galaxies. If the mean lifetime of these binaries against spiraling together and coalescing is of order the age of the universe, then such coalescences in the halo of our own Milky Way galaxy would give bursts in the frequency domain $f \gtrsim 30$ Hz at the cherished belief level. For further discussion see Zimmermann and Thorne (1980), and Bond and Carr (1984).

While this scenario is unlikely, it is not totally implausible, and it serves to remind us that our best estimates of the waves bathing the earth could be grossly pessimistic.

9.4.2 Periodic sources

(a) *Characterization of the waves from periodic sources and the noise in a Detector searching for them*

The gravitational waves from a periodic source will be characterized by a discrete set of frequencies, and the waves at a given frequency will typically be right-hand or left-hand elliptically polarized; i.e. for some suitable choice of polarization axes $\hat{e}_{x'}$, $\hat{e}_{y'}$, they will have the form (similar to that of a decaying binary, equation (42), but with constant frequency)

$$h_+(t)=h_{0+}\cos 2\pi ft, \qquad h_\times(t)=\pm h_{0\times}\sin 2\pi ft. \tag{48}$$

If one wishes to quantify in a precise and standard manner all the properties of these waves, including the orientations of the 'preferred' x' and y' polarization axes, one might best do so using 'Stokes Parameters' analogous to those used in electromagnetic theory (Section 15 of Chandrasekhar, 1950). However, in this chapter we shall be concerned only with the frequencies f of the waves, and at a given emitted frequency, with a suitably defined characteristic amplitude h_c and a corresponding noise amplitude h_n in a detector searching for the waves.

As an aid in defining h_c and h_n, consider the following situation (analog of that for burst sources in Section 9.4.1(a)): a theorist tells us the frequency f, the phase, and the amplitudes $h_{0+}(\iota, \beta, r)$ and $h_{0\times}(\iota, \beta, r)$ to be expected from a specific model for a source, with orientation angles, ι, β and distance r. Suppose, further, that this type of source is distributed randomly throughout the universe and that the mean number of sources inside the distance r_0 is n_0. What, then, will be (on average) the signal-to-noise ratio for the n_0th brightest source that a detector, broad-band or narrow, will see in a search (at known frequency and phase) lasting a time $\hat{\tau}$? By virtue of equations (27) and by analogy with equation (33) for burst sources, the answer turns out to be

$$\frac{S}{N}\cong\frac{h_c}{h_n(f)}. \tag{49}$$

Here

$$h_c\equiv(2/3)^{\frac{1}{2}}\langle|h_{0+}(\iota, \beta, r_0)|^2+|h_{0\times}(\iota, \beta, r_0)|^2\rangle^{\frac{1}{2}} \tag{50}$$

(with $\langle\cdots\rangle$ denoting an average over ι and β) is the characteristic amplitude

of the periodic source (analog of (31b) for a burst source), and

$$h_n(f) \equiv \frac{[S_h(f)/\hat{\tau}]^{\frac{1}{2}}}{\langle |F_+(\theta, \phi, \psi)|^2 \rangle^{\frac{1}{2}}} \tag{51}$$

is the noise amplitude. (The h_c of equation (50) is $(\frac{4}{3})^{\frac{1}{2}}$ larger than one naively would expect from equation (27). This factor $(\frac{4}{3})^{\frac{1}{2}}$ is an approximate correction for the fact that the angle averages in S/N should not be over squares but, rather, over squares associated with the rotation of the earth during data collection, then over (squares)$^{\frac{3}{2}}$ covering the rest of the sky and the orientation of the source, followed by a $\frac{1}{3}$ power after averaging; cf. the discussion preceding equation (30).) Correspondingly, if experimenters wish to be 90% confident of having seen that n_0th brightest source after $\frac{1}{3}$ year of search, then S/N must exceed $1.655 \simeq 1.7$. (Gaussian probability distribution), and h_c must exceed

$$h_{3/\mathrm{yr}} = 1.7 h_n = \frac{1.7}{\langle |F_+|^2 \rangle^{\frac{1}{2}}} [S_h(f) \times 10^{-7}\ \mathrm{Hz}]^{\frac{1}{2}} \text{ if } f \text{ and phase are known.} \tag{52a}$$

In the case that theory and electromagnetic observation have failed to tell us in advance the phase and frequency of the source, except to within Δf, the experimenters must try $\sim f/\Delta f$ values of the frequency, and correspondingly the Gaussian statistics of the noise will produce

$$h_{3/\mathrm{yr}} \simeq [2 \ln(f/\Delta f)]^{\frac{1}{2}} h_n(f) = \frac{[2 \ln(f/\Delta f)]^{\frac{1}{2}}}{\langle F_+^2 \rangle^{\frac{1}{2}}} [S_h(f) \times 10^{-7}\ \mathrm{Hz}]^{\frac{1}{2}}$$

$$\text{if } f \text{ is known only to within } \Delta f \sim f. \tag{52b}$$

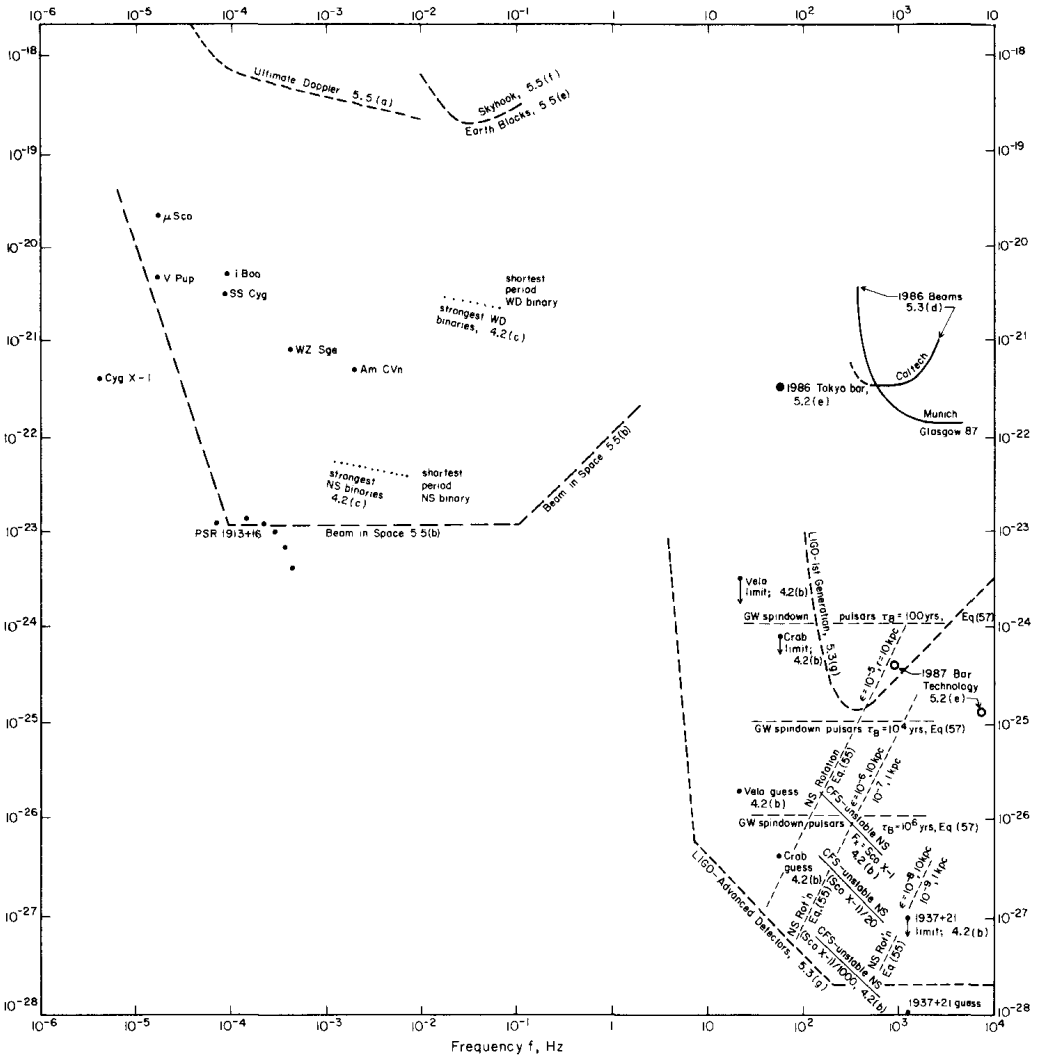
Fig. 9.6 shows h_c, for several postulated types of source (to be discussed in the following subsections), and correspondingly $h_{3/\mathrm{yr}}$ with f and the phase known, for several types of detectors (to be discussed in Section 9.5 below).

(b) *Rotating neutron stars*

A rotating neutron star (e.g. a pulsar) will emit gravitational waves at several frequencies as a result of deviations from symmetry around its rotation axis (deviations from 'axisymmetry'). The larger are those deviations and the more rapidly they rotate, the stronger will be the radiation.

Deviations from axisymmetry could arise in several ways: (i) The star's solid crust (well-established) or a solid core (not so well established) could support deformations that are residual remnants of the star's past history – a history that might be quite complex, including star quakes in which the crust or core cracks and deforms in much the manner of the solid earth in an earthquake; for detailed discussions see, e.g. Pandharipande, Pines and

Fig. 9.6. The characteristic amplitudes h_c (equation (50)) and frequencies f of waves from several postulated *periodic sources* (thin curves), and the sensitivities $h_{3/yr}$ of several existing and planned detectors (thick curves and circles) ($h_{3/yr}$ is the amplitude h_c of the weakest source detectable with 90% confidence in a $\frac{1}{3}$ yr $= 10^7$ s integration if the frequency and phase of the source are known in advance; equation (52a)). The sources shown in the high-frequency region, $f \gtrsim 10$ Hz, are all special cases of rotating, nonaxisymmetric neutron stars (Section 9.4.2(b)). The steeply sloping dotted lines labeled NS Rotation refer to rigidly rotating neutron stars with moment of inertial $I_{\bar{z}\bar{z}} = 10^{45}$ g cm^{-2}, and with various ellipticities ε and distances r labeled on the lines (equation (55)). The sources in the low-frequency region, $f \lesssim 0.1$ Hz, are all binary star systems in our galaxy (Section 9.4.2(c)): several specific, known binaries, which are indicated by name (μ Sco, V Pup, ...); the strongest six spectral lines from the famous binary pulsar PSR1913 + 16; and the estimated strengths of the strongest white-dward ('WD') and neutron-star ('NS') binaries in our galaxy. The detectors are discussed in detail in the indicated subsections of Section 9.5.



Smith (1976) and references therein. In old pulsars that have been spun up by accretion to near-millisecond rotation rates, theory and observational data suggest that the crust and core are quite well annealed into a nearly axisymmetric shape (Alpar and Pines, 1985); but in neutron stars that are only tens or hundreds or thousands of years old, it might well be otherwise (e.g. Zimmermann, 1978). (ii) The star's internal magnetic field, if sufficiently strong, could produce sufficient magnetic pressure to distort the star significantly (Zimmermann, 1978; Gal'tsov, Tsvetkov and Tsirulev, 1984). However, 'sufficiently strong' means, in the case, e.g., of the Crab Pulsar, ten times stronger than the star's measured surface field. (iii) If the star is rotating more rapidly than a critical rotation period, $P_{\rm crit} \cong 0.7$–1.7 ms (which depends on the star's structure and its temperature-dependent viscosity), then an instability driven by gravitational radiation reaction ('Chandrasekhar (1970)–Friedman–Schutz' (1978), or 'CFS' instability) will create and maintain significantly strong hydrodynamic waves in the star's surface layers and mantle, propagating in the opposite direction to the star's rotation; and these will radiate strongly. For detailed discussions see Wagoner (1984), Lindblom (1986, 1987), Friedman, Ipser and Parker (1986), Schutz (1987), Cutler and Lindblom (1987).

At present we are extremely ignorant of the degree of asymmetry in rotating neutron stars, and accordingly we are ignorant of the strengths of the periodic waves to be expected from them. Pessimists will note that there is no observational evidence in any observed pulsar for sufficient non-axisymmetry to produce interestingly strong waves. Pessimists will point, especially, to the extremely small slowdown rate of the 1.6 ms pulsar PSR1937+21, which implies such weak radiation reaction that the characteristic amplitude at earth cannot exceed 1×10^{-27} (Fig. 9.6), and the star's non-axisymmetric ellipticity cannot exceed 3×10^{-9}.

Optimists will also point to PSR1937+21 and some other millisecond pulsars, and note a reasonably likely scenario for their origin (van den Heuvel, 1984; Wagoner, 1984): that they were spun up long ago by accretion from a binary companion until they hit the CFS instability, that they remained just beyond the instability point for awhile, with the spinup torque of accretion being counterbalanced by gravitational radiation reaction, and that the accretion stopped long ago leaving the stars plenty of time to anneal and settle down into their presently observed, highly axisymmetric states. Given the extreme observational difficulty of finding by electromagnetic means evidence for rapid neutron-star rotation (see, e.g., Section IV of Reynolds and Stinebring (1984) for searches in the radio), it may well be that

there are a number of accreting neutron stars in our galaxy now in the CFS regime, radiating strong gravitational waves. For such a star the energy being radiated in gravitational waves and that being radiated as accretion-induced X-rays will both be proportional to the accretion rate; and consequently the characteristic amplitude of the gravitational waves at earth will be proportional to the square root of the X-ray flux arriving at earth, F_X:

$$h_c \simeq 2 \times 10^{-27} \left(\frac{300 \text{ Hz}}{f} \right)^{\frac{1}{2}} \left(\frac{F_X}{10^{-8} \text{ erg cm}^{-2} \text{ s}^{-1}} \right)^{\frac{1}{2}} \tag{53}$$

(Wagoner, 1984). The frequency f of the waves will be $f = lv_p/(2\pi R)$ where R is the star's radius, $l = 3$ or 4 or 5 is the spherical-harmonic order of the hydrodynamic wave, and v_p is the pattern speed of the wave as seen in the inertial frame of distant observers. The X-ray flux $F_X \cong 10^{-8}$ erg cm^{-2} s^{-1} is $\frac{1}{20}$ that of Sco X-1, the brightest quasi-steady source in the sky and itself a candidate for a CFS-unstable object. As Fig. 9.6 shows, stars with X-ray fluxes as low as $\frac{1}{1000}$ Sco X-1 could be interestingly strong sources of gravitational waves. In such a star the density waves in the surface layers should modulate the emitted X-rays, but the sensitivities of past X-ray telescopes have been too poor to detect such rapid and weak modulations. There is an interesting proposal (Wood *et al.*, 1986) for a new, more sensitive X-ray telescope designed to search for such modulations in Sco X-1 and other, weaker X-ray sources. Such a telescope, operated in coordination with gravitational-wave detectors, might one day give a wealth of new information about neutron stars.

Even the most rapidly rotating of neutron stars will be smaller than a reduced wavelength of its emitted gravitational waves and thus can be described with reasonable accuracy by a slow-motion formalism (Ipser, 1971). If the emitter is CFS density waves, the multipoles involved are $l = 3$, 4, or 5. If the emitter is solidly supported or magnetic-field supported deformations, the waves should be largely quadrupolar, $l = 2$. Let us focus now on the latter case, i.e. on a star with rotation period large enough that the CFS instability does not act. Although gravity inside the star is significantly non-Newtonian, one can still define for the star a moment-of-inertia tensor I_{jk} equal to the ratio of its angular momentum to its angular velocity (both being vectors defined in the weak-gravity region well outside the star; see Thorne and Gürsel, 1983). If the star rotates about a principal axis of this moment-of-inertia tensor, i.e. if its angular momentum and angular velocity are parallel, then it will not precess, and (assuming a rotation period sufficiently long that it is CFS stable), its gravitational waves

will be emitted at twice the rotation frequency. From the quadrupole variant of the slow-motion formalism we can then compute that the waves will have the standard periodic form of equation (48) with amplitudes

$$h_{0+} = 2(1+\cos^2 \iota)\frac{(\mathscr{I}_{\bar{x}\bar{x}} - \mathscr{I}_{\bar{y}\bar{y}})(\pi f)^2}{r},$$

$$h_{0\times} = 4\cos\iota\frac{(\mathscr{I}_{\bar{x}\bar{x}} - \mathscr{I}_{\bar{y}\bar{y}})(\pi f)^2}{r}. \tag{54}$$

Here ι is the angle between the neutron star's rotation axis and the line of sight from the earth, and $\mathscr{I}_{\bar{x}\bar{x}}$ and $\mathscr{I}_{\bar{y}\bar{y}}$ are the components of the star's quadrupole moment along the principal axes in its equatorial plane. The characteristic amplitude of these waves (equation (50)) is

$$h_c = 8\pi^2\left(\frac{2}{15}\right)^{\frac{1}{2}}\frac{\varepsilon I_{\bar{z}\bar{z}} f^2}{r} = 7.7\times 10^{-20}\varepsilon\left(\frac{I_{\bar{z}\bar{z}}}{10^{45}\text{ g cm}^2}\right)\left(\frac{f}{1\text{ kHz}}\right)^2\left(\frac{10\text{ kpc}}{r}\right), \tag{55}$$

where $I_{\bar{z}\bar{z}}$ is the moment of inertia of the star about its rotation axis (so $-\frac{1}{2}I_{\bar{z}\bar{z}}(\pi f)^2$ is its rotational energy), and

$$\varepsilon \equiv \frac{\mathscr{I}_{\bar{x}\bar{x}} - \mathscr{I}_{\bar{y}\bar{y}}}{I_{\bar{z}\bar{z}}} \tag{56}$$

is its 'gravitational ellipticity' in the equatorial plane. All neutron stars for which masses have been measured have M near $1.4M_\odot$; and depending on the equation of state these masses correspond to $3\times 10^{44}\text{ g cm}^2 \lesssim I_{\bar{z}\bar{z}} \lesssim 3\times 10^{45}\text{ g cm}^2$. The likely values of the ellipticity ε are far less clear.

The observed slow-down rates of the Crab ($f=60$ Hz, $r=2$ kpc), Vela ($f=22$ Hz, $r=500$ pc), and PSR1937+21 ($f=1.25$ kHz, $r=5$ kpc) pulsars, if due to gravitational radiation reaction (possible but not likely), correspond to $\varepsilon \simeq 6\times 10^{-4}$, 4×10^{-3}, and 3×10^{-9} respectively; and to $h_c \simeq 8\times 10^{-25}$, 3×10^{-24}, and 1×10^{-27}. Zimmermann (1978) argues that reasonable values for the Crab and Vela are $\varepsilon \sim 3\times 10^{-6}$ and 3×10^{-5} corresponding to $h_c \sim 4\times 10^{-27}$ and 2×10^{-26}. Alpar and Pines (1985) suggest reasonable values for PSR1937+21 (which is old and well annealed) in the range $\varepsilon \sim 4\times 10^{-10}$–$1\times 10^{-11}$ corresponding to $h_c \sim 1\times 10^{-28}$.

Blandford (1984) points out that, if there is a population of young pulsars (not yet discovered) that are spinning down by gravitational radiation reaction on a spin-down timescale τ_{GW}, then (i) the nearest will be at a distance $r \simeq R_G(\tau_B/\tau_{GW})^{\frac{1}{2}}$ (assumed $\leqslant \tau_{GW}$) is the mean time between births of these pulsars in our galaxy and R_G is the radius of the galaxy's disk;

(ii) the flux of gravitational-wave energy at earth from the nearest such pulsar $(3/64\pi)(2\pi f)^2 h_c^2$, will be equal to $[I_{\bar{z}\bar{z}}(\pi f)^2/\tau_{GW}](4\pi r^2)^{-1}$; and (iii) as a consequence, the characteristic amplitude of the waves from the nearest one will be

$$h_c \simeq \left[\frac{4}{3}\frac{I_{\bar{z}\bar{z}}}{r^2\tau_{GW}}\right]^{\frac{1}{2}} \simeq \left[\frac{4}{3}\frac{I_{\bar{z}\bar{z}}}{R_G^2\tau_B}\right]^{\frac{1}{2}} \sim 1.1\times 10^{-25}\left(\frac{10^4 \text{ years}}{\tau_B}\right)^{\frac{1}{2}}, \tag{57}$$

independently of its frequency and ellipticity.

Fig. 9.6 shows values of h_c and f corresponding to some of the above possibilities.

If the star does not rotate about a principal axis of its moment of inertia, then it will precess. When one idealizes the star's interior as rigid, then although the interior gravity is significantly non-Newtonian, the precession is still described by the classic equations of Euler (see Thorne and Gürsel, 1983, for a proof); and the resulting waves will have a form that typically will entail significant spectral components at three frequencies: twice the rotation frequency, and the rotation frequency plus and minus the precession frequency (Zimmermann, 1980; Zimmermann and Szedenits, 1979). In reality, pliability of the neutron-star material will cause significant deviations from rigid rotation; but these three frequencies may still be dominant. If the star is idealized as a fluid body deformed by the pressure of an off-axis internal magnetic field, then the star does not precess and the radiation is emitted at the rotation frequency and twice the rotation frequency (Gal'tsov, Tsvetkov and Tsirulev, 1984).

If and when gravitational waves from rotating neutron stars are detected, they may carry a wealth of information about the star's structure and dynamics in the amplitudes and relative phasings of their various spectral components. Especially interesting may be the evolution of the various spectral components after a star quake; together with electromagnetic timing of the post-quake rotation, these may give us new insights into the coupling of the solid crust or core to the fluid mantle.

(c) *Binary stars*

Ordinary binary star systems are the most reliably understood of all sources of gravitational waves. From the measured mass and orbital parameters of a binary and its estimated distance, one can compute with confidence the details of its waves. Unfortunately, ordinary binaries have orbital periods no shorter than about an hour and, correspondingly, gravitational-wave frequencies $f \lesssim 10^{-3}$ Hz. The shortest known binary of all is a white-dwarf/

neutron-star system with orbital period 11 minutes and gravitational-wave frequency $f \simeq 3 \times 10^{-3}$ Hz (Priedhorsky, Stella and White, 1986). Because of seismic noise, detectors in earth laboratories cannot hope to see waves of such low frequency. However, beam detectors in space, tentatively planned for the turn of the century, should see them with relative ease (cf. Section 9.5.5(b) below).

Detailed formulas for the waves from a binary star, including the effects of the eccentricity and inclination of its orbit, have been derived from the quadrupole formalism by Peters and Mathews (1963) and Wahlquist (1987). See also Wagoner and Will (1976), and Gal'tsov, Matiukhin and Petukhov (1980) for post-Newtonian corrections. By virtue of the eccentricity of the orbit, waves will be emitted in equally spaced 'spectral lines' at twice the orbital frequency and harmonics thereof. For eccentricity $\varepsilon \lesssim 0.2$ the line at $f = 2f_{\rm orb}$ is dominant; for $\varepsilon \simeq 0.5$ the lines at $f/f_{\rm orb} \simeq 2$ through 8 are all strong; for $\varepsilon \simeq 0.7$ the lines at $f/f_{\rm orb} \simeq 4$ through 20 are all strong. In the low-eccentricity case $\varepsilon \lesssim 0.2$ the waves have the form (42) with $f = 2f_{\rm orb}$, which corresponds to a characteristic amplitude (equation (50))

$$h_c = 8\left(\frac{2}{15}\right)^{\frac{1}{2}} \frac{\mu}{r} (\pi M f)^{\frac{2}{3}}$$

$$= 8.7 \times 10^{-21} \left(\frac{\mu}{M_\odot}\right)\left(\frac{M}{M_\odot}\right)^{\frac{2}{3}}\left(\frac{100\ \text{pc}}{r}\right)\left(\frac{f}{10^{-3}\ \text{Hz}}\right)^{\frac{2}{3}}, \tag{58}$$

where μ is the reduced mass and M the total mass of the system. This amplitude is plotted in Fig. 9.6 for a few of the most strongly radiating known binaries. For lists of the most strongly radiating binaries and their characteristics see Braginsky (1965) and Douglass and Braginsky (1979).

White-dwarf and neutron-star binaries should also be important emitters – and they should extend to higher frequencies than ordinary binaries; but there is a paucity of observational data on them and the example of shortest known period has f only 3×10^{-3} Hz (see above). From the data that do exist (e.g. Iben and Tutukov, 1984), Lipunov and Postnov (1986), Lipunov, Postnov and Prokhorov (1987), and Hils *et al.* (1987) have estimated the characteristic amplitudes of the strongest white-dwarf and neutron-star binaries; see Fig. 9.6. For a very detailed treatment of the white-dwarf case see Evans, Iben and Smarr (1987). The highest frequency to be expected for any white-dwarf binary in our galaxy is 0.06 Hz since mass transfer from the less massive star to the more massive begins at or before this frequency; the highest for any neutron-star binary is 0.007 Hz since any binary of higher frequency than this would coalesce in a time less than the mean interval between coalescences, $\sim 10^4$ years.

Gravitational radiation reaction plays an important role in driving the evolution of close binary systems; see Paczynski and Sienkiewicz (1981) for details.

9.4.3 Stochastic sources

(a) *Characterization of stochastic gravitational waves*

It is useful to think about stochastic gravitational waves in terms of traveling-wave normal modes of the gravitational field. As with the electromagnetic field, there are two modes (because of two polarization states) for each volume $(2\pi\hbar)^3$ in phase space; and correspondingly, one can easily show, the energy density per unit logarithmic interval of frequency divided by the critical energy density $\rho_{\rm crit} \sim 10^{-8}$ erg cm^{-3} to close the universe is

$$\Omega_{\rm GW}(f) \equiv \frac{dE^{\rm GW}/d^3x\, d\ln f}{\rho_{\rm crit}} \sim \frac{\bar{n}}{10^{37}} \left(\frac{f}{1\ {\rm kHz}}\right)^3. \tag{59}$$

Here $\bar{n}$ is the average number of quanta in all modes with frequencies of order f. Below we shall see that $\Omega_{\rm GW}(f)$ is likely to be $\gtrsim 10^{-14}$ at all frequencies of interest, and correspondingly the mean number of quanta in each mode is likely to be $\gtrsim 10^{20}$ – so large that a classical treatment is in order.

In TT coordinates the metric perturbation due to stochastic gravitational waves, evaluated at any chosen location x^i, will be a sum over contributions of all the modes of the field

$$h^{\rm TT}_{jk}(t, x^i) = \sum_K h^{\rm TT}_{Kjk}(t, x^i). \tag{60}$$

Here the index K labels modes of the field. For stochastic gravitational waves, the wave field $h^{\rm TT}_{Kjk}$ associated with mode K can be regarded as a 'random process' (i.e. a stochastically fluctuating function of time t); and the total field $h^{\rm TT}_{jk}$ at location x^i is the sum over all of the modes' random processes.

The field of a chosen mode K can be expressed as

$$h^{\rm TT}_{Kjk} = h_K(t, x^i) e^K_{jk}, \tag{61}$$

where $h_K(t)$ is its scalar wave function and e^K_{jk} is its constant polarization vector, so normalized that $e^K_{jk} e^K_{jk} = 2$ in Cartesian coordinates; cf. equation (7c). Then h_K is a scalar random process in time (at fixed x^i) and its statistical properties can be characterized by a spectral density $S_{h_K}(f)$.

I shall assume that the modes are defined in such a way that there is no significant correlation between their wave fields $h^{\rm TT}_{Kjk}$. As a result, when one averages the stress-energy tensor (9) of the waves at x^i over a sufficiently

long time, all cross-terms between different modes get washed out and one obtains for the time-averaged specific intensity I_f at location x^i

$$I_f(t, x^i)\,\Delta\Omega \equiv \frac{dE}{dA\,dt\,df\,d\Omega}\,\Delta\Omega = \sum_{K \text{ in } \Delta\Omega} \frac{\pi f^2}{4} S_{h_K}. \tag{62}$$

Here E denotes energy, A denotes area, Ω denotes solid angle, and the sum is over all modes K with propagation directions in the infinitesimal solid angle $\Delta\Omega$. The total energy per unit logarithmic interval (used in defining $\Omega_{GW}(f)$ above) can be expressed in terms of the specific intensity in the standard way

$$\Omega_{GW}(f)\rho_{crit} = \frac{dE}{d^3x\,d\ln f} = \int f I_f\,d\Omega = \sum_K \frac{\pi f^3}{4} S_{h_K}, \tag{63}$$

where the integral is over the entire sphere and the sum is over all modes.

As for burst and periodic waves, so also for stochastic, we shall introduce a single characteristic amplitude h_c that is tied to a specific experimental situation: the experimenters use two identical gravity-wave receivers (broad-band or narrow), separated by a distance $\ll \lambda\!\!\!^{-} = c/2\pi f$, to search for *isotropic*, stochastic waves in the neighborhood of frequency f. The search is performed by a standard technique (Bendat, 1958; Section 9 of Drever, 1983): the outputs, $h_1(t)$ and $h_2(t)$ of detectors 1 and 2 are passed through identical filters which admit only Fourier components in a bandwidth $\Delta f \lesssim f$ centered on frequencies $\pm f$. The filtered outputs $w_1(t)$ and $w_2(t)$ are then multiplied together and integrated for a time $\hat{\tau}$ to get a single number $W = \int_{t_0}^{t_0+\hat{\tau}} w_1(t) w_2(t)$. This number will consist of a signal due to the identical stochastic backgrounds contained in $w_1(t)$ and $w_2(t)$, and a Gaussian noise due to the independent noises in the two detectors (each with the same spectral density $S_h(f)$). It turns out (e.g. Chapter 7 of Bendat 1958) that the ratio of the signal to the root-mean-square noise is

$$\frac{S}{N} = \frac{h_c(f)}{h_n(f)}, \tag{64}$$

where

$$h_c(f) \equiv \left[\sum_K f S_{h_K}(f)\right]^{\frac{1}{2}} = \left[\frac{4}{\pi f}\int I_f\,d\Omega\right]^{\frac{1}{2}} = \left[\frac{4}{\pi f^2}\,\Omega_{GW}(f)\rho_{crit}\right]^{\frac{1}{2}}$$
$$= 1.3\times 10^{-18}\left(\frac{\rho_{crit}}{1.7\times 10^{-8}\ \text{erg cm}^{-3}}\right)^{\frac{1}{2}}\left(\frac{1\ \text{Hz}}{f}\right)[\Omega_{GW}(f)]^{\frac{1}{2}} \tag{65}$$

is the characteristic amplitude of the isotropic, stochastic waves, and

$$h_n(f) = \frac{1}{(\frac{1}{2}\hat{\tau}\Delta f)^{\frac{1}{4}}}\,\frac{[f S_h(f)]^{\frac{1}{2}}}{\langle F_+^2\rangle^{\frac{1}{2}}} \tag{66}$$

is the characteristic noise amplitude of the detectors. (Note: $\rho_{crit}/1.7\times$

10^{-8} erg cm$^{-3} = (H_0/100 \text{ km s}^{-1} \text{ Mpc}^{-1})^2$ where H_0 is the Hubble constant.) Correspondingly, if the experimenters wish to be 90% confident of having seen the stochastic background during a search of duration $\hat{\tau} = \frac{1}{3}$ year, the Gaussian probability distribution for S/N requires $S/N = 1.7$, which in turn means that h_c must exceed the noise level

$$h_{3/\text{yr}}(f) = 1.7 h_n(f) = 2.0 \left(\frac{\Delta f}{10^{-7} \text{ Hz}} \right)^{-\frac{1}{4}} \frac{[f S_h(f)]^{\frac{1}{2}}}{\langle F_+^2 \rangle^{\frac{1}{2}}}. \tag{67}$$

The characteristic amplitudes h_c of various possible stochastic sources, and the noise levels $h_{3/\text{yr}}$ of various detectors are shown in Fig. 9.7.

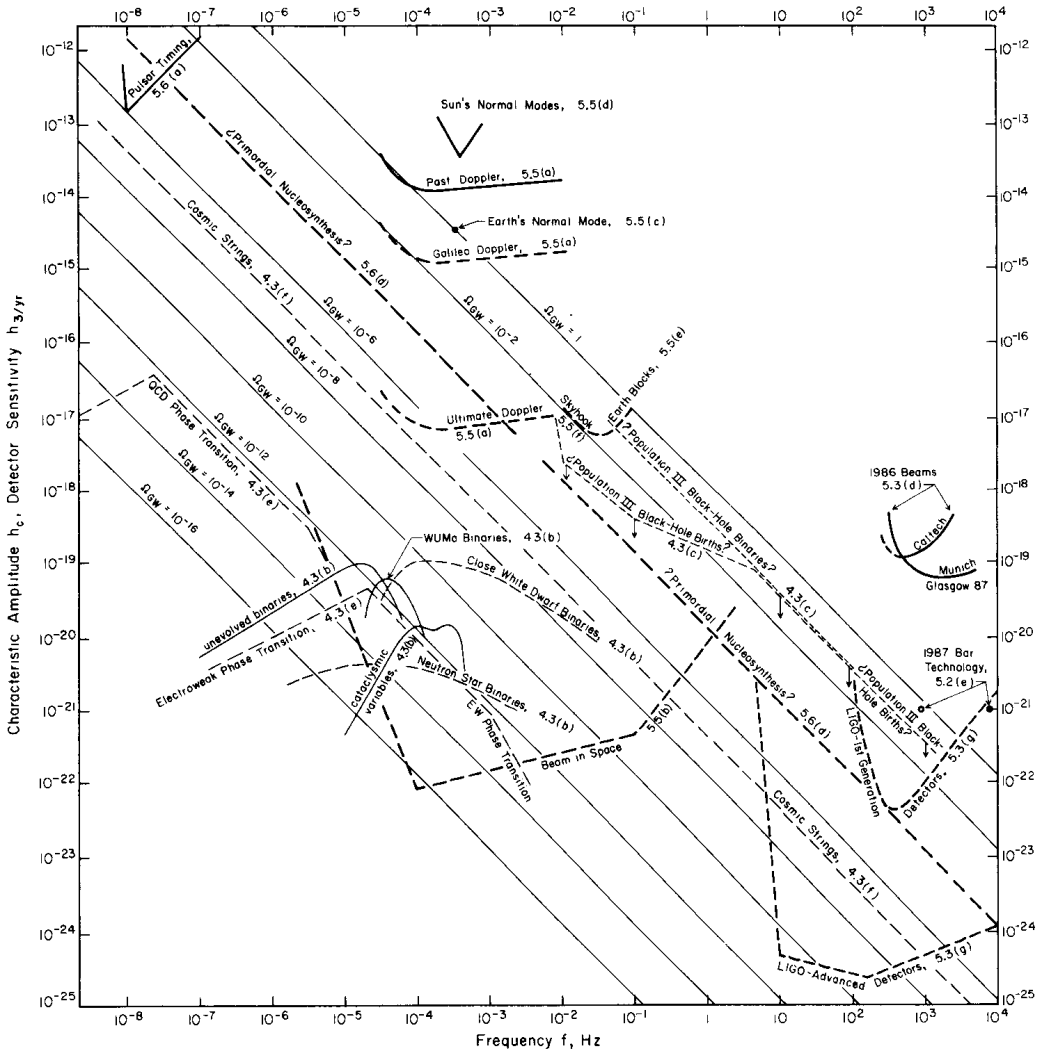
(b) *Binary stars*

So many binary stars in our galaxy and in other galaxies radiate in the frequency region $f \lesssim 0.03$ Hz that they should superpose to produce a strong stochastic background. Lipunov and Postnov (1986), Lipunov, Postnov and Prokhorov (1987) and Hils *et al.* (1987) have made careful calculations of the characteristic amplitude of this stochastic background as a function of frequency; the results of Hils *et al.* are shown in Fig. 9.7 for the contribution of our own galaxy (which should be concentrated in the galactic plane). The contributions of all other galaxies (which should be isotropic) should be down from those of our own galaxy by $(h_c)_{\text{other}}/(h_c)_{\text{us}} \sim 0.15$.

The binary stochastic background in Fig. 9.7 is broken up into contributions from various types of binaries. Those shown as solid curves (unevolved binaries, WUMa stars (first discussed by Mironovskii, 1966), and cataclysmic variables (white-dwarf/normal-star systems)) are rather firmly based on optical studies of the statistics of these types of stars, and thus are rather reliable. Those shown dashed (close white-dwarf binaries (see Evans, Iben and Smarr, 1987, and the above references), and neutron-star binaries) are based on so little observational data and so much theory that they are highly uncertain.

The binary background presents a serious potential obstacle to searches for other kinds of waves in the frequency band 0.03 Hz $\lesssim f \lesssim 10^{-5}$ Hz where space-based beam detectors will operate. A broad-band burst can be seen above this background only if it has $h_{c\,\text{burst}} > h_{c\,\text{background}}$ – which means, e.g., that a $1M_\odot$ star falling into a supermassive black hole in the Virgo cluster will not be discernible unless the hole's mass is $M < 3 \times 10^5 M_\odot$ (cf. Figs. 9.4 and 9.7). A periodic source can be seen, after an integration time $\hat{\tau}$, only if it has $h_{c\,\text{periodic}} > (f\hat{\tau})^{-\frac{1}{2}} h_{c\,\text{background}}$ – which means, e.g., that if the close white-dwarf binaries are as numerous as estimated, the binaries ι Boo and SS Cyg will be discernible only after integration times of $\hat{\tau} > 10^7$ s. For further discussion see Evans, Iben and Smarr (1987) and Hils *et al.* (1987).

Fig. 9.7. The characteristic amplitudes h_c (equation (65)) and frequencies f of waves from several postulated *stochastic sources* (thin curves), and the sensitivities $h_{3/yr}$ of several existing and planned detectors (thick curves and circles) ($h_{3/yr}$ is the amplitude of the weakest source that can be detected with 90% confidence in a $\frac{1}{3}$ yr $= 10^7$ s integration). The very thin diagonal lines indicate values of h_c corresponding to constant $\Omega_{GW}(f) =$ (gravity wave energy in bandwidth $\Delta f = f$)/(energy to close the universe if $H_0 = 100$ km s^{-1} Mpc^{-1}). The sources shown as solid curves (background from various types of binary stars in our galaxy; Section 9.4.3(c)) are rather firm predictions. The sources shown as dashed and dotted curves are much less firm; see Sections 9.4.3(c)–(f) for discussion of specific sources. Not shown are primordial waves from the big-bang (Section 4.3(d)), which could have Ω_{GW} as large as 1 or as small as 10^{-14} or less according to various plausible scenarios – but which are limited by observations as discussed in Section 9.5.6. The detectors are discussed in detail in the indicated subsections of Section 9.5.



(c) *Population III stars*

If there was a pre-galactic population of massive stars ('Population III stars'; Carr, 1986), the violent events that terminated their lives (supernovae and collapse to black holes) might have produced gravitational waves that we today would see as isotropic and stochastic. Whereas existing binary stars are a firm source of stochastic background, these Population III stars are a highly speculative one. Carr (1980) has derived an upper limit on the characteristic amplitude that could have been produced by the deaths of such stars under any reasonable scenario; it is shown in Fig. 9.7 (upper short-dashed curve) along with the maximum characteristic amplitude that could come from the remnants of these stars if they became black-hole binaries that decay by radiation reaction on a timescale $\tau_{GW} \simeq 10^{10}$ years (Bond and Carr, 1984).

(d) *Primordial gravitational waves*

Photons coming from the big bang last scattered off matter at a cosmological redshift $z \sim 1000$ when the universe was roughly one million years old; and neutrinos last scattered at $z \sim 10^{10}$ when it was about 0.1 s old. An order-of-magnitude calculation shows that gravitons, by contrast, last scattered at roughly the Planck time, i.e. during the first 10^{-43} seconds when spacetime was quantized and the laws of physics were exceedingly different from today (Section 7.2 of Zel'dovich and Novikov, 1983). (An unlikely exception occurs if, at the epoch when the waves' reduced wavelength was $\lambda\!\!\!^{-} \sim$ (horizon size), much of the universe's energy density was in relativistic particles with mean free paths of order $\lambda\!\!\!^{-}$; then non-negligible absorption may occur; see Vishniac (1982).) Thus, in studying primordial gravitational waves (waves created in the big-bang), one usually can ignore their subsequent interactions with matter.

Not so for their subsequent interactions with the background spacetime curvature of the universe. Grishchuk (1974, 1975*a*,*b*, 1977) has shown that, as the primordial perturbations that give rise to present-day waves 'come inside the cosmological horizon' – and also before they enter the horizon – they can be parametrically amplified by their interaction with the dynamical background spacetime curvature; in other words, they can trigger further graviton creation. In this way exceedingly small initial fluctuations can be amplified into an interestingly strong stochastic background today.

Just how much stochastic background is produced depends crucially on ill-understood aspects of the initial singularity and on the equation-of-state-

dependent and vacuum-dependent expansion rate in the very early universe. Some otherwise plausible models produce so much, $\Omega_{GW}(f) \gg 1$, as to be in violent conflict with the observed current state of the universe (e.g. p. 621 of Zel'dovich and Novikov, 1983). Other, equally plausible models can produce so little, $\Omega_{GW}(f) \ll 10^{-14}$, that there is no hope of detecting the waves in the foreseeable future.

In currently fashionable inflationary models of the universe vacuum fluctuations which initially are smaller than the horizon ($\lambda\!\!\!^{-} \ll \mathscr{R}_B =$ (background radius of curvature)) are driven outside the horizon ($\lambda\!\!\!^{-} \gg \mathscr{R}_B$) by the inflationary expansion. While outside the horizon, they are 'frozen' with constant amplitude h. Much later, after inflation ends, non-inflationary expansion brings them back inside the horizon. The number of quanta in each mode before entering and after leaving the horizon is

$$n \sim \left[\frac{1}{16\pi}\left(\frac{h}{\lambda\!\!\!^{-}}\right)^2\right](2\pi\lambda\!\!\!^{-})^3\left(\frac{1}{\hbar/\lambda\!\!\!^{-}}\right) \sim \frac{\pi}{4}\frac{(h\lambda\!\!\!^{-})^2}{\hbar},$$

where the first factor is the waves' energy density, the second is the volume occupied by each mode, and the third is 1/(energy of one graviton). Before entering the horizon $n \simeq \frac{1}{2}$; so the above relation says that upon leaving it

$$n_{out} \simeq n_{enter}\frac{\lambda\!\!\!^{-}_{leave}}{\lambda\!\!\!^{-}_{enter}} = \frac{1}{2}\frac{a_{leave}}{a_{enter}},$$

where a is the expansion factor of the universe. Thus, the epoch of amplitude freezing is actually an epoch of parametric amplification (stimulated creation of new gravitons); and the total number of gravitons created depends on the total amount of expansion that occurs while the waves are outside the horizon. (There will be additional parametric amplification as the waves emerge from the horizon, $\lambda\!\!\!^{-} \sim \mathscr{R}_B$, but in inflationary models that is generally small compared to the amplification during freezing, $\lambda\!\!\!^{-} \gg \mathscr{R}_B$.) The total amount of inflationary expansion differs from one inflationary model to another; and correspondingly, the models can give Ω_{GW} as large as unity or Ω_{GW} too small ($\ll 10^{-14}$) for there to be hope of detecting the waves.

For discussions of the influence of the equation of state in the early universe on the spectrum of the amplified waves, see Grishchuk (1977) and Fig. 4 of Grishchuk and Polnarev (1980). For calculations of the waves produced by specific inflationary scenarios see Starobinsky (1979), Rubakov, Sazhin and Veryaskin (1982), Abbott and Wise (1984), Halliwell and Hawking (1985), Mijic, Morris and Suen (1986) and references therein. Because the range of possible strengths of primordial waves is so great, we

do not bother to show it in Fig. 9.7 – aside from indicating the values of h_c corresponding to various values of $\Omega_{GW}(f)$.

(e) *Phase transitions*

During the early expansion of the universe, there may have been first-order phase transitions associated with QCD interactions and with Electroweak interactions. In each of these phase transitions the original phase would be supercooled, by the cosmological expansion, below the equilibrium temperature of the new phase. Bubbles of the new phase would then nucleate at isolated locations and expand at near-light velocity until they have compressed the original phase enough for the two phases to coexist in equilibrium. As Witten (1984) has pointed out, and Hogan (1986) has analyzed in detail, this 'cavitation' should have produced gravitational waves in two ways: (i) directly from expanding bubbles and the subsequent sound waves they generate, and (ii) subsequently from the inhomogeneities associated with the two co-existing phases (large-scale density inhomogeneities and corresponding inhomogeneities in the Hubble expansion rate). The resulting gravitational waves should possess a spectrum that peaks at wavelengths which were of order the horizon size when the cavitation occurred. Those wavelengths correspond to frequencies today $f_{max} \sim (2 \times 10^{-7}\ \text{Hz})(kT/1\ \text{GeV})$, where T is the temperature of the phase transition. Hogan's (1986) predicted spectra, shown in Fig. 9.7, thus peak at $f_{max} \sim 2 \times 10^{-8}$ Hz (QCD, $T \sim 100$ MeV) and $f_{max} \sim 2 \times 10^{-5}$ Hz (Electroweak, $T \sim 100$ Gev). The $f^{+\frac{1}{2}}$ shape of the spectra at frequencies $f > f_{max}$ is a firm prediction, but the shape f^{-1} at $f < f_{max}$ is not (it could well be f^{-p} with $p > 1$). The amplitude shown is a reasonable upper limit, unless the phase transition is unusually catastrophic with very strong supercooling.

(f) *Cosmic strings*

Long before the QCD and Electroweak phase transitions – i.e. nearer the initial singularity – there may have been a phase transition associated with the grand-unified interactions, and that transition may have created cosmic strings – one-dimensional 'defects' in the vacuum with mass per unit length estimated to be $\mu \sim 10^{-6}$ and with tension equal to mass per unit length (Zel'dovich, 1980; Vilenkin, 1981*a*). As the universe's horizon expands to uncover the stochastic inhomogeneities in a string's shape, those inhomogeneities should begin to vibrate with speeds up to the speed of light. By self-intersection of the string, closed loops should form; and those loops could have acted as seeds for the condensation of galaxies and galaxy

clusters (Zel'dovich, 1980; Vilenkin, 1981*a*; Turok and Brandenberger, 1986; Sato, 1987).

An unavoidable byproduct of this model for galaxy formation is huge amounts of stochastic gravitational waves produced by the vibrations of the closed loops (Vilenkin, 1981*b*). Detailed calculations by Vachaspati and Vilenkin (1985) (confirming earlier, less accurate calculations by many others) predict that, if the strings are not superconducting (for the superconducting case see Ostriker *et al.*, 1986),

$$\Omega_{\rm GW}(f) \sim 10^{-7}\left(\frac{\mu}{10^{-6}}\right)^{\frac{1}{2}} \quad \text{for all } f \gtrsim 10^{-8}\ \text{Hz}\left(\frac{10^{-6}}{\mu}\right). \tag{68}$$

(For the spectrum at lower frequencies see Hogan and Rees, 1984.) If μ is far less than 10^{-6} (i.e. if gravity-wave observations constrain $\Omega_{\rm GW}(f)$ to be $\ll 10^{-7}$), then the non-superconducting cosmic-string theory of galaxy formation will face severe difficulties. Fig. 9.7 shows the predicted waves (68). From that diagram and the corresponding discussions in Section 9.5 it is clear that several different observational techniques have the prospect of placing cosmic string theory in jeopardy – or, hopefully, of discovering string-produced waves. (The apparent disproof of $\Omega_{\rm GW} \sim 10^{-7}$ coming from 5°-scale anisotropy of the cosmic microwave radiation (Fig. 9.7 and Section 9.5.6(c)) does not in fact constrain cosmic strings, since this observational limit is sensitive only to waves that were present and had (reduced wavelength) ~ (horizon size) at the epoch of recombination – before the strings that produce this wavelength began to vibrate and radiate.)

9.5 Detection of gravitational waves

9.5.1 Methods of analyzing gravitational wave detectors

When analyzing the performance of a gravitational wave detector, it is important to pay attention to the size L of the detector compared with a reduced wavelength $\lambda\!\!\!^{-}$ of the waves it seeks.

If $L \ll \lambda\!\!\!^{-}$ then the detector can be contained entirely in the proper reference frame of its center, and the analysis can be performed using non-relativistic concepts augmented by the quadrupolar gravity-wave force field (3), (5). If one prefers, of course, one instead can analyze the detector in TT coordinates using general relativistic concepts and the spacetime metric (8). The two analyses are guaranteed to give the same predictions for the detector's performance, unless errors are made. However, errors are much more likely in the TT analysis than in the proper-reference-frame analysis, because our physical intuition about how experimental apparatus behaves is

proper-reference-frame based rather than TT-coordinate based. As an example, we intuitively assume that if a microwave cavity is rigid, its walls will reside at fixed coordinate locations x^j. This remains true in the detector's proper reference frame (aside from fractional changes of order $(L^2/\lambda\!\!\!^{-2})h$, which are truly negligible if the detector is small and which the proper-reference-frame analysis ignores). But it is not true in TT coordinates; there the coordinate locations of a rigid wall are disturbed by fractional amounts of order h, which are crucial to analyses of microwave-cavity-based gravity wave detectors.

Thus, for small detectors, $L \ll \lambda\!\!\!^{-}$, the proper-reference-frame analysis is much to be preferred.

For large detectors, $L \gtrsim \lambda\!\!\!^{-}$, one cannot introduce a proper reference frame that covers the entire detector. Such detectors can only be analyzed using general relativistic concepts in TT coordinates (usually the best) or in some other suitable coordinate system (rarely as good).

9.5.2 Resonant bar detectors

More effort has been put into resonant bars than into any other type of gravity-wave detector. Weber's original detectors were of the resonant-bar type; all but one of the first-generation (pre-1977) earth-based detectors were of this type; and eight of the world's twelve research groups now building and operating earth-based detectors are working with bars. Of the current bar efforts three are in the United States (the University of Maryland (Weber, 1986), Stanford University (Boughn *et al.*, 1982; Michelson, 1983) and Louisiana State University (Hamilton *et al.*, 1986)); two are in Europe (the University of Rome with its detector sited at CERN (Amaldi *et al.*, 1984) and Moscow University (Braginsky, 1983)); and three are in the Far East (The University of Western Australia in Perth (Blair, 1983), Tokyo University (Owa *et al.*, 1986) and Guangzhou, China (Hu *et al.*, 1986)). The improvements in resonant-bar sensitivities since Weber's first detector have been a factor of roughly 200 in amplitude, corresponding to 40 000 in energy; and significant further improvements are yet to come.

(a) *How a resonant-bar detector works*

Schematically (Fig. 9.8), a resonant-bar detector consists of a large, heavy, solid bar whose mechanical oscillations are driven by gravitational waves, a transducer that converts information about the bar's oscillations into an electrical signal, an amplifier for the electrical signal, and a recording

system. The transducer and amplifier together are sometimes called the sensor.

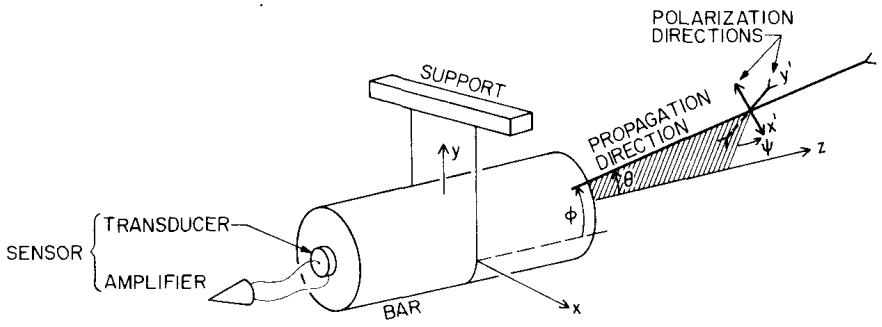
The transducer typically is mounted on one end of the bar (though other mountings are sometimes used), and it produces an output voltage or current proportional to the displacement $x(t)$ of the bar's end from equilibrium. Although $x(t)$ is a sum of contributions from all the $\sim 10^{29}$ normal modes of the bar, the transducer's output is filtered by the amplifier so that only the contribution of the bar's fundamental normal mode is passed on through. This is accomplished by a band-pass filter centered on the frequency f_0 of the fundamental mode, with bandwidth Δf somewhat smaller than the difference $f_1 - f_0$ between the bar's fundamental and its first harmonic. Thus, in effect, it is the fundamental mode of the bar that acts as the gravity-wave detector; and all the other normal modes are almost irrelevant.

Since the fundamental mode involves the relative in and out motion of the bar's left and right ends with just one node (at the bar's center), it corresponds to a standing sound wave with wavelength twice the length of the bar. Correspondingly, the bar's length must be

$$L \simeq \tfrac{1}{2} v_s / f_0, \tag{69}$$

where v_s is the speed of sound in the bar. Typical solid materials have longitudinal sound speeds of order 5 km s^{-1}; astrophysics suggests (Section 9.4) that 1 kHz is a reasonable frequency to search for gravitational waves; and correspondingly the lengths of typical resonant bar detectors are about 2 m and their masses are several tonnes. Notice that equation (69) gives for the ratio of the length of the bar to the reduced wavelength of the

Fig. 9.8. Schematic diagram of a *resonant-bar detector* for gravitational waves. The angles (θ, ϕ, ψ) characterizing the propagation and polarization directions of the waves relative to the detector are a specialization of the angles (θ, ϕ, ψ) shown in Fig. 9.2.



gravitational waves, $\lambda\!\!\!^{-}=c/2\pi f_0$,

$$L/\lambda\!\!\!^{-} \simeq \pi v_s/c \simeq 5\times 10^{-5}. \tag{70}$$

This justifies, with high accuracy, the use of a Newtonian-language, proper-reference-frame viewpoint in analyses of bar detectors. I shall adopt that viewpoint in this chapter.

The contribution of the fundamental mode to the displacement $x(t)$ of the bar's end can be expressed in the standard harmonic-oscillator form

$$x(t)=\text{Real}[X(t)\,e^{-i2\pi f_0 t}], \tag{71}$$

where

$$X(t)=X_1+iX_2 \tag{72}$$

is the mode's complex amplitude and f_0 is its eigenfrequency. It is actually the quantity $X(t)$ that the sensor monitors. To the extent that one can ignore forces from the fundamental mode's environment (e.g. very weak coupling to the other modes), only gravitational waves will produce time changes of $X(t)$. When waves hit, they drive $X(t)$ to evolve in a complicated, time-dependent way, and that evolution in principle can be deconvolved to reveal some details of the waveform $h^{TT}_{jk}(t)$.

Unfortunately, noise in the sensor is typically so severe that to control it the experimenter must use a bandwidth Δf that is far smaller than the frequency f_0 (typically $\Delta f/f_0 \sim 0.01$ in present detectors). Correspondingly, the sensor averages $X(t)$ for a time $\hat{\tau} \simeq 1/\Delta f$ that is long compared to the period $P_0 = 1/f_0$ of the gravitational waves being sought, before passing $X(t)$ on to the recording system. This means that for typical gravitational-wave bursts (e.g. from supernovae), which have durations of only a few times P_0, all that can be monitored is the total change ΔX in the complex amplitude from before the wave arrives until after it has passed. Only uncommonly long bursts, those lasting for more than $f_0/\Delta f \sim 100$ cycles, can be monitored in greater detail. In the future, however, there is hope of bringing the sensor noise under better control and thereby opening up the bandwidth to $\Delta f \simeq 0.2 f_0$ to permit detailed monitoring of much shorter bursts (Michelson and Taber, 1984). (For a description of several first-generation broad-band bars see Figure 2 of Drever, 1977 and associated text, and references therein.)

(b) *The sensitivity of bar detectors to short bursts*

A bar detector couples to the field (equation (26))

$$h(t)=F_+(\theta,\phi,\psi)h_+(t;\iota,\beta)+F_\times(\theta,\phi,\psi)h_\times(t;\iota,\beta), \tag{73}$$

where, if the bar is axially symmetric and the direction (θ,ϕ) and

polarization angle ψ are defined as in Fig. 9.8, the beam-pattern factors are

$$F_+ = \sin^2\theta \cos 2\psi, \qquad F_\times = \sin^2\theta \sin 2\psi. \tag{74}$$

(See Chapter 37 of MTW for the key elements of a derivation.) We shall presume, throughout this subsection, that $h(t)$ is a burst of such short duration $\Delta t \lesssim \hat{\tau} = 1/\Delta f$ that the optimal way to search for it is to measure the mean square change $|\Delta X|^2$ it produces in the fundamental mode's complex amplitude. In this case the general formula (29) for the ratio S^2/N^2 of the burst's squared signal to the mean-square Gaussian noise in the detector can be reduced to the simple form (see, e.g., Giffard, 1976; Pallotino and Pizella, 1981; Michelson and Taber, 1984)

$$\frac{S^2}{N^2} = \frac{\frac{1}{2}M_{\text{eff}}(2\pi f_0)|\Delta X|^2}{kT_n} \tag{75}$$

Here k is Boltzmann's constant, T_n is a 'noise temperature' which characterizes the overall noise in the detector, and M_{eff} is an 'effective mass' associated with the fundamental mode, so defined that $\frac{1}{2}M_{\text{eff}}|X|^2(2\pi f_0)^2$ is the total energy in the mode when it is vibrating with complex amplitude X. (Since X is actually the amplitude of motion of the end of the bar, for a bar that has uniform cross-section and is long compared to its diameter, the effective mass is $M_{\text{eff}} = \frac{1}{2}(1+\nu^2)M$ where ν is the Poisson ratio of the bar's material and M is the bar's mass.)

Because the net wave-induced change ΔX in complex amplitude is independent of the mode's initial complex amplitude, the numerator of equation (75) is the energy that the wave would have deposited in the mode if the mode had been initially unexcited. This deposited energy can conveniently be expressed in terms of the cross-section $\sigma_o(f)$ that the mode would present to the wave if the wave had hit it from an optimal direction (broadside, $\theta = \pi/2$) and with an optimal polarization (+ mode with $\psi = 0$):

$$\tfrac{1}{2}M_{\text{eff}}(2\pi f_0)^2|\Delta X|^2 = \int_0^\infty \frac{\pi}{2} f^2 |\tilde{h}(f)|^2 \sigma_o(f)\, df. \tag{76}$$

Here $\tilde{h}(f)$ is the Fourier transform of $h(t)$ (equation (73)), and for an optimal direction and polarization $(\pi/2)f^2|\tilde{h}(f)|^2$ would be the energy per unit area per unit frequency ($f \geq 0$) carried by the waves. Because $\sigma_o(f)$ is extremely sharply peaked around the resonant frequency f_0 (Section 37.5 of MTW), we can rewrite (76) and thence (75) in the form

$$\frac{S^2}{N^2} = \frac{\pi}{2} f_0^2 |\tilde{h}(f_0)|^2 \frac{\int \sigma_o\, df}{kT_n}. \tag{77}$$

This is a special narrow-band-detector version of the general equation (29)

for arbitrary detectors. Correspondingly, equation (30) for the strongest burst seen, on average, at the same rate as bursts occur inside the distance to our source, becomes

$$\frac{S^2}{N^2} \cong \frac{3}{2}\frac{\pi}{2} f_0^2 \langle |\tilde{h}_+(f_0)|^2 + |\tilde{h}_\times(f_0)|^2 \rangle \langle F_+^2 \rangle \frac{\int \sigma_o \, df}{kT_n}, \tag{78}$$

where, for the F_+ and $F_\times$ of (74),

$$\langle F_+^2 \rangle = \langle F_\times^2 \rangle = \frac{4}{15} = 0.267. \tag{79}$$

Equation (78) motivates us to define

$$h_c \equiv (3)^{\frac{1}{2}} f_0 \langle |\tilde{h}_+(f_0)|^2 + |\tilde{h}_\times(f_0)|^2 \rangle^{\frac{1}{2}} \tag{80}$$

as the *characteristic amplitude of the burst* (analog of (31b) for broad-band detectors) and

$$h_n \equiv \left[\frac{15}{\pi} \frac{kT_n}{\int \sigma_o \, df} \right]^{\frac{1}{2}} \cong 2.2 \left[\frac{G}{c^3} \frac{kT_n}{\int \sigma_o \, df} \right]^{\frac{1}{2}} \tag{81}$$

as the *detector's characteristic noise amplitude* (analog of (32) for a broad-band detector). Correspondingly, the characteristic amplitude $h_c = h_{3/yr}$ that a source must have to be detectable with 90% confidence in a search lasting $\frac{1}{3}$ year is

$$\begin{aligned} h_{3/yr} &\cong 5h_n \cong 11 \left[\frac{G}{c^3} \frac{kT_n}{\int \sigma_o \, df} \right]^{\frac{1}{2}} \\ &= 2.0 \times 10^{-16} \left[\frac{T_n/1K}{\int \sigma_o \, df / 10^{-21} \text{ cm}^2 \text{ Hz}} \right]^{\frac{1}{2}}. \end{aligned} \tag{82}$$

For a broad-band burst that is peaked near the detector's resonant frequency f_0 (so the f_c of equation (31a) is approximately f_0) and that lasts for a time not much longer than $\Delta t = 1/f_0$, the narrow-band characteristic amplitude (80) will be roughly equal to the broad-band characteristic amplitude (31b). For such bursts, and only for such bursts, it makes sense to plot a bar detector's narrow-band $h_{3/yr}$ on the same graph (Fig. 9.4) as a broad-band detector's $h_{3/yr}$. Inspiraling binaries do not belong to this class of bursts, so their detection by bars must be discussed separately from Fig. 9.4.

(c) *How to optimize the sensitivities of bar detectors*

Equation (82) shows that to optimize the sensitivity of a bar detector to short, broad-band bursts, one must achieve the largest possible frequency-

integrated cross section $\int \sigma_0 \, df$ and the smallest possible noise temperature T_n.

The integrated cross section $\int \sigma_0 \, df$ can be computed by analyzing the response of the bar's fundamental mode to the force field (5) produced by a sinusoidal wave with optimal direction and polarization, and by then integrating over the wave's frequency. See MTW Box 37.4 for details. For a cylindrical bar with length L somewhat greater than radius R (the usual situation), the result depends only on the bar's mass M and internal sound velocity $v_s = (E/\rho)^{\frac{1}{2}}$ with $E =$ (Young's modulus) and $\rho =$ (density)(Paik and Wagoner, 1976):

$$\int \sigma_0 \, df = \frac{8}{\pi} \frac{GMv_s^2}{c^3} [1 + \tfrac{1}{2}\nu(1-2\nu)(\pi R/L)^2 + \cdots]$$

$$= 1.6 \times 10^{-21} \text{ cm}^2 \text{ Hz} \left(\frac{M}{10^3 \text{ kg}}\right)\left(\frac{v_s}{5 \text{ km s}^{-1}}\right)^2 . \tag{83}$$

(Here ν is the Poisson ratio of the bar's material, and only the leading shape-dependent corrections are given.) Thus, it is desirable to build detectors that are as massive and have as high sound speed as possible.

The noise temperature T_n is determined by a combination of noise in the sensor and thermal noise in the bar.

The *thermal noise in the bar* is caused by weak coupling of the bar's fundamental mode to its environment – its $\sim 10^{29}$ other modes, the wire or cable or prongs that suspend the bar, and the residual gas in the vacuum chamber (Braginsky, Mitrofanov and Panov, 1985). If, as is normal, this environment is thermalized at some physical temperature T_b (subscript 'b' for bar or for thermal bath), then these couplings cause the mode's amplitude to execute a random walk (Brownian motion) in the domain $|X| \lesssim X_{th}$ corresponding to an energy kT_b:

$$X_{th} = \left[\frac{2kT_b}{M_{eff}(2\pi f_0)^2}\right]^{\frac{1}{2}} . \tag{84}$$

The fluctuation-dissipation theorem states that the timescale on which this random walk produces changes of order X_{th} is the same as the timescale $\tau^* = Q/\pi f_0$ for large-amplitude vibrations to be damped frictionally. (Here Q is the quality factor of the mode's large-amplitude oscillations: the number of radians of oscillation required for the energy to damp by a factor e.) Consequently, the mean square Brownian change in the mode's amplitude

during the sensor's averaging time $\hat{\tau} = 1/\Delta f$ is

$$|\Delta X_{\rm th}|^2 = X_{\rm th}^2 \frac{\hat{\tau}}{\tau^*} = X_{\rm th}^2 \frac{\pi}{Q} \frac{f_0}{\Delta f} \tag{85}$$

corresponding to

$$\tfrac{1}{2} M_{\rm eff} (2\pi f_0)^2 |\Delta X_{\rm th}|^2 = \pi \frac{kT_{\rm b}}{Q} \frac{f_0}{\Delta f}. \tag{86}$$

The details of the *sensor noise* depend on a variety of factors, including: the detailed structure of the transducer (for a review of many transducer structures see Section 4.1.5 of Amaldi and Pizella, 1979), the strength of coupling of the transducer to the bar (characterized by a dimensionless coupling constant β that is roughly equal to the number of cycles of oscillation required for all the fundamental mode's energy to be fed into the transducer); the impedance mismatch between the transducer and the amplifier; the noise temperature $T_{\rm a}$ of the amplifier (converted to an equivalent noise temperature $(f_0/f_{\rm a})T_{\rm a}$ at the bar's frequency f_0 if the amplifier operates at a different frequency $f_{\rm a}$ than the bar); and the sensor bandwidth Δf. Although the details vary from one sensor to another (see, e.g., Weiss, 1978; Pallotino and Pizella, 1980; Blair, 1983; Michelson and Taber, 1984), the spirit of the details is captured by the following approximate expression for the mean square change $|\Delta X_{\rm sensor}|^2$ that the experimenter would infer from the sensor's output if (i) only the sensor noise were present, (ii) back-action forces of the sensor on the bar were negligible (see subsection (f) below), and (iii) impedances were properly matched:

$$\tfrac{1}{2} M_{\rm eff} (2\pi f_0)^2 |\Delta X_{\rm sensor}|^2 \simeq \frac{kT_{\rm a}(f_0/f_{\rm a})}{\beta} \frac{\Delta f}{f_0}. \tag{87}$$

The sum of $\frac{1}{2} M_{\rm eff} (2\pi f_0)^2 |\Delta X_{\rm sensor}|^2$ (equation 87)) and $\frac{1}{2} M_{\rm eff} (2\pi f_0)^2 |\Delta X_{\rm th}|^2$ (equation (86)) is the mean square noise N^2 that appears in the denominator of equation (75); i.e., by definition, the detector's noise energy $kT_{\rm n}$ is this sum. If the experimenters choose too large a bandwidth Δf, then the sensor noise (87) will become inordinately large, producing a large $T_{\rm n}$ and masking the gravity-wave signal. If they choose too small a bandwidth, then the bar's thermal noise (86) will become inordinately large. There is an optimal bandwidth, typically $\Delta f \sim 10$ Hz in present detectors (corresponding to the averaging time $\hat{\tau} \sim 0.1$ s) at which roughly half the noise is from the sensor and half is from thermal motion of the bar. When the bandwidth is chosen optimally these two noise sources together produce a detector noise

temperature

$$T_n \simeq \left[\frac{\pi}{\beta}\frac{T_b}{Q}\left(T_a\frac{f_0}{f_a}\right)\right]^{\frac{1}{2}}. \tag{88}$$

In practice, the experimenters choose a bar early in their experiments, thereby fixing the integrated cross section $\int \sigma_0 \, df$; and they then struggle for many years to develop a sensor and its coupling to the bar, and a thermally cold environment, that will minimize the noise temperature T_n. Maximizing $\int \sigma_0 \, df$ is achieved by maximizing the bar's mass and its velocity of sound (and, to a small extent, optimizing its shape subject to other experimental constraints such as available cryostats.) Minimizing T_n is achieved according to equation (88) by (i) maximizing the fundamental mode's quality factor Q (i.e. minimizing its coupling to the rest of the world), (ii) cooling the bar to as low a physical temperature T_b as possible, (iii) maximizing the strength β of coupling of the transducer to the bar, (iv) using an amplifier with as low a 'noise number' $kT_a/(2\pi\hbar f_a)$ as possible (the Heisenberg uncertainty principle limits the noise number to be $\gtrsim 1$; Weber, 1959; Heffner, 1962; Caves, 1982), and (v) struggling to get good impedance matching of the transducer and amplifier (a requirement for equation (88) to be valid).

(d) *Parameters of first- and second-generation bar detectors*

The first-generation bar detectors (pre-1977) were all made of aluminum, weighed roughly 1.5 tonnes, and had $f_0 \simeq 1.6$ kHz and $Q \sim 10^5$; they all operated at room temperature, $T_b \cong 300$ K; and most used piezo-electric transducers – i.e. crystals glued to the bar which produce small voltages when squeezed. For most the coupling of the transducer to the bar was weak, $\beta \lesssim 10^{-4}$; but those in Britain achieved strong coupling, $\beta \simeq 0.2$ and hence wide bandwidth $\Delta f/f_0 \sim 1$ at the price of reducing the bar's Q from $Q \sim 10^5$ to $Q \sim 2000$. The best amplifiers that could be impedance matched to the piezo-electric transducers had rather large noise numbers. For these first-generation bars the integrated cross-sections were $\int \sigma_0 \, df \simeq 2 \times 10^{-21}$ cm^2 Hz, and the lowest detector noise temperatures were $T_n \simeq 4$ K corresponding to a minimum detectable burst amplitude with $\frac{1}{3}$ year of observation $h_{3/\mathrm{yr}} \cong 3 \times 10^{-16}$. Despite great effort in the early 70s by excellent experimenters, there was great room for improvement. (For a thorough review of the first-generation experiments see Amaldi and Pizella, 1979; for other reviews, see Drever, 1977, Tyson and Giffard, 1978 and Weber, 1986.)

In moving into the second generation, almost all the groups cooled their bars to liquid helium temperatures ($T_b = 1.5$–4 K rather than 300 K).

Several of the groups (Maryland, Stanford, LSU) constructed massive bars ($M \cong 2$–5 tonnes) from a new alloy of aluminum that the Tokyo group discovered has a Q 10 times higher than the old one (5×10^7 vs 5×10^6 at 10^4 K temperature; Suzuki, Tsubono and Hirakawa, 1978). Perth and Moscow, by contrast, chose to use exotic bar materials. Perth used a 1.5 tonne Niobium bar with $Q = 2 \times 10^8$, while Moscow initially used a 10 kg sapphire bar with $Q = 4 \times 10^9$, and later when difficulty with cracking of the sapphire occurred, Moscow switched to a 10 kg silicon crystal bar with $Q = 2 \times 10^9$ – silicon and sapphire because of their very large Qs, an advantage bought at the price of low mass.

In the second generation each of the groups deemphasized piezo-electric transducers and developed some variant of one or another of two new transducer concepts: Moscow, LSU, Perth and Tokyo developed *parametric transducers* in which the bar's vibrations with frequency $f_0 \sim 10^3$ Hz modulate the capacitance in a microwave cavity (or rf circuit), thereby moving microwave photons from the frequency $f_D \sim 10^9$ Hz, at which the cavity is driven, into side bands at $f = f_D \pm f_0$, at which the amplifier measures the signal. The number of photons moved is proportional to the amplitude of vibration of the bar. Stanford and Maryland developed *resonant transducers* in which a mechanical diaphragm, with a mechanical resonant frequency very nearly that of the bar's fundamental mode, is attached to the bar's end. The vibration energy is quickly transferred back and forth between the bar and the diaphragm, with the diaphragm's displacement amplitude amplified over that of the bar by $|X_{\text{diaphragm}}|/|X_{\text{bar}}| \sim [(\text{bar mass})/(\text{diaphragm mass})]^{\frac{1}{2}}$. The vibrating diaphragm modulates the inductance of a superconducting circuit, and a SQUID (superconducting quantum interference device) amplifier is used to monitor the current in the circuit. Rome developed a similar resonant transducer with the diaphragm replaced by a toad stool which was one plate of a capacitor and with a FET transducer used to read out oscillations of the toad stool's voltage.

At present (late 1986) Stanford, Rome and LSU are all on the air with systems that have resonant frequencies $f_0 \cong 900$ Hz, cross-sections $\int \sigma_0 \, df \cong (4 \text{ to } 8) \times 10^{-21}$ cm^2 Hz, and noise temperatures $T_n \cong (0.01$–$0.04)$ K. Correspondingly the weakest burst they can detect with $\frac{1}{3}$ year of observation is $h_{3/\text{yr}} \cong 1.0 \times 10^{-17}$. Maryland and Perth are both likely to be on the air with similar characteristics before this book is published; and all five groups are hoping to push their noise levels down to $h_{3/\text{yr}} \sim 2 \times 10^{-18}$ within several years by further straightforward refinements. The Guangzhou group, having only entered the field very recently, is still at

room temperature, but with a sensitivity $h_{3/\mathrm{yr}} \simeq 1.6 \times 10^{-16}$, two times better than that of any of the first-generation room-temperature bars (Hu *et al.* 1986). The Moscow group with its small silicon and sapphire bars operates at a much higher frequency than any of the other groups, $f_0 \cong$ 8 kHz; by the time this book is published they will likely be on the air with $h_{3/\mathrm{yr}} \sim 4 \times 10^{-17}$. The Tokyo group, on the other hand, has chosen a much lower frequency than the others, $f_0 \cong 60$ Hz and is now operating a narrow-band search for periodic waves from the Crab pulsar (see below). These sensitivities are shown in Fig. 9.4, along with those of other kinds of detectors. For details of the present detector configurations and near-term plans, see the gravitational-wave articles in the proceedings of recent conferences (Ruffini, ed., 1986; MacCallum, ed., 1987).

(e) *Sensitivies of bar detectors to periodic and stochastic waves*

Although I have discussed the detectors' performances entirely in the context of searching for short bursts of gravitational waves, bar detectors can be used also to search for periodic gravitational waves and a stochastic background at frequencies within the bandwidth Δf of their sensors.

In a search for periodic gravitational waves, the experimenters will typically use a bar with eigenfrequency f_0 slightly different from the expected frequency f of the waves. The waves then will drive the fundamental mode's complex amplitude X into oscillations with the beat frequency $f - f_0$. In searching for these oscillations the experimenters integrate for a long time; and correspondingly they can turn down the coupling strength β of the transducer to the bar, and/or narrow the bandwidth Δf, until the sensor's noise becomes negligible compared to the thermal noise of the bar (see, e.g., Michelson and Taber, 1981, 1984). (The maximum resulting bandwidth with the present Stanford detector would be $\Delta f \simeq 0.5$ Hz). When this is done, the general formulas of Section 9.4.2(a) are valid, with $S_h(f)$ the spectral density of the bar's thermal noise, converted into an equivalent gravity-wave spectral density

$$S_h(f) = \frac{4kT_\mathrm{b}}{\int \sigma_0 \, \mathrm{d}f} \frac{1}{f_0 Q}, \tag{89}$$

and with the beam-factor averages having the values (79). In particular, the detector's characteristic noise amplitude (equation (51)) is

$$h_\mathrm{n} = \left[15 \frac{G}{c^3} \frac{kT_\mathrm{b}}{\int \sigma_0 \, \mathrm{d}f} \frac{1}{Q} \frac{1}{f_0 \hat{\tau}} \right]^{\frac{1}{2}}, \tag{90}$$

and the brightest source that can be seen with 90% confidence in $\hat{\tau} = \frac{1}{3}$ year of

integration is

$$h_{3/\mathrm{yr}} = 1.7h_{\mathrm{n}} = 3.9\times 10^{-25}\left(\frac{T_{\mathrm{b}}}{1\,\mathrm{K}}\right)^{\frac{1}{2}}\left(\frac{10^{-21}\,\mathrm{cm^2\,Hz}}{\int \sigma_0\,\mathrm{d}f}\right)^{\frac{1}{2}}\left(\frac{1000\,\mathrm{Hz}}{f_0}\right)^{\frac{1}{2}}\left(\frac{10^7}{Q}\right)^{\frac{1}{2}}. \tag{91}$$

The Tokyo group is currently carrying out a search for gravitational waves from the Crab pulsar using the above technique (Owa *et al.*, 1986). Their 74 kg, cryogenically cooled antenna has $Q=2.1\times 10^7$, $T_{\mathrm{b}}=4$ K, $f_0=60$ Hz, and $\int \sigma_0\,\mathrm{d}f \simeq 2.2\times 10^{-27}\,\mathrm{cm^2\,Hz}$ (so low because for noncylindrical antennas with frequency f_0 lowered by special shaping, $\int \sigma_0\,\mathrm{d}f \propto f_0^2$; Hirakawa *et al.*, 1976). Correspondingly it has $h_{3/\mathrm{yr}} \sim 3\times 10^{-22}$. No other present bars are optimized for periodic waves, since there are no known sources in their frequency bands (~ 900 Hz and ~ 8 kHz). However, with the technology of present burst-optimized bars it should be possible to achieve the thermal-noise-limited sensitivity (91) with $T_{\mathrm{b}}=4$ K, $\int \sigma_0\,\mathrm{d}f \simeq 8\times 10^{-21}\,\mathrm{cm^2\,Hz}$, $f_0 \simeq 900$ Hz, and $Q\simeq 5\times 10^6$ corresponding to $h_{3/\mathrm{yr}} \simeq 4\times 10^{-25}$ (Stanford, LSU, Rome, Maryland); and $T_{\mathrm{b}}=4$ K, $\int \sigma_0 \mathrm{d}f \simeq 2\times 10^{-23}\,\mathrm{cm^2\,Hz}$, $f_0 \simeq 8$ kHz, and $Q\simeq 2\times 10^9$ corresponding to $h_{3/\mathrm{yr}} \simeq 1.4\times 10^{-25}$ (Moscow). See Fig. 9.6.

When searching for stochastic background it is desirable to open up the bandwidth Δf until the sensor noise becomes almost as large as the bar's thermal noise ($\Delta f \simeq 0.5$ Hz for the present Stanford bar). With $S_h(f)$ then given (approximately) by the thermal-noise spectral density (equation (89)), the noise amplitude h_{n} and the 90%-conficence $\frac{1}{3}$-year sensitivity $h_{3/\mathrm{yr}}$ for stochastic background become (equations (66) and (67))

$$h_{\mathrm{n}}(f) \simeq \left[15\frac{G}{c^3}\frac{kT_{\mathrm{b}}}{\int \sigma_0\,\mathrm{d}f}\frac{1}{Q}\frac{1}{\sqrt{(\frac{1}{2}\hat{\tau}\,\Delta f)}}\right]^{\frac{1}{2}}, \tag{92}$$

$$h_{3/\mathrm{yr}} = 1.7h_{\mathrm{n}} \cong 8\times 10^{-22}\left(\frac{T_{\mathrm{b}}}{1\,\mathrm{K}}\right)^{\frac{1}{2}}\left(\frac{10^{-21}\,\mathrm{cm^2\,Hz}}{\int \sigma_0\,\mathrm{d}f}\right)^{\frac{1}{2}}\left(\frac{10^7}{Q}\right)^{\frac{1}{2}}\left(\frac{1\,\mathrm{Hz}}{\Delta f}\right)^{\frac{1}{4}}. \tag{93}$$

This $h_{3/\mathrm{yr}}$ is shown in Fig. 9.7 for the parameters of 1987 bar technology (those given in the last paragraph, plus $\Delta f \simeq 1$ Hz). For details of searches for stochastic background that were carried out using first-generation bar detectors, see Hough *et al.* (1975) and Hirakawa and Narihara (1975). For discussions of sensitivity that are more detailed and sophisticated than the above sketch, see Hirakawa, Owa and Iso (1985), Weiss (1979) and references therein.

(f) *Quantum limit and quantum non-demolition*

Quantum mechanics constrains the sensitivity that can be achieved by bar

detectors using the present kinds of sensors: the fundamental mode of a bar, being highly decoupled from the rest of the world, can be regarded as a simple harmonic oscillator with mass $M_{\rm eff}$ and angular frequency $\omega_0 = 2\pi f_0$. As such, it is subject to the laws of quantum mechanics for oscillators: its generalized position x and momentum p must be regarded as hermitian operators that fail to commute, $[x, p] = i\hbar$; and, correspondingly, the real and imaginary parts of its complex amplitude,

$$X_1 = x \cos \omega_0 t - \left(\frac{p}{M_{\rm eff}\omega_0}\right) \sin \omega_0 t,$$

$$X_2 = x \sin \omega_0 t + \left(\frac{p}{M_{\rm eff}\omega_0}\right) \cos \omega_0 t, \tag{94}$$

are non-commuting hermitian operators with commutator

$$[X_1, X_2] = i \frac{\hbar}{M_{\rm eff}\omega_0}. \tag{95}$$

From this commutation relation we infer, via the Heisenberg uncertainty principle, an absolute limit on the variances of X_1 and X_2 in any quantum mechanical state (Thorne *et al.*, 1978):

$$\Delta X_1 \, \Delta X_2 \geq \frac{\hbar}{2M_{\rm eff}\omega_0}. \tag{96}$$

The principles of quantum mechanics guarantee that one can never measure X_1 and X_2 simultaneously with a precision that violates this uncertainty principle. In fact, it turns out (Caves, 1982; Yamamoto and Haus, 1986) that even the most ideal of measuring systems will introduce additional noise equal to (96), thereby producing a lower bound

$$\Delta X_1 \, \Delta X_2 \geq \frac{\hbar}{M_{\rm eff}\omega_0} \tag{97}$$

on the products of the rms noise in the measured values of X_1 and X_2.

The sensors used in the present generation of bar detectors measure X_1 and X_2 simultaneously with equal accuracies; and, correspondingly, in searches for short bursts they are subject to the 'standard quantum limit' (Braginsky and Vorontsov, 1974; Giffard, 1976)

$$T_{\rm n} = \tfrac{1}{2} M_{\rm eff}\omega_0^2 [(\Delta X_1)^2 + (\Delta X_2)^2]/k \geq \hbar\omega_0/k = 4.8 \times 10^{-8}\ {\rm K} \left(\frac{f_0}{1000\ {\rm Hz}}\right). \tag{98}$$

Notice that this standard quantum limit places the severe constraint

$$h_{3/\mathrm{yr}} \gtrsim 4.4 \times 10^{-20} \left(\frac{f_0}{1000\ \mathrm{Hz}} \right)^{\frac{1}{2}} \left(\frac{10^{-21}\ \mathrm{cm^2\ Hz}}{\int \sigma_0\, \mathrm{d}f} \right)^{\frac{1}{2}} \tag{99}$$

on the detector's burst sensitivity (82). It is fairly likely, though far from certain, that the strongest kilohertz-frequency bursts striking the earth three times per year have characteristic amplitudes $h_c < 10^{-20}$ (see Section 9.4.1); and, correspondingly, it may turn out to be crucial for bar detectors of the future to circumvent the standard quantum limit (98), (99).

The uncertainty principle (96) suggests a promising method for circumventing the standard quantum limit (Thorne *et al.*, 1978; Braginsky, Vorontsov and Khalili, 1978): one should devise a new kind of sensor that measures X_1 with high accuracy, while giving up accuracy on X_2. Such sensors, called 'back-action-evading sensors' (a special case of 'quantum non-demolition sensors'), are now under development in a number of laboratories (Braginsky, 1983; Bocko and Johnson, 1984; Oelfke, 1983; Blair, 1982); and they may make possible bar sensitivities in the 1990s that will beat the standard quantum limit by modest factors. For a detailed review of quantum non-demolition measurements – i.e. measurements that do not change the quantum state of the system being measured – see Caves (1983).

Although a back-action-evading sensor gives up accuracy on one of the wave's two quadrature components, that accuracy can be regained by looking at the same wave with two different detectors: on one detector, with complex amplitude $X = X_1 + iX_2$, measure X_1 with high accuracy and X_2 with poor; on the other, with complex amplitude $Y = Y_1 + iY_2$, measure Y_2 with high accuracy and Y_1 with poor. From X_1 infer the detailed evolution of one of the wave's two quadrature components; from Y_2 infer the evolution of the other. In this way, in principle, the quantum mechanical properties of the detector can be completely circumvented and the only constraints of principle on the accuracy of measurement are associated with quantization of the waves themselves. For further discussion and details see the reviews by Caves *et al.* (1980); Caves (1983); Braginsky, Vorontsov and Thorne (1980).

It is worth noting that a back-action-evasion measurement, ideally performed, should drive the bar's fundamental mode into a 'squeezed state' (Hollenhorst, 1979). Squeezed states have been studied extensively in recent years in the context of quantum optics (see, e.g. Schumaker, 1986; and Walls, 1983); and we shall return to them in Section 9.5.3(f) below when discussing beam detectors.

(g) *Looking toward the future*

It may be that bar detectors in the distant future will find their greatest applications at much lower frequencies than are now common: at lower frequencies the bar can be much longer and more massive, and correspondingly more sensitive; and good bandwidth, $\Delta f \sim f_0$ will correspond to a longer averaging time and thus will be easier to achieve. For discussion, see Michelson (1986).

Although individual bar detectors in the foreseeable future will be limited to moderately small bandwidths $\Delta f/f \lesssim 0.2$, and therefore will be able to acquire only very limited information about the wave form $h_{jk}^{\mathrm{TT}}(t)$ of a gravity-wave burst, once waves are being detected in profusion it may be possible cheaply to replicate the detectors with a variety of sizes and hence a variety of fundamental-mode frequencies, thereby creating a 'xylophone' of networked detectors with a large overall bandwidth (Michelson and Taber. 1984).

9.5.3 Beam detectors

(a) *A brief history of beam-detector research*

The germ of the idea of a laser-interferometer gravitational-wave detector ('beam detector') can be found in Pirani (1956); but – so far as I am aware – the first explicit suggestion of such a detector was made by Gertsenshtein and Pustovoit (1962). In the mid-1960s Joseph Weber, unaware of the Gertsenshtein–Pustovoit work, reinvented the idea but left it lying in his laboratory notebook unpublished and unpursued. In 1970 Rainer Weiss at MIT, unaware of Gertsenshtein–Pustovoit or Weber, reinvented the idea and carried out a detailed design and feasibility study (Weiss, 1972) in which many of the techniques now being used were conceived. Unfortunately. Weiss was unable to obtain funding to push forward with a significant experimental effort.

Robert Forward at Hughes Research Laboratories in Malibu, California. having learned the concept of the beam detector from Weber (his former thesis advisor), was motivated indirectly by Weiss in 1971 to construct a prototype detector with funding from Hughes. By 1972 Forward and his colleagues at Hughes were operating the world's first prototype beam detector – an instrument that demonstrated the idea could really work, and that was remarkably sensitive considering the modest effort put into it: $[S_h(f)]^{\frac{1}{2}} \simeq 2 \times 10^{-16}\ \mathrm{Hz}^{-\frac{1}{2}}$ between 2500 Hz and 25000 Hz, corresponding to $h_{3/\mathrm{yr}} \simeq 1 \times 10^{-13}$ for 2500 Hz bursts (Moss, Miller and Forward, 1971;

Forward and Moss, 1972; Forward, 1978). Regrettably, Forward could not obtain funds to move from this first prototype to a more sophisticated instrument; so his project was shut down.

With the completion of the first generation of bar detectors in 1975, each experimental group that decided to stay in the field looked carefully at a variety of possibilities for sensitivity improvement. While most groups decided to stick with bars, two switched to beam detectors: Munich, led by H. Billing, and Glasgow, led by Ronald Drever with Jim Hough second in command. The Munich group was strongly influenced by a proposal to develop beam detectors that Weiss had submitted to NSF, and that NSF had refused to fund; and so Munich pushed forward (Winkler, 1977) along the lines that Weiss had hoped to follow, using a Michelson interferometer design (see below). The Glasgow group first built a small Michelson interferometer (Drever *et al.*, 1977), then switched in 1977 to a new Fabry–Perot design invented by Drever (Drever *et al.*, 1980).

In 1979 Caltech managed to attract Drever away from Glasgow (part-time at first, full-time later), leaving Hough as the Glasgow leader. At Caltech Drever started up a beam-detector project; and NSF, finally recognizing that beam detectors were worth funding, agreed to support both Weiss at MIT to develop his original idea of a Michelson system and Drever at Caltech to develop his Fabry–Perot system. More recently, in 1983, Alain Brillet initiated a beam-detector effort in Orsay, France (Brillet and Tourrenc, 1983; Brillet, 1985).

Munich, Glasgow, Caltech and MIT all now have working beam detectors with amplitude sensitivities ~ 2000 times better than that of Forward's first prototype but ~ 5 times worse than the best bars. These detectors are small-scale (1–40 m) prototypes for the full-scale (several kilometer) beam detectors that will be required for real success. Design and costing studies are now underway for the full-scale systems (called 'Laser Interferometer Gravity Wave Observatories' or LIGOs); for details of these studies, see Linsay *et al.* (1983), Drever *et al.* (1985), Maischberger *et al.* (1985), Winkler *et al.* (1986), Hough *et al.* (1986). There is hope that full-scale LIGOs will be constructed in the late 1980s and early 1990s and will be operating in the mid- to late-1990s with sensitivities in the region where gravity waves are expected.

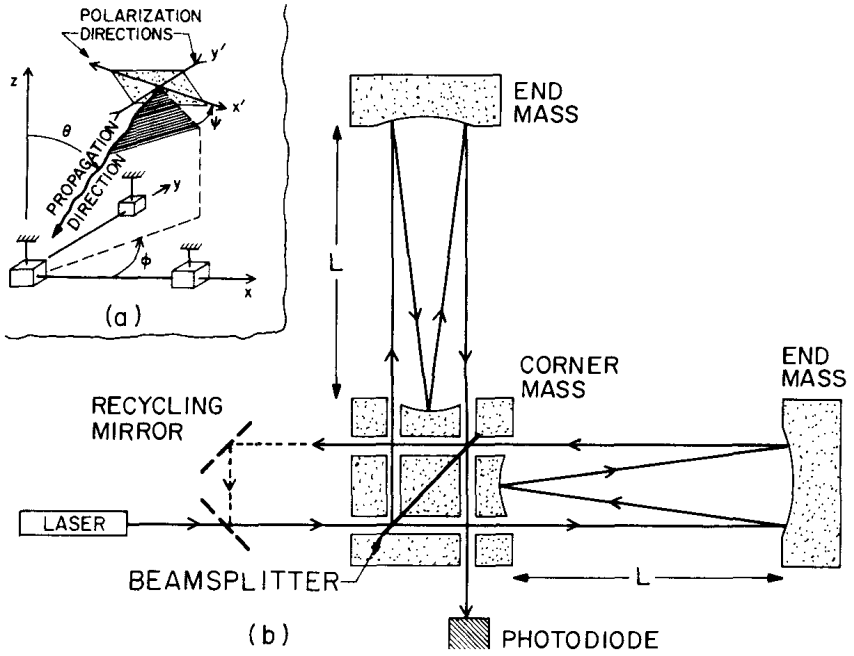
(b) *How a beam detector works*

The current and planned beam detectors are designed to operate at frequencies below 10 kHz because astrophysical arguments suggest that the

waves will be weak above this frequency (see page 158 of Thorne, 1978); and they have their best sensitivities at frequencies below 1 kHz. Correspondingly, the waves they seek all have reduced wavelengths $\lambda\!\!\!^{-} > 5$ km and most have $\lambda\!\!\!^{-} > 50$ km. Since the planned detectors all have sizes $L \leqslant 4$ km, the condition $L \ll \lambda\!\!\!^{-}$ for use of a 'proper-reference-frame analysis' is satisfied, though only marginally in extreme cases. I shall use such an analysis in the discussion below. For an outline of the alternative, TT analysis, see, e.g., Exercise 37.6 of MTW.

A beam *detector* consists of one or more *receivers* that are operated simultaneously, with cross-correlated outputs – the cross-correlation, as usual, being the key to removing spurious, non-Gaussian noise. A simple version of a Michelson-type receiver is shown in Fig. 9.9, three-dimensionally in part (*a*) and as seen from above in part (*b*). (Ignore for the moment the propagation and polarization pieces of part (*a*).) The receiver consists of three masses which hang on wires from overhead supports and swing like pendula. The masses are arranged at the ends and corner of a right-angle L. When a gravitational wave propagates vertically through the receiver with polarization axes along the L ('+' polarization), its

Fig. 9.9. Schematic diagram of a *Michelson-type beam receiver* for gravitational waves (part (*b*)), and of the waves' propagation and polarization angles (θ, ϕ, ψ) relative to the receiver (part (*a*); cf. Fig. 9.2).



quadrupolar force field (5) pushes together the masses on one arm of the L while pushing apart the masses on the other arm. In the next half-cycle of the wave, the directions of the pushes are reversed. Since the waves being sought have frequencies f far above the 1 Hz swinging frequency of the pendula, the pendular restoring forces have no opportunity to make themselves felt: the masses respond to the gravity-wave pushes as though they were free. With the origin of the proper reference frame placed on the central mass, the central mass is left unaffected while the end masses oscillate longitudinally with displacements

$$\delta x(t)=\tfrac{1}{2}Lh_+(t) \qquad \text{for mass on } x \text{ axis,} \tag{100a}$$

$$\delta y(t)=-\tfrac{1}{2}Lh_+(t) \qquad \text{for mass on } y \text{ axis} \tag{100b}$$

(equation (6)). Here L is the (approximately equal) length of each arm. Correspondingly, there is an oscillation in the difference $l(t)$ of the arm lengths, $\delta l(t)=\delta x(t)-\delta y(t)$, given by

$$\delta l(t)=h_+(t)L. \tag{101}$$

It is straightforward to show that in the more general case of a wave which impinges from a direction (θ, ϕ) on the sky with polarization axes rotated at an angle ψ relative to the constant-ϕ plane (Fig. 9.9(a)), the difference in arm lengths l oscillates as

$$\delta l(t)=h(t)L, \tag{102}$$

where $h(t)$ has the standard form (26)

$$h(t)=F_+(\theta, \phi, \psi)h_+(t; \iota, \beta)+F_\times(\theta, \phi, \psi)h_\times(t; \iota, \beta), \tag{103}$$

with beam-pattern factors (cf. Forward, 1978; Rudenko and Sazhin, 1980; Estabrook, 1985; Schutz and Tinto, 1987)

$$F_+(\theta, \phi, \psi)=\tfrac{1}{2}(1+\cos^2\theta)\cos 2\phi \cos 2\psi-\cos\theta \sin 2\phi \sin 2\psi, \tag{104a}$$

$$F_\times(\theta, \phi, \psi)=\tfrac{1}{2}(1+\cos^2\theta)\cos 2\phi \sin 2\psi+\cos\theta \sin 2\phi \cos 2\psi. \tag{104b}$$

The difference of arm lengths $l(t)$ is monitored by Michelson interferometry: a beam splitter and two mirrors are attached to the corner mass as shown in Fig. 9.9(b), and one mirror is attached to each end mass. A laser beam shines through a hole in the corner mass and onto the beam splitter, which directs half the beam toward each end mass. The mirrors on the end masses reflect the beams back toward the corner-mass mirrors, which in turn reflect the beams back to the end masses, which reflect the beams back through holes in the corner-mass mirrors and onto the beam

splitter where they are recombined. Part of the recombined beam goes out one side of the beam splitter toward the laser (ignore for now the 'recycling mirror' in (*b*), it is absent in the simple version of the receiver being discussed here); the other part of the recombined beam goes out the other side toward a photodetector. Oscillations in the arm-length difference $\delta l(t)$ produce oscillations in the relative phases of the recombining light, and thence oscillations in the fraction of the light which goes to the photodetector versus that which goes back toward the laser. The photodetector, by monitoring the oscillations in received intensity, in effect is monitoring the oscillations $\delta l(t)$ of arm-length difference and thence the gravity-wave oscillations $h(t)$.

In practice the laser beams are made to bounce back and forth in the arms not just twice as shown in Fig. 9.9(*b*), but rather a large number of round-trip times B, making B distinct spots on each end mirror. In the simple case that the gravity-wave-induced arm-length difference $\delta l = Lh$ does not change much during these many round trips (see subsection (e) below for the case of large change), the bouncing light beam will build up during its B trips a total phase delay

$$\Delta\Phi = \frac{2B\,\delta l}{\lambda\!\!\!^{-}_{e}} = \frac{2BL}{\lambda\!\!\!^{-}_{e}}\,h \tag{105}$$

where $\lambda\!\!\!^{-}_{e} = \lambda_e/2\pi$ is the reduced wavelength of the light ($\lambda\!\!\!^{-}_{e} = 0.0818$ microns for the light from the argon ion lasers currently being used). This phase delay can be monitored, by the photodetector, with a precision $\Delta\Phi = 1/(N_\gamma \eta)^{\frac{1}{2}}$, where N_γ is the total number of photons that the laser puts out during the time $\hat{\tau}$ over which the photodetector intensity is averaged, and η is the photon counting efficiency of the photodetector ($\eta \sim 0.4$–0.9). When searching for a gravity-wave burst with characteristic frequency f, it is optimal to average the photodetector intensity for half a gravity-wave period, $\hat{\tau} \cong 1/2f$; and correspondingly the phase delay can be inferred with a photon-counting-noise ('shot-noise') precision

$$\Delta\Phi_{\text{shot}} = \frac{1}{(N_\gamma \eta)^{\frac{1}{2}}} \simeq \left(\frac{\hbar c/\lambda\!\!\!^{-}_{e}}{I_0 \eta (1/2f)}\right)^{\frac{1}{2}}, \tag{106}$$

where I_0 is the laser output power. By comparing equations (105) and (106) we obtain a rough estimate of the amplitude of a gravity-wave burst that produces a signal of the same strength as the rms shot noise

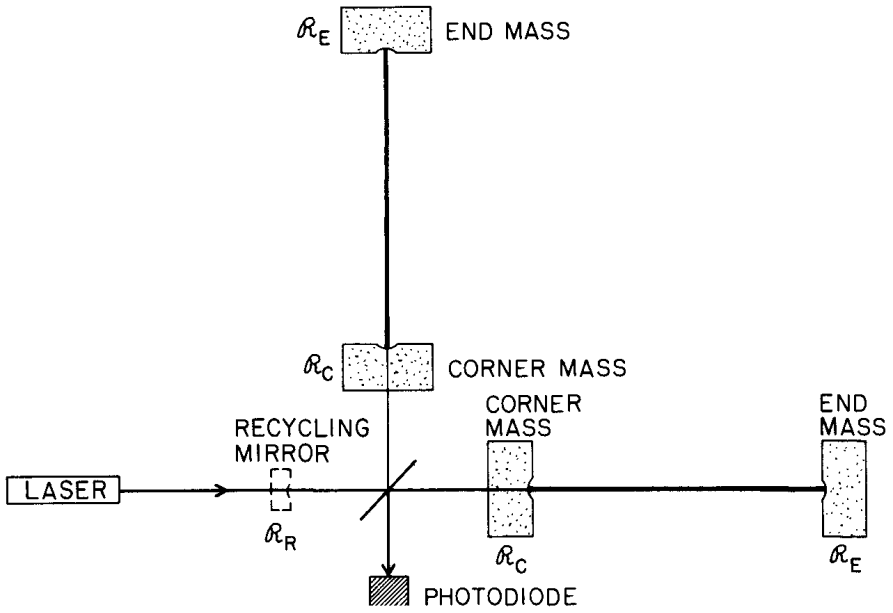
$$h_{\text{shot}} \cong \left[\frac{2\hbar c \lambda\!\!\!^{-}_{e}}{I_0 \eta} \frac{f}{(2BL)^2}\right]^{\frac{1}{2}}$$

$$\cong 7.2 \times 10^{-21} \frac{50}{B} \frac{1\,\text{km}}{L} \left(\frac{\lambda\!\!\!^{-}_{e}}{0.082\,\mu\text{m}}\right)^{\frac{1}{2}} \left(\frac{10\,\text{Watts}}{I_0 \eta}\right)^{\frac{1}{2}} \left(\frac{f}{1000\,\text{Hz}}\right)^{\frac{1}{2}}. \quad (107)$$

This formula and these numbers give some indication of the potential sensitivities of beam receivers.

Fabry–Perot beam receivers have essentially the same potential sensitivities as Michelson receivers. Fig. 9.10 shows a Fabry–Perot receiver viewed from above. Here, by contrast with Fig. 9.9, the corner mass has been broken into three separate pieces, one carrying each mirror and one carrying the beam splitter. This breaking up of the corner mass, initiated in Munich, reduces spurious forces on the mirrors; since 1985 it has been standard in all beam receivers. In the Fabry–Perot system of Fig. 9.10 each arm is operated as a resonant Fabry–Perot cavity: light from the laser is split at the beam splitter and enters the two cavities through the backs of the corner masses' partially transmitting mirrors. The two arms are arranged to have equilibrium lengths very nearly equal to a half-integral multiple of the wavelength $2\pi\lambda\!\!\!^{-}_{e}$ of the laser light. Consequently, the entering light finds itself in resonance with a mode of each cavity; and it gets resonantly

Fig. 9.10. Schematic diagram of a *Fabry–Perot-type beam receiver* for gravitational waves.



trapped in the cavity, building up to high intensity before exiting back toward the beam splitter.

Slight changes in the length of each cavity drive the cavity slightly off resonance and thereby produce sharp changes in the phase of the exiting light. Consequently, when the exiting light beams from the two cavities recombine at the beam splitter, their relative phase is highly sensitive to slight modulations δl of the two cavities' length difference l; and correspondingly the intensity of light onto the photodetector is highly sensitive to δl. If the corner mirrors have a probability for reflecting photons $\mathscr{R}_C$ and a transmission probability $1-\mathscr{R}_C$ (and no scattering), and if the end mirrors reflect much more efficiently, then this Fabry–Perot sensitivity is described by the same formulas (105)–(107) as for a Michelson receiver, with the number of round-trips B in each Michelson arm replaced by $4/(1-\mathscr{R}_C)$:

$$B \to 4/(1-\mathscr{R}_C). \tag{108}$$

It is also possible (and, in fact, is current practice) to operate a Fabry–Perot receiver in an alternative mode where, instead of recombining and interfering the beams, the laser's frequency is locked to the eigenfrequency of one arm and the difference between the laser's frequency and that of the other arm is the gravity-wave signal; see, e.g. Hough *et al.* (1983) or Spero (1986*a*) for details. This mode of operation is technically easier than beam recombination, but the ease is bought at the price of some debilitation in the ultimately achievable shot noise.

(c) *Noise in beam detectors*

Photon shot noise is but one of many noise sources that plague beam detectors. Almost always, thus far, the other noise sources are so strong that the effects of shot noise are lost amidst them. Typically the experimenters struggle for a long time to reduce the other noises sufficiently that shot noise shows up; then they improve the shot noise somewhat by increasing the laser power; then they begin a long struggle once again with the other noise sources. In this way the overall gravity-wave amplitude noise at kilohertz frequencies has been reduced during the period 1980–86 by a factor ~ 1000 (10^6-fold improvement in energy noise).

Among the most serious of the other noise sources are the following: amplitude and phase fluctuations in the laser output; imperfect matching of the laser beam into the Fabry–Perot cavities or the Michelson mirror system (imperfections in beam direction, beam shape, and beam wavelength); imperfect matching of the wave fronts of the recombining beams at the beam

splitter; fluctuations in the index of refraction of the gas inside the arms (even though the arms are inside vacuum pipes, residual gas can cause problems); scattering of light from one part of the optical system into another; thermal noise (thermal vibrations) in the end and corner masses and in the wires that suspend them; imperfect alignment of the mirrors; and seismic and acoustic noise from the outside world, which cause vibrations of the wires from which the masses hang. Almost all of these effects produce a displacement noise δl that is independent of the arm length L; and, correspondingly, their effects on gravity-wave amplitude sensitivities scale as $h \propto 1/L$. It is this scaling that motivates the move from short prototypes to long LIGOs.

A beam receiver is intrinsically broad band: its output, the intensity of the light into the photodetector $I_{\rm pd}(t)$, is averaged (by filtering) over the shortest timescale, $\hat{\tau} \sim 10^{-4}$ s, that gravity waves are likely to contain, and then is recorded for future analysis. The future analysis can include searching in the data for signals of all frequencies $f \lesssim 1/\hat{\tau} \sim 10^4$ Hz, and for signals with complicated time dependences that embody a broad range of frequencies. Correspondingly, a beam receiver's net noise depends on the frequency f at which one studies the recorded output.

More specifically, the noise (excluding non-Gaussian, spurious events which are removed by coincident operation of two independent receivers) is characterized by the spectral density $S_l(f)$ of the output-inferred armlength difference l; or, equally well, by the spectral density $S_h(f)$ of the gravity wave $h(t)$ (equation (103)). Because the effect of the gravity wave on the arm-length difference is $\delta l(t) = Lh(t)$, $S_h(f)$ is also called the spectral density of strain, and these two spectral densities are related by

$$S_h(f) = (1/L^2) S_l(f). \tag{109}$$

It is $S_h(f)$ or its square root, or $S_l(f)$ or its square root that experimenters generally quote when discussing the overall performance of their beam detectors.

In Sections 9.4.1(b), 9.4.2(a) and 9.4.3(a) we derived expressions in terms of $S_h(f)$ for the minimum-amplitude signal $h_{3/\rm yr}$ that can be detected with 90% confidence in a $\frac{1}{3}$ year search using a detector composed of two identical cross-correlated receivers. Those expressions involve averages over the receiver's beam-pattern factors. For the L-shaped beam receivers of Figs. 9.9 and 9.10, with beam-pattern factors (104), the appropriate averages are

$$\langle F_+^2 \rangle = \langle F_\times^2 \rangle = \tfrac{1}{5}, \tag{110}$$

and, correspondingly,

$$h_{3/\mathrm{yr}}(f_c) = 11[f_c S_h(f_c)]^{\frac{1}{2}} \text{ for bursts (equation (34)),} \quad (111)$$

$$h_{3/\mathrm{yr}}(f) = 3.8[S_h(f) \times 10^{-7}\ \mathrm{Hz}]^{\frac{1}{2}}$$

for periodic waves (equation (52a)), (112)

$$h_{3/\mathrm{yr}}(f) = 4.5\left(\frac{\Delta f}{10^{-7}\ \mathrm{Hz}}\right)^{-\frac{1}{4}} [f S_h(f)]^{\frac{1}{2}}$$

for stochastic waves (equation (67)). (113)

Here in the burst equation the numerical factor in equation (34) has been taken to be 5 corresponding to a characteristic frequency $f_c \sim 10$ Hz to 10^4 Hz – the largest band that earth-based receivers are likely ever to operate in; and in the periodic equation it is assumed that the frequency and phase are known in advance (equation (52a) rather than (52b)). Expressions (111)–(113) are the bases of the beam-detector noise amplitudes shown in Figs. 9.4, 9.6 and 9.7.

When the two receivers that make up a detector do not lie in the same plane (because of curvature of the earth between them) or do not have their arms parallel, their joint sensitivity is somewhat debilitated. Detailed analyses of this have been carried out by Whitcomb and Saulson (1984) and by Schutz and Tinto (1986).

(d) *The noise in the present prototypes*

At present the MIT group is developing Michelson receivers using a prototype with $L = 1.5$ m; the Munich group is developing Michelsons using $L = 30$ m (Shoemaker *et al.*, 1986); the Glasgow group is developing Fabry–Perot receivers using $L = 10$ m; and the Caltech group is developing Fabry–Perots using $L = 40$ m. Although the arm lengths and optics of these four prototypes are very different, their displacement sensitivities are roughly the same – and have been so throughout the past six years, during which all have improved roughly in step by about three orders of magnitude. Throughout these improvements Munich has maintained a slight (factor 2 or 3 in amplitude) lead over the other groups. (*Note added in proof.* Since this was written Glasgow has forged ahead of Munich by a factor 3 in displacement sensitivity, making them equal in h sensitivity.) Fig. 9.11 shows the spectral density of strain noise for the Munich prototype as of February 1986 (Schilling, 1986). The noise is due, predominantly, to seismic noise (inadequate isolation) below 250 Hz, probably seismic noise between 250 and 1000 Hz, photon shot noise between 1000 and 6000 Hz, and thermal

noise in the mirrors and pockels cells above 6000 Hz. In Figs. 9.4, 9.6 and 9.7 (upper right) are shown the sensitivities $h_{3/\mathrm{yr}}$ (equations (111)–(113)) for the Munich and Caltech prototypes in 1986 (Schilling, 1986; Spero, 1986*b*) and Glasgow in early 1987. As an illustration of the sensitivity progress, Fig. 9.4 also shows $h_{3/\mathrm{yr}}$ for bursts in the Munich and Caltech prototypes a few months after each was first turned on (1980 and 1983).

(e) *Spectral density of shot noise for simple, recycling and resonating receivers*

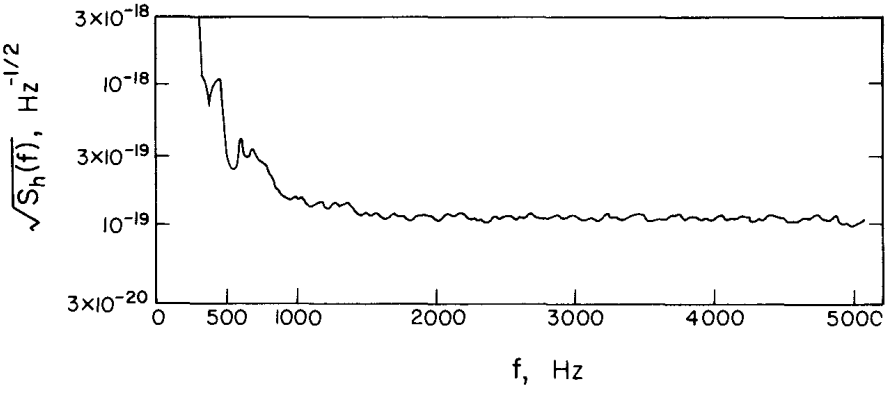
In the present prototypes it is advantageous to store the light in the arms as long as possible, thereby building up the largest possible phase shift and gravity-wave sensitivity; cf. the B-dependence in equations (105) and (107). However, in a kilometer-scale LIGO one easily can store the light longer than half a gravity-wave period, i.e. for more than $B = 75(1\ \mathrm{km}/L) \times (1000\ \mathrm{Hz}/f)$ round-trip traverses of an arm. Such long storage is self-defeating: the phase shift built up so laboriously during the first half-period of the wave gets removed during the second half-period because $h(t)$ reverses sign.

This shows up clearly in the spectral densities of shot noise $S_h(f)$ for the idealized Michelson and Fabry–Perot receivers of Figs. 9.9 and 9.10 (still without the 'recycling mirrors' in place). Assuming as above that the Michelson mirrors have negligible losses during B round trips in each arm, and the Fabry–Perot end mirrors have negligible transmission compared to the corner mirrors

$$1 - \mathscr{R}_{\mathrm{E}} \ll 1 - \mathscr{R}_{\mathrm{C}} \equiv 4/B \tag{114}$$

(cf. equation (108)), the spectral densities of shot noise are (cf. Gürsel *et al.*,

Fig. 9.11. Square root of spectral density of noise $[S_h(f)]^{\frac{1}{2}}$ plotted against frequency f for the Munich Michelson-type beam detector with 30 m arms, as of February 1986.

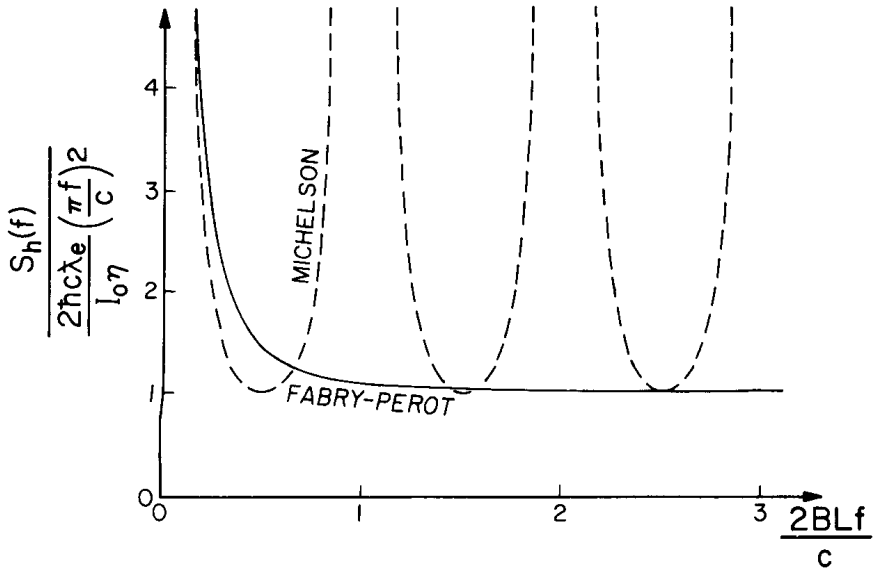


1983; Meers, 1983; Brillet and Meers, 1987)

$$S_h(f)=\frac{2\hbar c\lambda_e}{I_0\eta}\left(\frac{1}{2BL}\right)^2\times\begin{cases}1+(2\pi BLf/c)^2 & \text{for Fabry–Perot} \quad (115a)\\ \left[\dfrac{2\pi BLf/c}{\sin(2\pi BLf/c)}\right]^2 & \text{for Michelson.} \quad (115b)\end{cases}$$

These spectral densities are shown in Fig. 9.12 as a function of $2BLf/c=$ (light storage time)/(gravity-wave period), with (gravity-wave period) $=1/f$ held fixed. When (storage time)/(period) $\ll 1$ the two receivers have the same shot noise, and that shot noise improves with increasing storage time as $S_h(f)\propto$ (storage time)$^{-2}$. When (storage time)/(period) $\gtrsim 1$ the Fabry–Perot shot noise stops improving, while the Michelson shot noise undergoes a series of oscillations with minima equal to the Fabry–Perot shot noise. Note that the minima of the Michelson oscillations occur when $2BLf/c=$ (light-storage time)/(gravity-wave period) $=0.5,\ 1.5,\ 2.5,\ldots$, while their

Fig. 9.12. The improvement of photon shot noise with increasing number of bounces of light in a simple beam receiver. The receiver is of the Fabry–Perot (Fig. 9.10) or Michelson (Fig. 9.9) type with recycling mirrors absent, with $1-\mathscr{R}_E\ll 1-\mathscr{R}_C=4/B$ in the Fabry–Perot case and with B round trips in each arm in the Michelson case. Plotted vertically is spectral density of shot noise in units proportional to f^2 but independent of the light-storage time $2BL/c$. Plotted horizontally is (light-storage time)/(gravity-wave period) $=(2BL/c)f$. The Fabry–Perot shot noise is given by the solid curve (equation (115a)); the Michelson by the dashed curve (equation (115b)).



maxima ($S_h = \infty$) occur when the light is stored for an integral number of gravity-wave periods. This is just what we would expect from the fact that $h(t)$ reverses sign every half-period, thereby removing during a second half-period the signal put onto the light during a first half-period. In the Fabry–Perot receiver there are no oscillations of $S_h(f)$ because different photons experience different storage times – i.e. because of the probabilistic nature of the reflectivity $\mathscr{R}_C$.

Drever (1983) has devised a method for improving the sensitivity of either a Michelson or a Fabry–Perot when mirror reflectivities permit storing light for much longer than a half-period. The basic idea is to extract the light after a half-period, when further storage is self-defeating, and then reinsert it into the cavity along with and in phase with new laser light. More specifically, for the Michelson of Fig. 9.9(*b*), one adjusts the relative arm lengths so that very little of the recombined light emerges from the beam splitter toward the photodiode where the gravity-wave signal is read out (a mode of operation that optimizes the sensitivity, it turns out). Then most of the recombined light emerges toward the laser; and 'recycling mirrors' are inserted to direct that emergent light back into the beam splitter along with fresh laser light. For the Fabry–Perot of Fig. 9.10 the same effect is achieved with a single recycling mirror.

As an aid in quantifying the recycling-induced improvement in shot noise, consider the (realistic) situation in which the technology of mirror coatings limits the mirror reflectivities to some maximum value $\mathscr{R}_{max}$ (0.9999 at present; perhaps 0.999 99 a few years from now). It obviously is optimal to place at the end of each arm a mirror with this maximum, so $\mathscr{R}_E = \mathscr{R}_{max}$. Consider, for concreteness, the Fabry–Perot receiver of Fig. 9.10. If the corner mirrors are chosen also to have the maximum reflectivity, $\mathscr{R}_C = \mathscr{R}_E$, then essentially all the unused light leaks out the end mirrors and nothing is gained by recycling. In this case of a simple, non-recycling Fabry–Perot with $\mathscr{R}_C = \mathscr{R}_E = \mathscr{R}_{max}$ the spectral density of shot noise is (cf. Gürsel *et al.*, 1983; Meers, 1983; Brillet and Meers, 1987)

$$S_h(f) = S_o[1 + (f/f_o)^2], \tag{116a}$$

where

$$S_o = \frac{2\hbar c \lambda_e}{I_o \eta}\left(\frac{1-\mathscr{R}_E}{2L}\right)^2$$
$$= \left[\frac{3.6\times 10^{-25}}{(\text{Hz})^{\frac{1}{2}}}\left(\frac{\lambda_e}{0.0818\ \mu\text{m}}\right)^{\frac{1}{2}}\left(\frac{100\ \text{W}}{I_o\eta}\right)^{\frac{1}{2}}\left(\frac{1-\mathscr{R}_E}{10^{-4}}\right)\left(\frac{1\ \text{km}}{L}\right)\right]^2, \tag{116b}$$

$$f_o = \frac{(1-\mathscr{R}_E)c}{4\pi L} = 2.4\ \text{Hz}\left(\frac{1-\mathscr{R}_E}{10^{-4}}\right)\left(\frac{1\ \text{km}}{L}\right). \tag{116c}$$

(Note that, because so much light is lost out the end mirrors, this noise is worse than the $\mathscr{R}_C \gtrsim \mathscr{R}_E$ limit of equation (115a) – worse by a factor 4 at frequencies $f \gg f_o$ and by a factor 16 at $f \ll f_o$.) This non-recycled ('simple') Fabry–Perot noise is shown as a solid curve in Fig. 9.13.

An experimenter who wishes to improve on this noise level by recycling must choose a frequency $f_k \gg f_o$ near which the noise level is to be minimized. The minimal noise level near $f = f_k$ will then be achieved by setting

$$1-\mathscr{R}_C = 8\pi L f_k/c, \tag{117a}$$

so the effective storage time in each arm is $2BL/c = 8Lc^{-1}/(1-\mathscr{R}_C) = (\pi f_k)^{-1} = (1/\pi) \times$ (period of a gravity wave with the optimal frequency f_k). Minimal noise also requires a special choice for the reflectivity $\mathscr{R}_R$ of the recycling mirror (Fig. 9.10)

$$1-\mathscr{R}_R = \frac{4(1-\mathscr{R}_E)}{(1-\mathscr{R}_C)}. \tag{117b}$$

With these choices of reflectivity, the light-recycling Fabry–Perot receiver of Fig. 9.10 has shot noise (cf. Gürsel *et al.*, 1983; Meers, 1983; Brillet and Meers, 1987)

$$S_h(f) = \frac{f_k}{2f_o} S_o\left[1+\left(\frac{f}{f_k}\right)^2\right]. \tag{117c}$$

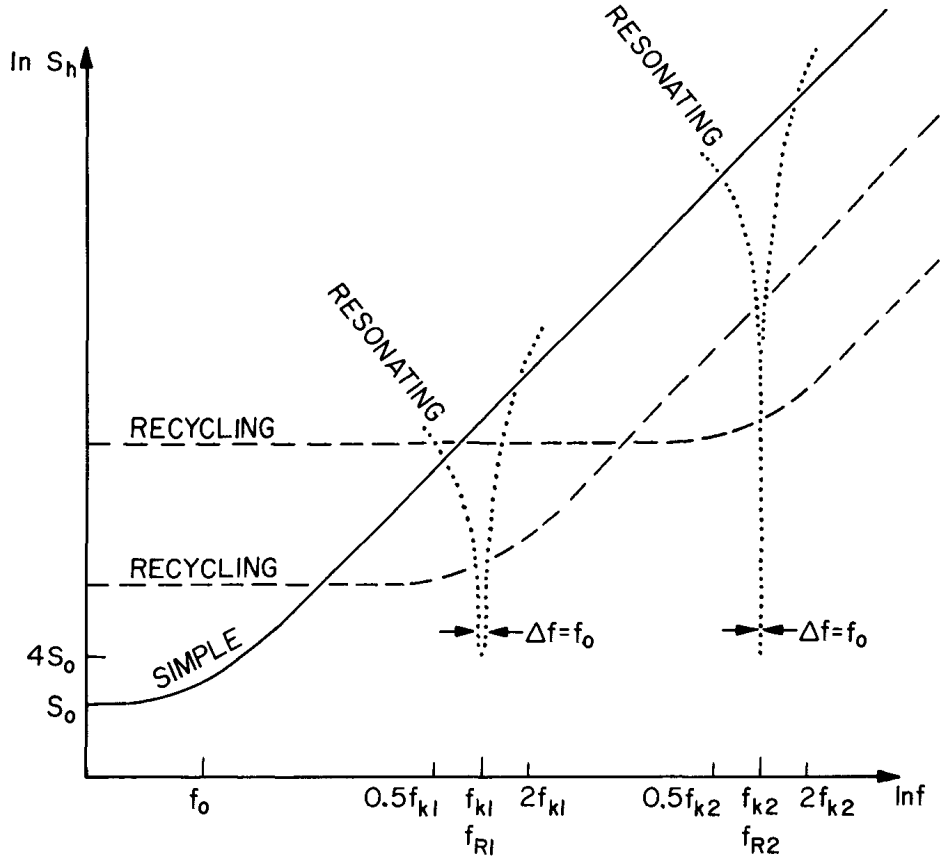
This noise is depicted in Fig. 9.13 for two choices of f_k (dashed curves). Note that this noise has a 'knee' at the frequency f_k; for this reason f_k is called the *knee frequency*. Note further that recycling produces an overall improvement in $S_h(f)$, at frequencies $f \gtrsim f_k$, by a factor $f_o/2f_k$ relative to a non-recycled Fabry–Perot with equal reflectivities for corner and end mirrors (equations (116)) and an improvement by $2f_o/f_k$ relative to the very best non-recycled Fabry–Perot – one with $1-\mathscr{R}_{max} = 1-\mathscr{R}_E \ll 1-\mathscr{R}_C \ll 8\pi L f_k/c$ (equation (115a)). This improvement at $f \gtrsim f_k$ is bought at the price of worsened noise at $f \lesssim (f_k f_o/2)^{\frac{1}{2}}$. Note further that the improvement factor, over an optimal non-recycled Fabry–Perot at $f \gtrsim f_k$, is

$$\frac{S_h^{\text{recycled}}}{S_h^{\text{non-recycled}}} = \frac{2f_o}{f_k} = \frac{1-\mathscr{R}_E}{1-\mathscr{R}_C}, \tag{118}$$

which is 1/(the mean number of times that the light can be recycled before it is lost by leakage through the high-reflectivity end mirrors). This is just what physical intuition should suggest.

For the recycled Michelson receiver of Fig. 9.9, recycling produces essentially the same $S_h(f)$ as for the Fabry–Perot of Fig. 9.10 at frequencies $f \lesssim f_k$. However, above the knee frequency the recycled Michelson exhibits a

Fig. 9.13. Spectral density of shot noise for Fabry–Perot beam receivers operated in various optical configurations. The vertical and horizontal scales are both logarithmic. The solid curve is for a simple Fabry–Perot with all mirrors – corner and end – having the maximum achievable reflectivity, $\mathscr{R}_C = \mathscr{R}_E = \mathscr{R}_{max}$ (equations (116)). The dashed curves are for a Fabry–Perot with light recycling (equation (117c)), in which the end mirrors have the maximum achievable reflectivity $\mathscr{R}_E = \mathscr{R}_{max}$, the corner mirrors are adjusted to produce the desired knee frequency (f_{k1} and f_{k2} for the two curves shown; equation (117a)), and the recycling mirror is adjusted to minimize the noise at the knee frequency (equation (117b)). The dotted curves are for a Fabry–Perot with light resonating (equation (119c); Fig. 9.14) in which the end mirrors have the maximum achievable reflectivity $\mathscr{R}_E = \mathscr{R}_{max}$; the corner mirrors are adjusted to produce the desired resonant frequency for gravity waves (f_{R1} and f_{R2} for the two curves shown; equation (119a)); and the recycling mirror is adjusted to minimize the noise at the resonant frequency (equation (119b)).

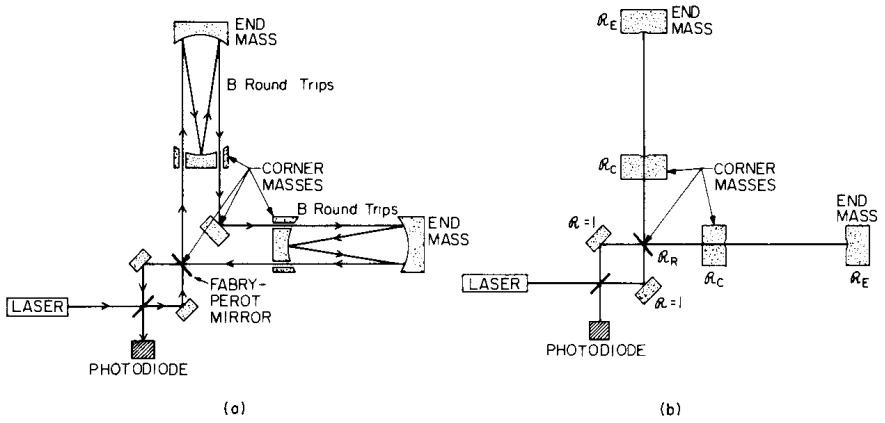


sequence of noise peaks and troughs analogous to those for a non-recycled Michelson (equation (115b) and Fig. 9.12).

Of all optical configurations yet invented, recycling is the best when one is searching for broad-band gravitational waves (waves with $\Delta f \gtrsim f$). However, when one is searching for narrow-band waves (waves with $\Delta f \ll f$, e.g. the periodic waves from a pulsar), recycling gives worse noise than a configuration called *light resonating* (invented by Drever, 1983), which is shown in Fig. 9.14. As usual, the method is conceptually simplest in the Michelson case but is described by the simplest formulas in the Fabry–Perot case.

The basic idea, as shown for a Michelson in Fig. 9.14(*a*), is to store the light in an arm for one half-cycle of the gravity-wave frequency of interest f_R (i.e. for $B = c/4f_R L$ round trips); then, instead of extracting it through the beam splitter and reinjecting it, simply move it into the other arm using a high-reflectivity, Fabry–Perot-type mirror – i.e. exchange the light between the arms. Then during the next half-cycle, with arms interchanged and the sign of the gravity wave reversed, each piece of light will experience a phase shift in the same direction as during the first half-cycle. At the end of the second half-cycle, interchange the light in the two arms, and repeat as many times as mirror losses will permit. The arrows in Fig. 9.14(*a*) show the light, as a result of these exchanges, circulating around the interferometer in a clockwise direction, but there is an equal amount of light circulating in a counterclockwise direction. The light circulating one way experiences a continual increase in phase shift throughout its circulation; the light

Fig. 9.14. Schematic diagrams of optical configurations that could be used for *resonating light* in Michelson-type (*a*) and Fabry–Perot-type (*b*) beam receivers (Dreaver, 1983).



circulating oppositely experiences a continual decrease in phase shift; and these oppositely circulating beams, upon re-emerging through the back of the Fabry–Perot-type mirror and recombining at the beam splitter, produce a much enhanced interference signal in the photodetector.

For the Fabry–Perot receiver of Fig. 9.14(*b*) a single 'resonating mirror', denoted $\mathscr{R}_R$, produces the resonant interchange of the light between the two arms: the lengths of the cavities are so adjusted as to produce in each cavity a resonance at a frequency f_e very near the laser frequency; the reflectivities $\mathscr{R}_C$ of the corner mirrors, and the path lengths between them and the resonating mirror, are then so adjusted as to produce two resonant modes of the coupled cavities at light frequencies $f_e \pm f_R/2$ – which requires

$$1-\mathscr{R}_C = 4\pi L f_R/c; \tag{119a}$$

the laser frequency is then adjusted so that it drives the mode at $f_e - f_R/2$; and the gravity waves with frequency f_R, by wiggling the end mirrors, then upconvert photons into the mode at $f_e + f_R/2$. In order to maximize the resulting gravity-wave-induced signal at the photodetector, the reflectivity $\mathscr{R}_R$ of the resonating mirror must be adjusted to

$$1-\mathscr{R}_R = 2\frac{1-\mathscr{R}_E}{1-\mathscr{R}_C} = 2\frac{f_o}{f_R}. \tag{119b}$$

With these optimizations the spectral density of shot noise near f_R is (cf. Gürsel *et al.*, 1983; Meers, 1983; Vinet, 1986; Brillet and Meers, 1987)

$$S_h(f) = 4S_o\left[1+\left(\frac{f-f_R}{f_o}\right)^2\right] \quad \text{for } |f-f_R| \ll f_o. \tag{119c}$$

As is seen in Fig. 9.13 (dotted curves), this represents an enormous improvement in noise over either a simple Fabry–Perot or a recycling Fabry–Perot. However, this improvement is bought at the price of an enormous narrow-banding of the receiver response: at frequencies $f \sim f_R/2$ and $f \sim 2f_R$ the noise is comparable to that in a simple Fabry–Perot; and at $f \ll f_R/2$ and $f \gg 2f_R$ it is far worse.

For an optimally configured, resonating Michelson the sensitivy near resonance, $|f-f_R| \ll f_o$, is similar to that of a Fabry–Perot (equation (119c)).

Neither recycling nor resonating is potentially useful in present prototypes because present mirror reflectivities limit the prototypes to short storage times. However, in the planned full-scale LIGOs where 100-fold longer arms permit 100-fold greater storage times, recycling and resonating both give promise of substantial improvements in beam-detector sensitivities. The first attempts to implement recycling on a prototype were

carried out by the Munich group in 1986, with moderate success. Resonating has not yet been attempted.

(f) *Quantum limit, quantum non-demolition, and squeezed vacuum*

A beam receiver is similar to a 'Heisenberg microscope', in which a photon is used to measure the position of a particle. The analog of the Heisenberg photon is the laser beam and the analogs of the Heisenberg particle are the end and corner masses. Just as the Heisenberg photon kicks the particle it is trying to measure thereby enforcing the Heisenberg uncertainty principle, $\Delta x\,\Delta p \geqslant \hbar/2$, so fluctuations in the beam's light pressure kick the end and corner masses thereby enforcing a quantum limit on the sensitivity of the beam receiver. As I shall discuss below, these quantum fluctuations have their ultimate source in the vacuum fluctuations of quantum electrodynamics (Caves, 1980, 1981).

Caves (1987*b*) has used Feynman path-integral techniques (Caves, 1986, 1987*a*) to derive a pseudospectral density of displacement noise that characterizes the quantum limit of any harmonic oscillator (such as a beam receiver's swinging masses) when it is studied at frequencies high above the resonant frequency f_0, using techniques that produce the minimum possible noise:

$$S_x(f)=\frac{2\hbar}{m(2\pi f)^2}, \tag{120}$$

where m is the oscillator's mass and x is its displacement from equilibrium. For a beam receiver, whose swinging masses have resonant frequencies $f_0 \sim 1$ Hz far below the region of interest, this pseudospectral density for their displacements can be translated into a pseudospectral density for the gravity-wave field $h(t)$ that one reads off the receiver's output:

$$S_h(f)=\frac{8\hbar}{m(2\pi f)^2L^2}. \tag{121}$$

Here each of the masses on which a mirror is attached is assumed to have the same mass m. This pseudospectral density can be used, in the same manner as an ordinary classical spectral density (subsection (c) above), to compute the standard quantum limit on the performance of a beam detector in various situations.

As for a bar detector, so also for a beam detector, there should be methods of circumventing this standard quantum limit. Unfortunately nobody has yet found a remotely practical way of doing so, except in one special case: Unruh (1982) has invented a method, which Caves' path-integral formalism

says should work (Caves, 1987*b*), for beating the standard quantum limit when performing a narrow-band measurement of periodic gravitational waves. The key idea (which was invented independently and earlier in the classical domain by Gordienko, Gusev and Rudenko, 1977) is to place a spring between each mirror and the companion mass on which it rides, thereby turning the mirror and companion into a two-mode system. By setting the ratio of the spring frequency over the gravity-wave frequency equal to the square root of the mass of the big companion over the mass of the little mirror, one can force the laser beam's fluctuating light pressure to act on the motion of the big mass and not the mirror, while the interferometer reads out the mirror's motion. The result is an improved immunity of the receiver to Heisenberg's quantum noise over a narrow frequency band. While this scheme is clever and conceptually satisfactory, whether it can be implemented in a practical manner is not yet clear.

Although, for broad-band measurements with beam receivers, there as yet is no known way of beating the standard quantum limit, Caves (1981) has invented a clever scheme for getting closer to the quantum limit when inadequate laser power causes the shot noise to exceed it. The key to Caves' idea is his discovery that the ultimate source of the photon shot noise and the fluctuating radiation-pressure noise is *not*, as people previously thought, fluctuations in the laser output. Rather, it is fluctuations in the quantum-electrodynamic vacuum state ('vacuum fluctuations') that enter the beam splitter at right-angles to the incoming laser light and superpose on the laser light as it heads toward the two arms. As Caves has shown, by 'squeezing the vacuum' (i.e. reducing the vacuum fluctuations in the $\cos[(ct-x)/\lambda\!\!\!^{-}_{e}]$ part of the light while increasing them in the $\sin[(ct-x)/\lambda\!\!\!^{-}_{e}]$ part; achievable in principle by sending the vacuum through a pumped, non-linear medium), one can reduce the beam receiver's shot noise at the expense of increasing the fluctuations in its light-pressure noise. The net result is the same as if one were using a laser of higher power: in the typical case, where the actual power is too low to permit achieving the quantum limit, one improves the sensitivity and moves toward the quantum limit.

Recently Caves (1987*c*) has elucidated the ultimate sensitivities achievable when one combines his squeezed-vacuum technique with light recycling and resonating. He finds that because, in a resonating system, noise associated with losses in the end mirrors is as important on resonance as vacuum fluctuations entering the beam splitter, squeezing of the vacuum is not useful. By contrast, when combined with light recycling, squeezed-vacuum techniques can reduce $S_h(f)$, over a broad band of frequencies $\Delta f \gtrsim f$, to S_0

or to the quantum limit, whichever is larger:

$$S_h(f)=\max\begin{cases} S_o=\dfrac{2\hbar c\lambda\!\!\!^{-}_e}{I_o\eta}\left(\dfrac{1-\mathscr{R}_E}{2L}\right)^2 \\ \dfrac{8\hbar}{m(2\pi f)^2L^2}\end{cases} \tag{122}$$

Squeezed states of light have been produced by experimenters recently (Slusher *et al.*, 1985; Wu *et al.*, 1986), triggering great excitement in the field of non-linear quantum optics. This achievement has triggered an effort by the Munich group to implement Caves' squeezed-vacuum technique – an effort that is likely to succeed within the next few years.

(g) *Anticipated sensitivities of full-scale LIGO receivers*

The present Munich burst sensitivity $h_{3/yr}\cong 4\times 10^{-17}$ at $f\sim 1000$ Hz, when scaled from the prototype arm length of 30 m to the planned Munich LIGO arm length of 3 km, would become $h_{3/yr}\cong 4\times 10^{-19}$. Although scaling up is not straightforward (mirror diameters must be increased a factor ten, for example), the experimenters are quite confident of doing somewhat better than this scaled-up number by the early 1990s when the first LIGOs start operating.

The upper thick, dashed curves in Figs. 9.4, 9.6 and 9.7 (earlier in this chapter) show possible burst, periodic and stochastic sensitivities (based on a $\frac{1}{3}$-year search time and 90% confidence levels), for a first generation of detectors in the full-scale LIGOs. These curves correspond, at frequencies above ~ 500 Hz, to receivers that are shot-noise-limited, without recycling or resonating, with Argon-ion laser light $\lambda\!\!\!^{-}_e=0.0818$ μm, with (laser power) $\times$ (photodetector efficiency) $=I_o\eta=10$ W, and with light storage times of a half-gravity-wave period or longer, so [cf. equation (115) and Fig. 9.12]

$$[S_h(f)]^{\frac{1}{2}}=\left[\frac{2\hbar c\lambda\!\!\!^{-}_e}{I_o\eta}\left(\frac{\pi f}{c}\right)^2\right]^{\frac{1}{2}}=(2.4\times 10^{-22}\ \mathrm{Hz}^{-\frac{1}{2}})\left(\frac{f}{1000\ \mathrm{Hz}}\right); \tag{123a}$$

and, correspondingly (equations (111)–(113))

$$h_{3/yr}=11(f_cS_h)^{\frac{1}{2}}=8\times 10^{-20}\left(\frac{f_c}{1000\ \mathrm{Hz}}\right)^{\frac{3}{2}} \quad \text{for bursts,} \tag{123b}$$

$$h_{3/yr}=3.8(S_h\times 10^{-7}\ \mathrm{Hz})^{\frac{1}{2}}=2.9\times 10^{-25}\left(\frac{f}{1000\ \mathrm{Hz}}\right)$$

for periodic sources, (123c)

$$h_{3/\mathrm{yr}} = 4.5\left(\frac{f}{10^{-7}\,\mathrm{Hz}}\right)^{-\frac{1}{4}}(fS_h)^{\frac{1}{2}} \tag{123c}$$

$$= 1.1\times 10^{-22}\left(\frac{f}{1000\,\mathrm{Hz}}\right)^{\frac{5}{4}} \quad \text{for stochastic sources,} \tag{123c}$$

where the search for stochastic waves is presumed to use a bandwidth $\Delta f = f$. Below ~ 500 Hz it is presumed that seismic noise debilitates the performance of these first-generation LIGO receivers.

During the years following the first gravity-wave searches in the LIGOs, the experimenters plan to run a sequence of detectors with ever improving sensitivities, pushing the sensitivities ever downward, and improving the seismic isolation so the detectors can operate at ever decreasing minimum frequencies. A reasonable goal by the end of the 1990s is to approach the lower-most thick-dashed curves in Figs. 9.4, 9.6 and 9.7. These curves correspond to *advanced detectors* with the following characteristics:

$$\begin{aligned}
&L = (\text{arm length}) = 4\text{ km},\\
&\lambda\!\!\!^{-}_{\mathrm{e}} = 0.0818\ \mu\text{m (Argon-ion laser light)},\\
&I_{\mathrm{o}}\eta = (\text{laser power})\times(\text{photodetector sensitivity}) = 100\text{ W},\\
&\mathscr{R}_{\mathrm{E}} = \mathscr{R}_{\max} = (\text{maximum mirror reflectivity}) = 0.9999,\\
&m = (\text{mirror mass}) = 1000\text{ kg},\\
&\qquad \text{photon shot noise dominant at } 100\text{ Hz} \lesssim f \lesssim 10^4\text{ Hz},\\
&\qquad \text{quantum-limit noise dominant at } 10\text{ Hz} \lesssim f \lesssim 100\text{ Hz},\\
&\qquad \text{seismic noise dominant at } f \lesssim 10\text{ Hz}.
\end{aligned} \tag{124}$$

It is presumed that light recycling is used for burst searches, with the knee frequency adjusted to equal the frequency of interest, so the photon shot noise (equations (111), (117c) and (116b,c) with $f_{\mathrm{k}} = f_{\mathrm{c}}$) is

$$h_{3/\mathrm{yr}} = 11(f_{\mathrm{c}}S_h)^{\frac{1}{2}} = 11(S_{\mathrm{o}}f_{\mathrm{c}}^2/f_{\mathrm{o}})^{\frac{1}{2}} = 11\left[\frac{2\pi\hbar\lambda\!\!\!^{-}_{\mathrm{e}}}{I_{\mathrm{o}}\eta}\,\frac{(1-\mathscr{R}_{\mathrm{E}})}{L}\,f_{\mathrm{c}}^2\right]^{\frac{1}{2}}$$

$$= 1.3\times 10^{-21}\left(\frac{f_{\mathrm{c}}}{1\,\mathrm{kHz}}\right) \quad \text{for bursts;} \tag{125a}$$

and it is presumed that light resonating is used for periodic and stochastic searches, with the resonant frequency adjusted to equal the frequency of interest, so the photon shot noise (equations (112), (113), (119c) and (116b,c) with $f_{\mathrm{R}} = f$) is

$$h_{3/\mathrm{yr}} = 3.8(4S_{\mathrm{o}}\times 10^{-7}\ \mathrm{Hz})^{\frac{1}{2}} = 3.8\left[\frac{2\hbar c\lambda\!\!\!^{-}_{\mathrm{e}}}{I_{\mathrm{o}}\eta}\times 10^{-7}\ \mathrm{Hz}\right]^{\frac{1}{2}}\left(\frac{1-\mathscr{R}_{\mathrm{E}}}{L}\right)$$

$$= 2.1\times 10^{-28} \quad \text{for periodic waves,} \tag{125b}$$

$$h_{3/\mathrm{yr}}=4.5\left(\frac{f_0}{10^{-7}\,\mathrm{Hz}}\right)^{-\frac{1}{4}}(4S_0 f)^{\frac{1}{2}}=4.5\left[\frac{2\hbar c\lambda_e}{I_0\eta}f\right]^{\frac{1}{2}}\left[\frac{4\pi}{c}\left(\frac{1-\mathscr{R}_E}{L}\right)^3\times 10^{-7}\,\mathrm{Hz}\right]^{\frac{1}{4}}$$

$$=5.2\times 10^{-25}\left(\frac{f}{1\,\mathrm{kHz}}\right)^{\frac{1}{2}}\quad\text{for stochastic waves.}\tag{125c}$$

Here it is assumed that the stochastic search is restricted to the narrow bandwidth $\Delta f=f_0$ over which the resonating receiver has good performance. The quantum-limit noise, which exceeds these shot noise levels at $f\lesssim 100$ Hz, is given by (equations (121) and (111)–(113))

$$h_{3/\mathrm{yr}}=11\left[\frac{2}{\pi^2}\frac{\hbar}{mL^2}\frac{1}{f}\right]^{\frac{1}{2}}=1.3\times 10^{-23}\left(\frac{1000\,\mathrm{Hz}}{f}\right)^{\frac{1}{2}}\quad\text{for bursts,}\tag{126a}$$

$$h_{3/\mathrm{yr}}=3.8\left[\frac{2}{\pi^2}\frac{\hbar}{mL^2}\frac{10^{-7}\,\mathrm{Hz}}{f^2}\right]^{\frac{1}{2}}$$

$$=4.4\times 10^{-29}\left(\frac{1000\,\mathrm{Hz}}{f}\right)\quad\text{for periodic waves,}\tag{126b}$$

$$h_{3/\mathrm{yr}}=4.5\left[\frac{f_0(S_{\mathrm{QL}}/S_0)^{\frac{1}{2}}}{10^{-7}\,\mathrm{Hz}}\right]^{-\frac{1}{4}}\left[\frac{4}{3}fS_{\mathrm{QL}}\right]^{\frac{1}{2}}$$

$$=4.5\left(\frac{4}{3\pi}\right)^{\frac{1}{2}}\left(\frac{4\hbar}{mL^2}\right)^{\frac{3}{8}}\left[\frac{\hbar\lambda_e}{I_0\eta c}\left(\frac{10^{-7}\,\mathrm{Hz}}{f}\right)^2\right]^{\frac{1}{8}}$$

$$=1.5\times 10^{-25}\left(\frac{1000\,\mathrm{Hz}}{f}\right)^{\frac{1}{4}}\quad\text{for stochastic waves.}\tag{126c}$$

Here for stochastic waves S_{QL} is the quantum-limit spectral density (equation (121)), and the chosen bandwidth $\Delta f=f_0(S_{\mathrm{QL}}/S_0)^{\frac{1}{2}}$ is that which makes shot noise (equation (119c) integrated over Δf, with $f_0<\Delta f<f_R$) equal to $\frac{1}{3}$ the quantum noise and minimizes their sum. At frequencies $f\lesssim 10$ Hz, seismic noise is presumed to strongly debilitate the receiver performances (cf. Saulson, 1984).

Figs. 9.4, 9.6 and 9.7 show, together with the above receiver sensitivities, the characteristic strengths of the gravity waves from a variety of sources that were discussed in Section 9.4 above, and the present and projected sensitivities of other kinds of detectors. As these diagrams suggest, the prospects for successful detection of gravity waves with the planned LIGOs are high: at the sensitivities of the 'advanced detectors' one could see the coalescence of neutron-star binaries at $\frac{1}{4}$ the Hubble distance where the event rate is estimated to be several per day, the coalescence of $10M_\odot$ black-hole binaries throughout the universe, supernovae in our galaxy if they put out $10^{-9}M_\odot$ of energy near 1000 Hz frequency and in the Virgo cluster if

they put out $10^{-3}M_{\odot}$, millisecond pulsars whose frequencies are known in advance from electromagnetic measurements if they have ellipticities $10^{-8.5}$ or larger, CFS-unstable neutron stars with X-ray luminosities as small as $\frac{1}{1000}$ that of Sco X-1, and a stochastic background at $f \sim 30$ Hz with Ω_{GW} as small as 10^{-10}. It seems likely to me that such sensitivities are more than adequate for success; waves are likely to be detected at more modest sensitivities than these – somewhere between those of the 'first-generation' detectors and those of the 'advanced detectors' (Figs. 9.4, 9.6 and 9.7).

(h) *Ideas for other types of beam detectors*

The Michelson and Fabry–Perot configurations, on which almost all experimental work to date has focussed, are not the only possible types of beam receivers. A number of others have been conceived of, but have not been pursued vigorously. Since the LIGOs are intended to house several different receivers simultaneously and to have lifetimes of $\gtrsim 20$ years, during which a number of generations of receivers will be built and operated, some of these other configurations might one day operate in a LIGO. These other configurations include: (i) A *frequency-tagged interferometer* or *Michelson with overlapping beams* (Drever and Weiss, 1983), in which the light beam in each arm is made to shift in frequency with each round-trip pass, so that the beams of successive passes can overlap each other but not interfere with each other; such an interferometer is basically a Michelson but with small, Fabry–Perot-size mirrors. (ii) *Active interferometers* (Bagaev *et al.*, 1981; Brillet and Tourrenc, 1983), in which an active medium (atoms or molecules with transitions near the frequency of the laser light) resides in the arms (or arm – there might be only one) of the interferometer. (iii) *Spectroscopic detectors* (Nesterikhin, Rautian and Smirnov, 1978; Brillet and Tourrenc, 1983; Borde *et al.*, 1983), in which laser spectroscopy is used to monitor the frequency shifts produced by the gravitational waves.

Although in principle these alternative types of beam detectors can achieve sensitivities comparable to those of the standard (Michelson and Fabry–Perot) types, practical issues make the active and spectroscopic detectors look substantially less promising (Brillet and Tourrenc, 1983); and the frequency-tagged detector has not yet been pursued in sufficient depth to venture a tentative verdict.

9.5.4 Other types of earth-laboratory detectors

In addition to bar detectors and beam detectors, a number of other types of gravity-wave detectors that could operate in an earth-bound laboratory

have been conceived of. Although none of these has looked sufficiently promising to justify substantial experimental effort, some are question marks (because they have not been pursued far enough for a clear verdict) rather than discards. Because I have not looked at most of them in enough detail, I shall not venture to sort out the question marks from the total rejects.

(a) *Electromagnetically coupled detectors*

One type of transducer used on bar detectors (e.g. by the Moscow and Perth groups) is a re-entrant microwave-cavity resonator whose capacitance is modulated by the bar's vibrations (Section 10 of Braginsky, Mitrofanov and Panov, 1985; Blair, 1983). There is an obvious similarity of this to a Fabry–Perot beam receiver in which each arm is an optical cavity with length modulated by the motions of the swinging masses. In fact, one can imagine a continuous sequence of detector configurations that leads from one to the other, by splitting the bar into pieces and gradually moving them apart, with the microwave cavity gradually being distorted and expanded until it becomes the optical cavity of the Fabry–Perot (Caves, 1978).

This argument leads to the recognition that bar detectors and beam detectors are but two examples of a large class of possible 'electromagnetically coupled detectors', in which gravitational waves drive the motions of masses, and electromagnetic fields measure those motions; or, when the detector gets larger than a reduced wavelength, gravity waves drive vibrations of both the electromagnetic fields and the masses, which are coupled together.

Other earth-laboratory-scale electromagnetically coupled configurations, besides resonant bars and optical beams, that have been considered theoretically and show some promise but have not been pursued in a serious experimental way, include: (i) large microwave cavities in which wall motion, driven by gravitational waves with $\lambda\!\!\!^{-} \gg L$, upconverts microwave quanta from one mode to another mode of slightly higher frequency (Pegoraro, Picasso and Radicati, 1978; Caves, 1979); (ii) optical or microwave cavities with $L \lesssim \lambda\!\!\!^{-}$, intended for detection of high-frequency waves $f \gg 10^4$ Hz, in which the gravitational waves interact directly with the resonating electromagnetic field to move quanta from one mode into another, or interact directly with a DC electric or magnetic field to create quanta at the gravity-wave frequency (Braginsky *et al.*, 1973); (iii) as a specific, much studied example of such a cavity: an optical or microwave ring resonator in which circularly polarized gravitational waves

propagating orthogonal to the resonator plane resonate with the circulating electromagnetic field, producing a linearly growing phase shift in the field. This scheme, proposed by Braginsky and Menskii (1971), was incorrectly analyzed by them and by me (Box 37.6 of MTW) – a source of some embarrassment. My error was in thinking I could cover the detector with a single proper reference frame, when its diameter L is larger than the reduced wavelength $\lambda\!\!\!^{-}$ of the gravitational waves it sees. My incorrect conclusion, and that of Braginsky and Menskii (1971) was a quadratically growing phase shift, $\Delta\Phi \propto h(ct/\lambda\!\!\!^{-})(ct/\lambda\!\!\!^{-}_e)$. The correct conclusion (Linet and Tourrenc, 1976) is a linearly growing phase shift, $\Delta\Phi \propto h(ct/\lambda\!\!\!^{-}_e)$, which makes this scheme no better in principle than a standard beam detector. WARNING: The literature is full of similarly incorrect analyses of the various detectors described in this section, 9.5.4; (iv) an optical cavity filled with an isotropic medium, or an optical fiber, in which gravity-wave-produced strains induce optical effects such as birefringence (Iacopini *et al.*, 1979; Vinet, 1985); (v) detectors using the Mossbauer effect (Kauffman, 1970).

For a general review of electromagnetically coupled detectors, see Grishchuk (1983), and for general analyses of them, see Tourrenc and Grossiord (1974), Tourrenc (1978) and Teissier du Cros (1985).

Analyses of such detectors sometimes produce wildly overoptimistic conclusions because they overlook the mundane issue of thermal noise in the mechanical parts of the detector (e.g. the walls of an electromagnetic cavity). For an example of an analysis that *does* take thermal noise into account properly, see Caves (1979).

(b) *Superfluid interferometers and superconducting circuits*

Anandan (1981), Chiao (1982) and Anandan and Chiao (1982) have suggested that *if* it is possible to construct a superfluid weak link, analogous to the superconducting weak link (Josephson junction) on which SQUIDS are based, then such a link could form the basis for a superfluid ring interferometer that would be sensitive to gravitational waves. Unfortunately, such weak links do not yet exist. Schrader (1984) and independently Anandan (1985, 1986) have suggested a gravity-wave detector based on the direct interaction between the gravitational wave and the magnetic field in superconducting solenoids, with the resulting current change monitored by a SQUID.

9.5.5 Low-frequency detectors (10–10^{-5} Hz)

As one goes to lower and lower frequencies it becomes harder and harder to

isolate a gravity-wave detector in an earth-bound laboratory from seismic and acoustic vibrations and from fluctuating gravity gradients due to people, animals, trucks, etc. Ultimately, somewhere around 10 Hz, isolation will become impossibly difficult (see, e.g., Saulson, 1984). Thus, to operate in the 'low-frequency region' $10\,\text{Hz} \gg f \gtrsim 10^{-5}\,\text{Hz}$ will require putting detectors in space, or using normal modes of the earth or the sun as the detectors.

A number of space-borne or earth- or sun-normal-mode detectors have been conceived of for operation at low frequencies; and several have been constructed and have produced interesting limits on cosmic gravitational waves. In this section I shall describe these briefly.

(a) *Doppler tracking of spacecraft*

At periods between a few minutes and a few hours the best present gravity wave detector is the doppler tracking of spacecraft.

In doppler tracking a highly stable clock on earth ('master oscillator') is used to control the frequency of a monochromatic radio wave, which is transmitted from earth to the spacecraft. This 'uplink' radio wave is received by the spacecraft and 'transponded' back to the earth; i.e. the uplink wave is used by the spacecraft to control, in a phase-coherent way, the frequency of the 'downlink' radio wave that it transmits back to earth. When the down link radio wave is received at earth, its frequency is compared with that of the master oscillator; and from that comparison a doppler shift is read out.

When gravitational radiation sweeps through the solar system – most interestingly with a reduced wavelength of order the earth-spacecraft distance so it must be analyzed by TT methods – it perturbs the earth, the spacecraft, and the propagating radio wave. The net result is to produce fluctuations in the measured doppler shift with magnitude $\delta\nu/\nu \sim h_{jk}^{\text{TT}}$. Each feature in the gravitational waveform $h_{jk}^{\text{TT}}(t)$ shows up three times in the doppler shift, in a manner that can be regarded as due to interaction of the radio wave with the gravity wave at the events of emission from earth, transponding by spacecraft, and reception at earth (Estabrook and Wahlquist, 1975). This triplet structure can be used to help distinguish the effects of the gravitational wave from noise in the doppler data.

The use of doppler data for gravity-wave searches was first proposed by Braginsky and Gertsenshtein (1967), and was first pursued with preexisting data by Anderson (1971). The response of doppler tracking to a gravity wave was worked out by Davies (1974) for special cases and by Estabrook and Wahlquist (1975) for the general case. Experimental feasibility and noise

thresholds were first discussed by Wahlquist *et al.* (1977). Experimental results have been obtained with the Viking spacecraft (Armstrong *et al.*, 1979), the Voyager spacecraft (Hellings *et al.*, 1981), Pioneer 10 (Anderson *et al.*, 1984), and Pioneer 11 (Armstrong *et al.*, 1987). A future experiment will entail coordinated observations with the Galileo and Ulysses spacecraft (Estabrook, 1987).

In all past experiments the most serious noise source was fluctuations in the interplanetary plasma (solar wind), which cause fluctuations in the dispersion of the radio waves and thence fluctuations in the doppler shift. These fluctuations have limited the sensitivity under optimal circumstances to $[fS_h(f)]^{\frac{1}{2}} \sim 3 \times 10^{-14}$ in the band $10^{-4}\,\text{Hz} \lesssim f \lesssim 10^{-2}\,\text{Hz}$. Fortunately, dispersion in an ionized plasma becomes less serious when one moves to higher radio-wave frequencies – a move that is also desired to produce higher bit rates for information transfer between the spacecraft and earth. Some future spacecraft, including the Galileo mission, will use X-band radio signals (10 GHz frequency) on both the uplink and the downlink, rather than S-band (2 GHz) up and X-band down as in the past; and correspondingly their gravity-wave sensitivities should improve to $[fS_h(f)]^{\frac{1}{2}} \sim 3 \times 10^{-15}$. In fact, on such spacecraft the ionized-plasma dispersion will be sufficiently low that fluctuations in dispersion due to water vapor in the earth's atmosphere may become the most serious noise source (Armstrong and Sramek, 1982). Plans for monitoring the water vapor as a means of reducing this noise are being developed (Resch *et al.*, 1984). In the more distant future, doppler tracking may be improved by using dual-frequency X-band/K-band (30 GHz) tracking to reduce and monitor the ionized plasma dispersion, by installing a highly stable clock on board the spacecraft (Smarr *et al.*, 1983) and/or by moving the earth-based antenna into earth orbit. These improvements might ultimately produce $[fS_h(f)]^{\frac{1}{2}}$ as low as 10^{-17} (Estabrook, 1987).

Figs. 9.4, 9.6 and 9.7 show the sensitivities $h_{3/\text{yr}}$ corresponding to these noise levels (equations (34), (52a) and (67) with $\Delta f = f$).

At frequencies $f \sim 3$ Hz, far above the regime 10^{-2}–10^{-4} Hz of normal doppler tracking, but below the regime of earth-based detectors, one might hope to use doppler tracking of spacecraft in earth orbit (distances $\sim c/4f \sim 25\,000$ km). The best such spacecraft are those of the Global Positioning System (GPS). Unfortunately, even with the best projected improvements of the GPS we cannot anticipate $[fS_h(f)]^{\frac{1}{2}}$ better than $\sim 10^{-14}$ corresponding to a burst sensitivity $h_{3/\text{yr}} \sim 10^{-13}$; see Fig. 9.4 and see Hansen, Chiu and Chao (1986) for a detailed study.

(b) *Beam detectors in space*

Several conceptual designs for optical beam detectors in space were suggested by Weiss *et al.* (1976) and are discussed in Weiss (1979). More recently Faller and Bender (1981), Bender *et al.* (1984) and Faller *et al.* (1985) have been carrying out a detailed design and feasibility study for such detectors; and it is possible they will fly sometime around the turn of the century. As now conceived (Faller *et al.*, 1985), such a detector might consist of three 'drag-free' satellites, one at the corner and two at the ends of an L. The satellites might be in the same solar orbit as the earth–moon system, but far from the earth and moon; and their arm length might be $L \sim 1$ million km. Each satellite might carry its own 100 mW laser, with the end lasers phase locked to the light that arrives from the corner laser. With 50 cm diameter telescopes on each satellite to focus the laser beams, this could result in a one-pass ($B=1$) Michelson interferometer system with shot-noise-limited sensitivity $[S_h(f)]^{\frac{1}{2}} = 10^{-20}\,\mathrm{Hz}^{-\frac{1}{2}}$ over the range $10^{-1}\,\mathrm{Hz} \gtrsim f \gtrsim 10^{-4}\,\mathrm{Hz}$. Above 10^{-1} Hz the sensitivity would be degraded because of too-long an arm length ($L > \lambda\!\!\!^{-}$). Below 10^{-4} Hz stochastic non-gravitational forces might degrade the sensitivity. This projected sensitivity translates into the amplitude levels $h_{3/\mathrm{yr}}$ (equations (34), (52a) and (67) with $\Delta f = f$) shown in Figs. 9.4, 9.6 and 9.7.

(c) *Earth's normal modes*

Forward *et al.* (1961) and Weber (1967) pioneered the use of the earth's quadrupolar normal modes as 'resonant-bar' gravity-wave detectors. More recently Boughn and Kuhn (1984) have given a careful formulation of the method for inferring, from earth-normal-mode observations, limits on any stochastic gravitational waves that might be exciting the earth. Using their method on data from the IDA seismic network, Boughn and Kuhn (1987) have derived a slightly stronger limit on stochastic background than the early Weber (1967) limit: they find $\Omega_{\mathrm{GW}}(f) \lesssim 1$ at $f = 0.31$ mHz (Fig. 9.7). They expect, further, that a larger amount of data may permit this limit to be lowered to $\Omega_{\mathrm{GW}} \lesssim 0.1$.

(d) *Sun's normal modes*

Walgate (1983) stimulated people to think about the sun's quadrupolar normal modes as gravity-wave detectors, when he speculated (falsely; see, e.g., Kuhn and Bough, 1984, and Anderson *et al.*, 1984) that an observed mode was being excited by monochromatic gravitational waves from the binary star Geminga.

Boughn and Kuhn (1984) have carried out a detailed analysis of the gravity-wave implications of observed solar pulsations. From the data of Isaac (1981) on low-order p- and g-modes they find $\Omega_{GW}(f) \lesssim 100$ at $f \simeq 3 \times 10^{-4}$ Hz; and they argue that better observational data may produce orders of magnitude improvement, bringing $\Omega_{GW}(f)$ well below unity.

(e) *Vibrations of blocks of the earth's crust*

Braginsky *et al.* (1985) have recently pointed out that blocks of the earth's crust with sizes $50 \text{ km} \lesssim L \lesssim 70$ km could function as resonant-bar detectors for waves of frequency $f \simeq 0.03$ Hz. They propose monitoring such a block with an array of seismic stations and cross-correlating the data to pull out the seismic motions associated with the block's quadrupolar modes. Their calculations suggest a possible sensitivity $[fS_h(f)]^{\frac{1}{2}} \sim 2 \times 10^{-17}$ corresponding to $h_{3/\text{yr}} \sim 2 \times 10^{-16}$ for bursts.

(f) *Skyhook*

Braginsky and Thorne (1985) have suggested an earth-orbiting 'skyhook' gravity-wave detector that would operate in the 0.1–0.01 Hz region with sensitivity $[fS_h(f)]^{\frac{1}{2}} \sim 3 \times 10^{-17}$, which is much better than present or near-future doppler tracking and roughly comparable to that hoped for from blocks of the earth's crust (see Figs. 9.4, 9.6 and 9.7). The skyhook would consist of two masses, one on each end of a long thin cable with a spring at its center. As it orbits the earth, the cable would be stretched radially by the earth's tidal gravitational field. Gravitational waves would pull the masses apart and push them together in an oscillatory fashion; their motion would be transmitted to the spring by the cable; and a sensor would monitor the spring's resulting motion.

If it ever flies, the skyhook's role will be to provide, with a simple and inexpensive device, a moderate-sensitivity coverage of the 0.1–0.01 Hz region during the epoch before far more sensitive beam detectors are built and installed in space.

9.5.6 *Very-low-frequency detectors (frequencies below 10^{-5} Hz)*

At frequencies below about 10^{-5} Hz the only sources of gravitational waves are probably stochastic background from the early universe (Sections 9.4.3(d,e,f)); and the best detectors involve use of distant astronomical bodies.

(a) *Pulsar timing*

Remarkably good limits on very-low-frequency gravitational waves have come from the timing of pulsars (Taylor, 1987). The basic idea for pulsar timing as a gravity-wave detector (Sazhin, 1978; Detweiler, 1979) is this: the rotation of the pulsar's underlying neutron star is a highly stable clock, and pulsar timing compares that clock's ticking rate with the ticking of the very best atomic clocks in earth-bound laboratories. When gravity waves sweep over the pulsar, they affect the ticking rate of the pulsar's clock relative to clocks elsewhere in the universe, including earth; and those effects show up as fluctuations in the pulsar timing data. Similarly, when gravity waves sweep over the earth, they affect the ticking rates of all our clocks relative to those elsewhere in the universe; and those effects show up in pulsar timing. To the extent, then, that the pulsar timing data are free of fluctuations, one can infer that gravity waves of a given strength are not sweeping over either the pulsar or the earth. (See Blandford, Narayan and Romani, 1984, for a discussion of how to extract the gravity-wave information from the timing data and the pitfalls one encounters in doing so.) If fluctuations are seen in the timing data and are actually due to gravity waves sweeping over the pulsar, we probably will never be able to confirm that the source is gravity waves rather than fluctuations in the neutron star's rotation. However, if fluctuations are seen and are due to gravity waves sweeping over the earth, we might be able to learn their cause and study the waves' direction of propagation, polarization and wave form by simultaneously timing several pulsars on different parts of the sky.

In response to the Sazhin–Detweiler idea, Hellings and Downs (1983) and Romani and Taylor (1983) used timing data on four especially quiet pulsars (which Downs and Reichley (1983) had tracked for over a decade with the Goldstone antenna of NASA's deep space tracking network) to place a limit $\Omega_{GW}(f) \lesssim 1 \times 10^{-3}(f/10^{-8}\ \text{Hz})^4$ (90% confidence) on any stochastic background in the frequency range $4 \times 10^{-9} \lesssim f \lesssim 10^{-7}$ Hz; Bertotti, Carr and Rees (1983) used data from Helfand *et al.* (1980) to obtain a similar limit.

Intrinsic noise in all the pulsars used in these analyses would have made it difficult to improve on these limits at fixed frequency (though, by further observations the region covered could have been extended to lower frequencies; the lowest being of order 1/(total time since the measurements began)). Fortunately, while these analyses were in process, they were put out of business by the discovery (Backer *et al.*, 1982) of a pulsar far quieter than any previously known: the millisecond pulsar PSR 1937+21. From three

years of 1937+21 timing data taken with the Arecibo radio telescope, Davis *et al.* (1985) and Taylor (1987) have now placed the limit

$$\Omega_{GW}(f) \leqslant 1 \times 10^{-6}\left(\frac{f}{10^{-8}\text{ Hz}}\right)^4 \quad \text{for } f \gtrsim f_{\min} = 10^{-8}\text{ Hz} \qquad (127)$$

on any isotropic stochastic background: see Fig. 9.7.

This level of sensitivity is so good that further progress at fixed frequency is limited by the long-term frequency stability of the world's best atomic clocks. Thus, unless clocks improve, we can expect the coefficient in (127) to improve at best as t^{-1}, while the lower frequency limit $f_{\min}$ decreases as t^{-1} producing $\Omega_{GW}(f_{\min}) \propto t^{-5}$ (with $t=0$ in 1982). Ultimately, when clocks have improved by one or two orders of magnitude, noise due to interstellar scintillation may become a problem (Armstrong, 1984).

Recently two other quiet, fast pulsars PSR 1855+09 and PSR 1953+29 have been discovered (Segelstein *et al.*, 1986). Together with PSR1937+21 and others that we can hope for, they may one day form a network for gravitational-wave searches and observations. Such a network would alleviate problems with interstellar scintillations and atomic clock fluctuations.

(b) *The timing of orbital motions*

A gravitational wave with period long compared to observation times will produce a gradual secular change in the relative ticking rates of clocks in and out of the wave: $\dot{\omega}/\omega \sim (f/2\pi)h$, where f is the gravity-wave frequency, h is its amplitude, and ω is the ticking rate of the clock in the wave as monitored by any clock that is outside of the wave. For stochastic waves this gives $\dot{\omega}/\omega \sim [\Omega_{GW}(f)]^{\frac{1}{2}}(10^{10}\text{ yr})^{-1}$. Because the physical torques on pulsars are so large ($\dot{\omega}/\omega \gg (10^{10}\text{ yr})^{-1}$), pulsars cannot be useful in searches for waves with periods longer than the observation time. However (Mashhoon, Carr and Hu, 1981; Bertotti, Carr and Rees, 1983), the orbit of the binary pulsar is such a good clock, with such a well-understood slow down, that it can be beat against earth-based clocks to give interesting limits on waves with frequencies $10^{-8}\text{ Hz} \lesssim f \lesssim 10^{-13}\text{ Hz}$. (The low-frequency limit, 10^{-13} Hz, corresponds to waves with λ of order the distance between the earth and the pulsar.) The most recent observational data from Weisberg and Taylor (1984) give the limit

$$\int_{10^{-8}\text{ Hz}}^{10^{-13}\text{ Hz}} \Omega_{GW}(f) f^{-1}\, df \leqslant 0.5 H_{100}^{-2}, \qquad (128)$$

where H_{100} is the Hubble constant in units of 100 km/s Mpc^{-1}.

Orbital motions are also useful as detectors of waves with frequencies f of order the orbital frequency. The orbiting bodies respond to such waves in the same manner as does a resonant-bar detector. (After all, the orbit is, in a sense, nothing but a two-dimensional oscillator.) The idea of using a lunar or planetary or binary-star orbit in this way as a gravity-wave detector has been proposed or discussed by Braginsky and Gertsenshtein (1967). Anderson (1971), Bertotti (1973), Rudenko (1975) and many others. Mashhoon, Carr and Hu (1981) argue that Viking doppler data on the relative orbits of Mars and Earth place a limit $\Omega_{GW}(f) \lesssim 0.1$ at $f \simeq 3 \times 10^{-8}$ Hz (period of one year), and that laser ranging data on the moon's orbit – which now give $\Omega_{GW}(f) \lesssim 10$ – have the potential in some years to give $\Omega_{GW}(f) \lesssim 0.1$ at $f \simeq 3 \times 10^{-7}$ Hz (period of one month).

(c) *Anisotropies in the temperature of the cosmic microwave radiation*

Large-scale (quadrupolar) anisotropies in the cosmic microwave radiation would be produced by gravitational waves in the vicinity of the earth today (Sachs and Wolfe, 1967; Dautcourt, 1969). Present observational limits on such anisotropies imply $\Omega_{GW}(f) \lesssim 10^{-6}(f/3 \times 10^{-18}\ \text{Hz})^2$ for any f, even $f \lesssim 1/(\text{Hubble time})$ (Grishchuk and Zel'dovich, 1978). Small-scale anisotropies (angular scales of order 5 degrees) would have been produced by gravitational waves during the epoch of plasma recombination, when the microwave radiation we now see last interacted with matter. The limit from these is remarkably good: $\Omega_{GW}(f) \lesssim 10^{-13}$ in a narrow frequency window at $f \simeq 10^{-16}$ Hz (Carr, 1980; Zel'dovich and Novikov, 1983; Starobinsky, 1985); but this limit is much less firmly based than others. It presumes we understand thoroughly the propagation of the cosmic microwave radiation during recombination and from the epoch of recombination to the present. Note: this limit does not constrain waves from cosmic strings or any other source that emitted most of its waves later than the epoch of recombination; see Traschen, Turok and Brandenberger (1986) for the effects of strings on the microwave anisotropy.

(d) *Other astronomical observations*

In addition to the timing of pulsars and of orbital motions, and microwave anisotropies, several other astronomical observations can be used as probes of low-frequency gravitational waves: (i) *the observed velocities of galaxies and clusters of galaxies*, i.e. deviations from the Hubble flow (Rees, 1971; Burke, 1975; Dautcourt, 1977). These give $\Omega_{GW}(f) \lesssim (f/3 \times 10^{-17})^2$ for $10^{-17}\ \text{Hz} \lesssim f \lesssim 10^{-15}$ Hz (Carr, 1980). (ii) *Peculiarities in primordial nucleosynthesis*, produced by the influence of gravitational wave energy on the expansion rate of the universe during nucleosynthesis (the 'first three

minutes') (Schvartzman, 1969). These produce a limit $\Omega_{GW}(f) \lesssim 10^{-4}$ at $f \gtrsim 10^{-10}$ Hz, which is not fully firmly based because it presumes we thoroughly understand conditions in the first few minutes of the universe. Note that this limit is irrelevant for waves emitted since the epoch of nucleosynthesis.

For detailed reviews and references on these and other 'astronomical detectors' of gravity waves, see Dautcourt (1974), Chapter 17 of Zel'dovich and Novikov (1983), and Section 4 of Carr (1980).

9.6 Conclusion

As I look back over this review of gravitational waves, I am struck by the enormous changes in our theoretical understanding – or at least in theoretical fashion – that have occurred over the past 5, 10 and 15 years; and I am impressed even more by the progress that experimenters have made in the quest to invent, design and build detectors of ever greater sensitivity. That the quest ultimately will succeed seems almost assured. The only question is when, and with how much further effort. Five years ago Jerry Ostriker and I made a bet, to wit:

> *Whereas both Jeremiah P. Ostriker and Kip S. Thorne believe that Einstein's equations are valid*
>
> *And both are convinced that these equations predict the existence of gravitational waves*
>
> *And both are confident that Nature will provide what physical law predicts*
>
> *And both have faith that scientists can ultimately observe whatever Nature does supply*
>
> *Nevertheless, they differ on the likely strengths of natural sources and on the probability of a near-future and verifiable detection.*
>
> *Therefore they agree to wager one case of good red wine (JPO to supply French wine, KST to supply California) on the detection of extraterrestrial gravitational waves before the next Millennium (January 1, 2000). KST wins the wager if at least two experimental groups observe phenomena which they agree are gravitational waves. If not, JPO wins.*
>
> *Signed and officially sealed*
> *this sixth day of May* 1981
>
> *Jeremiah P. Ostriker*
> *Kip S. Thorne*

I expect to win – but I won't guarantee it.

Acknowledgments

For helpful comments on the manuscript of this chapter I thank Thibault Damour, Ron Drever, John Armstrong, Peter Bender, Jiři Bičak, David Blair, Roger Blandford, Herman Bondi, Carlton Caves, Frank Estabrook, Charles Evans, Sam Finn, Craig Hogan, Jim Hough, Ed Leaver, Brian Meers, Roger Romani, David Schoemaker, Dan Stinebring, and Kimio Tsubono.

References

Abbott, L. and Wise, M. (1984). *Nuclear Physics B*, **244**, 541.
Alpar, M. A. and Pines, D. (1985). *Nature*, **314**, 334.
Amaldi, E. and Pizella, G. (1979). In *Relativity, Quanta, and Cosmology in the Development of the Scientific Thought of Einstein*, Volume 1, 9.
Amaldi, E., Coccia, E., Cosmelli, C., Ogawa, Y., Pizzella, G., Rapagnani, P., Ricci, F., Bonifazi, P., Castellano, M. G., Vannaroni, G., Bronzoni, F., Carelli, P., Foglietti, V., Cavallari, G., Habel, R., Modena, I. and Pallottino, G. V. (1984). *Il Nuovo Cimento*, **7C**, 338.
Anandan, J. (1981). *Physical Review Letters*, **47**, 463.
Anandan, J. (1985). *Physics Letters*, **105A**, 280.
Anandan, J. (1986). In Ruffini, ed. (1986).
Anandan, J. and Chiao, R. Y. (1982). *General Relativity and Gravitation*, **14**, 515.
Anderson, A. J. (1971). *Nature*, **229**, 547.
Anderson, J. D., Armstrong, J. W., Estabrook, F. B., Hellings, R. W., Law, E. K. and Wahlquist, H. D. (1984). *Nature*, **308**, 158.
Anderson, J. L. and Hobill, D. W. (1986). In *Dynamical Spacetimes and Numerical Relativity*, ed. J. M. Centrella, p. 389. Cambridge University Press.
Anderson, J. L., Kates, R. E., Kegeles, L. S. and Madonna, R. G. (1982). *Physical Review D*, **25**, 2038.
Armstrong, J. W. (1984). *Nature*, **307**, 527.
Armstrong, J. W., Estabrook, F. B. and Wahlquist, H. D. (1987). *Astrophysical Journal* (in press).
Armstrong, J. W. and Sramek, R. A. (1982). *Radio Science*, **17**, 1579.
Armstrong, J. W., Woo, R. and Estabrook, F. B. (1979). *Astrophysical Journal*, **230**, 570.
Ashtekar, A. (1983). In Deruelle and Piran, eds. (1983), p. 421.
Ashtekar, A. (1984). In *General Relativity and Gravitation*, ed. B. Bertotti *et al.*, p. 37. Reidel: Dordrecht.
Backer, D. C., Kulkarni, S. R., Heilis, C., Davis, M. M. and Gross, W. M. (1982). *Nature*, **300**, 615.
Bagaev, S. N., Chebotaev, V. P., Dychkov, A. S. and Goldort, V. G. (1981). *Applied Physics*, **25**, 161.
Bardeen, J. M. and Press, W. H. (1973). *Journal of Mathematical Physics*, **14**, 7.
Bardeen, J. M. and Piran, T. (1983). *Physics Reports*, **96**, 205.
Begelman, M. C., Blandford, R. D. and Rees, M. J. (1980). *Nature*, **287**, 307.
Bender, P. L., Faller, J. E., Hall, J. L., Hils, D. and Vincent, M. A. (1984). Abstract for *Fifth International Laser Ranging Instrumentation Workshop*; Herstmonceaux, 10–14 September, 1984.

Bendat, J. S. (1958). *Principles and Applications of Random Noise Theory*, Wiley: New York.
Bertotti, B. (1973). *Astrophysical Letters*, **14,** 51.
Bertotti, B., Carr, B. J. and Rees, M. J. (1983). *Monthly Notices of the Royal Astronomical Society*, **203,** 945.
Bethe, H. (1986). In *Highlights in Modern Astrophysics*, eds. S. L. Shapiro and S. A. Teukolsky, p. 45. Wiley: New York.
Bicak, J. (1968). *Proceedings of the Royal Society of London A*, **302,** 201.
Bicak, J. (1985). In *Galaxies, Axisymmetric Systems, and Relativity*, ed. M. A. H. MacCallum, Cambridge University Press.
Bicak, J., Hoenselaers, C. and Schmidt, B. G. (1983). *Proceedings of the Royal Society of London A*, **390,** 411.
Blair, D. (1982). *Physics Letters A*, **91,** 197.
Blair, D. G. (1983). In Deruelle and Piran (1983), p. 339.
Blanchet, L. (1987a). *Proceedings of the Royal Society of London A*, **409,** 383.
Blanchet, L. (1987b). Paper in preparation.
Blanchet, L. and Damour, T. (1984). *Physics Letters*, **104A,** 82.
Blanchet, L. and Damour, T. (1986). *Philosophical Transactions of the Royal Society* (in press).
Blandford, R. D. (1979). In Smarr, ed. (1979), p. 191.
Blandford, R. D. (1984). Private communication.
Blandford, R. D., Narayan, R. and Romani, R. W. (1984). *Journal of Astrophysics and Astronomy*, **5,** 369.
Blandford, R. D. and Thorne, K. S. (1979). In *General Relativity: An Einstein Centenary Survey*, eds. S. W. Hawking and W. Israel, Cambridge University Press.
Blinnikov, S. I., Novikov, I. D., Perevodchikova, T. V. and Polnarev, A. G. (1984). *Soviet Astronomy Letters*, **10,** 177.
Bocko, M. F. and Johnson, W. W. (1984). *Physical Review A*, **30,** 2135.
Bond, J. R. and Carr, B. J. (1984). *Monthly Notices of the Royal Astronomical Society*, **207,** 585.
Bondi, H. (1957). *Nature*, **179,** 1072.
Bondi, H. (1960). *Nature*, **186,** 535.
Bondi, H., Pirani, F. A. E. and Robinson, I. (1959). *Proceedings of the Royal Society of London A*, **251,** 519.
Bondi, H., van der Burg, M. G. J. and Metzner, A. W. K. (1962). *Proceedings of the Royal Society of London A*, **269,** 21.
Bonnor, W. B. (1959). *Philosophical Transactions of the Royal Society of London A*, **251,** 233.
Bontz, R. J. and Haugan, M. P. (1981). *Astrophysics and Space Space Science*, **78,** 204.
Bontz, R. J. and Price, R. H. (1979). *Astrophysical Journal*, **228,** 560.
Bordé, C. J., Sharma, J., Tourrenc, P. and Damour, T. (1983). *Journal de Physique-Lettres*, **44,** L983.
Boughn, S. P. and Kuhn, J. R. (1984). *Astrophysical Journal*, **286,** 387.
Boughn, S. P. and Kuhn, J. R. (1987). Paper in preparation.
Boughn, S. P., Fairbank, W. M., Giffard, R. P., Hollenhorst, J. N., Mapoles, E. R., McAshan, M. S., Michelson, P. F., Paik, H. J. and Taber, R. C. (1982). *Astrophysical Journal*, **261,** L19.
Braginsky, V. B. (1965). *Uspekhi Fizicheskikh Nauk*, **86,** 433.
Braginsky, V. B. (1983). In Deruelle and Piran (1983), p. 387.
Braginsky, V. B. and Gertsenshtein, M. E. (1967). *Soviet Physics – JETP Letters*, **5,** 287.

Braginsky, V. B. and Grishchuk, L. P. (1985). *Zhurnal Eksperimentalnoi i Teoreticheskoi Fiziki*, **89,** 744. English translation: *Soviet Physics – JETP*, **62,** 427.
Braginsky, V. B. and Menskii, M. B. (1971). *Soviet Physics – JETP Letters*, **13,** 417.
Braginsky, V. B. and Nazarenko, V. S. (1971). In *Proceedings of the Conference on Experimental Tests of Gravitational Theories, November 11–13, 1970, California Institute of Technology*, ed. R. W. Davies. Jet Propulsion Laboratory Technical Memorandum, 33–499: Pasadena, California.
Braginsky, V. B., Grishchuk, L. P., Doroshkevich, A. G., Zel'dovich, Ya. B., Novikov, I. D. and Sazhin, M. V. (1973). *Soviet Physics – JETP*, **38,** 865.
Braginsky, V. B., Gusev, A. V., Mitrofanov, V. P., Rudenko, V. N. and Yakimov, V. N. (1985). *Uspekhi Fizicheskikh Nauk*, **147,** 422.
Braginsky, V. B., Mitrofanov, V. P. and Panov, V. I. (1985). *Systems with Small Dissipation*. University of Chicago Press: Chicago.
Braginsky, V. B. and Thorne, K. S. (1985). *Nature*, **316,** 610.
Braginsky, V. B. and Thorne, K. S. (1987). *Nature* (in press).
Braginsky, V. B. and Vorontsov, Yu. I. (1974). *Soviet Physics—Uspekhi*, **17,** 644.
Braginsky, V. B., Vorontsov, Yu. I. and Khalili, F. Ya. (1978). *Soviet Physics – JETP Letters*, **27,** 276.
Braginsky, V. B., Vorontsov, Yu. I. and Thorne, K. S. (1980). *Science*, **209,** 547.
Brill, D. R. and Hartle, J. B. (1964). *Phys. Rev. B*, **135,** 271.
Brillet, A. (1985). *Annales de Physique*, **10,** 219.
Brillet, A. and Meers, B. (1987). Paper in preparation.
Brillet, A. and Tourrenc, P. (1983). In Deruelle and Piran (1983).
Burke, W. L. (1969). Unpublished Ph.D. thesis, Caltech.
Burke, W. L. (1975). *Astrophysical Journal*, **196,** 329.
Burke, W. L. (1979). In *Isolated Gravitating Systems in General Relativity*, ed. J. Ehlers, p. 220. North Holland: Amsterdam.
Carr, B. J. (1980). *Astronomy and Astrophysics*, **89,** 6.
Carr, B. J. (1986). In *Inner Space/Outer Space*, ed. E. W. Kolb, University of Chicago Press: Chicago.
Carter, B. and Quintana, H. (1977). *Physical Review D*, **16,** 2928.
Caves, C. M. (1978). Private communication.
Caves, C. M. (1979). *Physics Letters*, **80B,** 323.
Caves, C. M. (1980). *Physical Review Letters*, **45,** 75.
Caves, C. M. (1981). *Physical Review D*, **23,** 1693.
Caves, C. M. (1982). *Physical Review D*, **26,** 1817.
Caves, C. M. (1983). *Foundations of Quantum Mechanics*, ed. S. Kamefuchi *et al.*, p. 195. Physical Society of Japan: Tokyo.
Caves, C. M. (1986). *Physical Review D*, **33,** 1643.
Caves, C. M. (1987*a*). *Physical Review D*, **35,** 1815.
Caves, C. N. (1987*b*). In *Quantum Measurement and Chaos*, ed. E. R. Pike. Plenum: New York.
Caves, C. M. (1987*c*). Paper in preparation.
Caves, C. M., Thorne, K. S., Drever, R. W. P., Sandberg, V. D. and Zimmermann, M. (1980). *Reviews of Modern Physics*, **52,** 341.
Chandrasekhar, S. (1950). *Radiative Transfer*. Oxford University Press: London.
Chandrasekhar, S. (1970). *Physical Review Letters*, **24,** 611.
Chandrasekhar, S. (1975). *Proceedings of the Royal Society of London A*, **343,** 289.
Chandrasekhar, S. (1983). *The Mathematical Theory of Black Holes*. Oxford University Press: Oxford.

Chandrasekhar, S. and Detweiler, S. L. (1975). *Proceedings of the Royal Society of London*, **344,** 441.
Chandrasekhar, S. and Detweiler, S. L. (1976). *Proceedings of the Royal Society of London A*, **350,** 165.
Chandrasekhar, S. and Esposito, F. P. (1970). *Astrophysical Journal*, **160,** 153.
Chandrasekhar, S. and Friedman, J. L. (1972). *Astrophysical Journal*, **176,** 745.
Chandrasekhar, S. and Friedman, J. L. (1973). *Astrophysical Journal*, **181,** 481.
Chandrasekhar, S. and Xanthopoulos, B. C. (1986). *Proceedings of the Royal Society of London*, **408,** 175.
Chiao, R. Y. (1982). *Physical Review B*, **25,** 1655.
Christodoulou, D. and Schmidt, B. G. (1979). *Commun. Math. Phys.*, **68,** 275.
Chrzanowski, P. L. (1975). *Physical Review D*, **11,** 2042.
Clark, J. P. A., van den Heuvel, E. P. J. and Sutantyo, W. (1979). *Astronomy and Astrophysics*, **72,** 120.
Clark, J. P. A. and Eardley, D. M. (1977). *Astrophysical Journal*, **215,** 315.
Cohen, J. M. and Kegeles, L. S. (1975). *Physics Letters*, **54A,** 5.
Cole, J. D. (1968). *Perturbation Methods in Applied Mathematics*. Blaisdell: Waltham, Mass.
Crowley, R. J. and Thorne, K. S. (1977). *Astrophysical Journal*, **215,** 624.
Cunningham, C. T., Price, R. H. and Moncrief, V. (1979). *Astrophysical Journal*, **230,** 870.
Cutler, C. and Lindblom, L. (1987). *Astrophysical Journal* (in press).
Cyranski, J. F. and Lubkin, E. (1974). *Annals of Physics*, **87,** 205.
Damour, T. (1983). In Deruelle and Piran (1983). p. 59.
Damour, T. (1986). In *Proceedings of the Fourth Marcel Grossman Meeting on the Recent Developments of General Relativity*, ed. R. Ruffini. North Holland: Amsterdam
Damour, T. (1987). In *Gravitation in Astrophysics*, eds. B. Carter and J. B. Hartle. Plenum: New York.
Damour, T. and Deruelle, N. (1986). *Annales Institut Henri Poincaré (Physique Théorique)*, **44,** 263.
Dautcourt, G. (1969). *Monthly Notices of the Royal Astronomical Society*, **144,** 255.
Dautcourt, G. (1974). In *Confrontation of Cosmological Theories with Observational Data*, Proceedings of IAU Symposium 63, ed. M. S. Longair, p. 299. Reidel: Dordrecht.
Dautcourt, G. (1977). *Astronomische Nachrichten*, **298,** 81.
Davies, R. W. (1974). In *Colloque Internationaux CNRS No.* 220, '*Ondes et Radiations Gravitationelles*'. Institut Henri Poincaré: Paris, p. 33.
Davis, M. M., Taylor, J. H., Weisberg, J. M. and Backer, D. C. (1985). *Nature*, **315,** 547.
De Logi, W. K. and Kovacs, S. J. (1977). *Physical Review D*, **16,** 2331.
Deruelle, N. and Piran, T. (1983). *Gravitational Radiation*. North Holland: Amsterdam.
de Sabbata, V. and Weber, J. eds. (1977). *Topics in Theoretical and Experimental Gravitation Physics*. Plenum: London.
Detweiler, S. L. (1975). *Astrophysical Journal*, **197,** 203.
Detweiler, S. L. (1977). *Proceedings of the Royal Society of London A*, **352,** 381.
Detweiler, S. L. (1978). *Astrophysical Journal*, **225,** 687.
Detweiler, S. L. (1979). *Astrophysical Journal*, **234,** 1100.
Detweiler, S. L. (1980). *Astrophysical Journal*, **239,** 292.
Detweiler, S. L. and Ipser, J. R. (1973). *Astrophysical Journal*, **185,** 685.
Detweiler, S. L. and Lindblom, L. (1985). *Astrophysical Journal*, **292,** 12.

Detweiler, S. L. and Szedenits, E. (1979). *Astrophysical Journal*, **231,** 211.
DeWitt, B. S. (1967*a*). *Phys. Rev.*, **160,** 1113.
DeWitt, B. S. (1967*b*). *Phys. Rev.*, **162,** 1239.
Douglass, D. H. and Braginsky, V. B. (1979). In *General Relativity, an Einstein Centenary Survey*, eds. S. W. Hawking and W. Israel, Cambridge University Press.
Downs, G. S. and Reichley, P. E. (1983). *Astrophysical Journal Supplement Series*, **53,** 169.
Drever, R. W. P. (1977). *Quarterly Journal of the Royal Astronomical Society*, **18,** 9.
Drever, R. W. P. (1983). In Deruelle and Piran (1983). p. 321.
Drever, R. W. P., Hough, J., Edelstein, W., Pugh, J. R. and Martin, W. (1977). In *Gravitazione Sperimentale*, ed. B. Bertotti, p. 365. Accademia Nazionale dei Lincei: Rome.
Drever, R. W. P., Ford, G. M., Hough, J., Kerr, I., Munley, A. J., Pugh, J. R., Robertson, N. A. and Ward, H. (1980). *Proceedings of the Ninth International Conference on General Relativity and Gravitation*, ed. E. Schmutzer, p. 265. VEB Deutscher Verlag der Wissenschaften: Berlin.
Drever, R. W. P. and Weiss, R. (1983). Unpublished work reported briefly in Appendix B.2(c) of Drever *et al.* (1985).
Drever, R. W. P., Weiss, R., Linsay, P. S., Saulson, P. R., Spero, R. and Schutz, B. F. (1985). A Detailed Engineering Design Study and Development and Testing of Components for a Laser Interferometer Gravitational Wave Observatory. Caltech: Pasadena, California, and MIT: Cambridge, Massachusetts (unpublished).
Dymnikova, I. G., Popov, A. K. and Zentsova, A. S. (1982). *Astrophysics and Space Science*, **85,** 231.
Dyson, F. J. (1969). *Astrophysical Journal*, **156,** 529.
Eardley, D. M. (1983). In Deruelle and Piran (1983). p. 257.
Eardley, D. M., Lee, D. L. and Lightman, A. P. (1973). *Physical Review D*, **8,** 3308.
Eardley, D. M., Lee, D. L., Lightman, A. P., Wagoner, R. V. and Will, C. M. (1973). *Physical Review Letters*, **30,** 884.
Eddington, A. S. (1924). *The Mathematical Theory of Relativity*, 2nd edn, Cambridge University Press; see especially supplementary note 8.
Edelstein, L. A. and Vishveshwara, C. V. (1970). *Physical Review D*, **1,** 3514.
Ehlers, J. and Kundt, W. (1962). In *Gravitation: An Introduction to Current Research*, ed L. Witten. Wiley: New York.
Ehlers, J., Rosenblum, A., Goldberg, J. N. and Havas, P. (1976). *Astrophysical Journal Letters*, **208,** L77.
Einstein, A. (1916). *Preuss. Akad. Wiss. Berlin, Sitzungsberichte der physikalisch-mathematischen Klasse*, p. 688.
Einstein, A. (1918). *Preuss. Akad. Wiss. Berlin, Sitzungsberichte der Physikalisch-mathematischen Klasse*, p. 154.
Einstein, A. and Rosen, N. (1936). *Journal of the Franklin Institute*, **223,** 43.
Epstein, R. and Wagoner, R. V. (1975). *Astrophysical Journal*, **197,** 717.
Esposito, F. P. (1971*a*). *Astrophysical Journal*, **165,** 165.
Esposito, F. P. (1971*b*). *Astrophysical Journal*, **168,** 495.
Estabrook, F. B. (1985). *General Relativity and Gravitation*, **17,** 719.
Estabrook, F. B. (1987). *Acta Astronautica* (in press).
Estabrook, F. B. and Wahlquist, H. D. (1975). *General Relativity and Gravitation*, **6,** 439
Evans, C. R. (1984). PhD thesis, University of Texas at Austin, unpublished.
Evans, C. R. (1986). In *Dynamical Spacetimes and Numerical Relativity*, ed. J. M. Centrella, p. 3. Cambridge University Press.

Evans, C. R. and Abrahams, A. M. (1987). *Physical Review D* (in preparation).
Evans, C. R., Iben, I. and Smarr, L. (1987). *Astrophysical Journal*, submitted.
Faller, J. E. and Bender, P. L. (1981). Abstract for the *International Conference on Precision Measurements and Fundamental Constants*, 8–12 June 1981.
Faller, J. E., Bender, P. L., Hall, J. L., Hils, D. and Vincent, M. A. (1985). In *Proceedings of the Colloquium 'Kilometric Optical Arrays in Space'*, Cargese (Corsica) 23–5 October 1984. ESA SP-226.
Feynman, R. P. (1963). *Acta Physica Polonica*, **24,** 697: also published in *Proceedings of the 1962 Warsaw Conference on the Theory of Gravitation*. PWN-Editions Scientifiques de Pologne: Warsaw (1964).
Finn, L. S. (1986). *Monthly Notices of the Royal Astronomical Society*, **222,** 393.
Fock, V. A. (1959). *Theory of Space, Time, and Gravitation*, Section 87. Pergamon: London.
Forward, R. L. (1978). *Physical Review D*, **17,** 379.
Forward, R. L. and Moss, G. E. (1972). *Bulletin of the American Physical Society*, **17,** 1183(A).
Forward, R. L., Zipoy, D., Weber, J., Smith, S. and Benioff, H. (1961). *Nature*, **189,** 473.
Friedman, J. L., Ipser, J. R. and Parker, L. (1986). *Astrophysical Journal*, **304,** 115.
Friedman, J. L. and Schutz, B. F. (1975*a*). *Astrophysical Journal (Letters)*, **199,** L157.
Friedman, J. L. and Schutz, B. F. (1975*b*). *Astrophysical Journal*, **200,** 204.
Friedman, J. L. and Schutz, B. F. (1978). *Astrophysical Journal*, **222,** 281.
Futamase, T. and Schutz, B. F. (1983). *Physical Review D*, **28,** 2363.
Gal'tsov, D. V., Matiukhin, A. A. and Petukhov, V. I. (1980). *Physics Letters*, **77A,** 387.
Gal'tsov, D. V., Tsvetkov, V. P. and Tsirulev, A. N. (1984). *Soviet Physics – JETP*, **59,** 472.
Geroch, R., Held, A. and Penrose, R. (1973). *Journal of Mathematical Physics*, **14,** 874.
Gertsenshtein, M. E. (1962). *Soviet Physics – JETP*, **14,** 84.
Gertsenshtein, M. E. and Pustovoit, V. I. (1962). *Soviet Physics – JETP*, **16,** 433.
Gilden, D. L. and Shapiro, S. L. (1984). *Astrophysical Journal*, **287,** 728.
Giffard, R. P. (1976). *Physical Review D*, **14,** 2478.
Gomez, R., Isaacson, R. A., Welling, J. S. and Winicour, J. (1986). In *Dynamical Spacetimes and Numerical Relativity*, ed. J. M. Centrella, p. 236. Cambridge University Press.
Gordienko, N. V., Gusev, A. V. and Rudenko, V. N. (1977). *Vestnik Moskovskovo Universiteta*, **18,** 48.
Grishchuk, L. P. (1974). *Soviet Physics – JETP*, **40,** 409.
Grishchuk, L. P. (1975*a*). *Lettere al Nuovo Cimento*, **12,** 60.
Grishchuk, L. P. (1975*b*). *Soviet Physics – JETP*, **40,** 409.
Grishchuk, L. P. (1977). *Annals of the New York Academy of Sciences*, **302,** 439.
Grishchuk, L. P. (1983). In *Proceedings of the Ninth International Conference on General Relativity and Gravitation*, ed. E. Schmutzer. Cambridge University Press.
Grishchuk, L. P. and Polnarev, A. G. (1980). In *General Relativity and Gravitation*, vol. 2, ed. A. Held, p. 393. Plenum: New York.
Grishchuk, L. P. and Zel'dovich, Ya. B. (1978). *Soviet Astronomy – AJ*, **22,** 125.
Gürsel, Y., Linsay, P., Saulson, P., Spero, R., Weiss, R. and Whitcomb, S. (1983). Unpublished Caltech/MIT manuscript.
Gusev, A. V. and Rudenko, V. N. (1976). *Radiotekhnika i Electronika*, **3,** 1865.
Haensel, P., Zdunik, J. L. and Schaeffer, R. (1986). *Astronomy and Astrophysics*, **160,** 251.
Halliwell, J. J. and Hawking, S. W. (1985). *Physical Review D*, **31,** 1777.

Halpern, L. E. and Desbrandes, R. (1969). *Annales Institut Henri Poincaré*, **9,** 309.
Hamilton, W. O., Xu, B.-X., Solomonson, N., Mann, A. G. and Sibley, A. (1986). Performance of the LSU Tuned Bar Gravitational Wave Detector. Unpublished technical memorandum, Louisiana State University.
Hansen, P. M., Chiu, Y. T. and Chao, C.-C. (1986). Aerospace Report No. ATR-86(8421)-2. The Aerospace Corporation: El Segundo, CA.
Hawking, S. W. (1985). *Physics Letters*, **150B,** 339.
Haugan, M. P., Shapiro, S. L. and Wasserman, I. (1982). *Astrophysical Journal*, **257,** 283.
Heffner, H. (1962). *Proceedings of the IRE*, **50,** 1604.
Helfand, D. H., Taylor, J. H., Backus, R. R. and Cordes, J. M. (1980). *Astrophysical Journal*, **237,** 206.
Hellings, R. W. and Downs, G. S. (1983). *Astrophysical Journal (Letters)*, **265,** L39.
Hellings, R. W., Callahan, P. S., Anderson, J. D. and Moffett, A. T. (1981). *Physical Review D*, **23,** 844.
Hils, D. L., Bender, P., Faller, J. E. and Webbink, R. F. (1987). Paper in preparation.
Hirakawa, H. and Narihara, K. (1975). *Physical Review Letters*, **35,** 330.
Hirakawa, H., Narihara, K. and Fujimoto, M.-K. (1976). *Journal of the Physical Society of Japan*, **41,** 1093.
Hirakawa, H., Owa, S. and Iso, K. (1985). *Journal of the Physical Society of Japan*, **54,** 1270.
Hogan, C. J. (1986). *Monthly Notices of the Royal Astronomical Society*, **218,** 629.
Hogan, C. J. and Rees, M. J. (1984). *Nature*, **311,** 109.
Hollenhorst, J. N. (1979). *Physical Review D*, **19,** 1669.
Hough, J., Meers, B. J., Newton, G. P., Robertson, N. A., Ward, H., Schutz, B. F. and Drever, R. W. P. (1986). A British Long Baseline Gravitational Wave Observatory, unpublished report submitted by Glasgow University to the SERC.
Hough, J., Pugh, J. R., Bland, R. and Drever, R. W. P. (1975). *Nature*, **254,** 498.
Hough, J., Drever, R. W. P., Munley, A. J., Lee, S.-A., Spero, R., Whitcomb, S. E., Pugh, J., Newton, G., Meers, B., Brooks, E. and Gursel, Y. (1983). In *Quantum Optics, Experimental Gravity, and Measurement Theory*, eds. P. Meystre and M. O. Scully, p. 515. Plenum: New York.
Hu, B. L. (1978). *Physical Review D*, **18,** 969.
Hu Enke, Guan Tongren, Yu Bo, Tang Mengxi, Chen Shusen, Zheng Qingzhang, Michelson, P. F., Moskowitz, B. E., McAshan, M. S., Fairbank, W. M. and Bassan, M. (1986). *Chinese Physics Letters* No. 11–12, 1.
Iacopini, E., Picasso, E., Pegoraro, F. and Radicati, L. A. (1979). *Physics Letters*, **73A,** 140.
Iben, I. and Tutukov, A. V. (1984). *Astrophysical Journal Supplements*, **54,** 335.
Ipser, J. R. (1971). *Astrophysical Journal*, **166,** 175.
Ipser, J. R. (1986). Private communication.
Ipser, J. R. and Managan, R. A. (1984). *Astrophysical Journal*, **282,** 287.
Ipser, J. R. and Thorne, K. S. (1973). *Astrophysical Journal*, **181,** 181.
Isaac, G. R. (1981). *Solar Physics*, **74,** 43.
Isaacson, R. A. (1968*a*). *Physical Review*, **166,** 1263.
Isaacson, R. A. (1968*b*). *Physical Review*, **166,** 1272.
Isaacson, R. A., Welling, J. S. and Winicour, J. (1983). *Journal of Mathematical Physics*, **24,** 1824.
Isaacson, R. A., Welling, J. S. and Winicour, J. (1984). *Physical Review Letters*, **53,** 1870.
Kafka, P. (1977). In de Sabbata and Weber (1977), p. 161.
Kahn, K. and Penrose, R. (1971). *Nature*, **229,** 185.

Kaufmann, W. J. (1970). *Nature*, **227,** 157.
Kellerman, K. I. and Sheets, B. (1983). *Serendipitous Discoveries in Radio Astronomy*. National Radio Astronomy Observatory: Green Bank, West Virginia.
Kojima, Y. and Nakamura, T. (1984*a*). *Progress of Theoretical Physics*, **71,** 79.
Kojima, Y. and Nakamura, T. (1984*b*). *Progress of Theoretical Physics*, **72,** 494.
Kovacs, S. J. and Thorne, K. S. (1977). *Astrophysical Journal*, **217,** 252.
Kovacs, S. J. and Thorne, K. S. (1978). *Astrophysical Journal*, **224,** 62.
Kuhn, J. R. and Boughn, S. P. (1984). *Nature*, **308,** 164.
Landau, L. D. and Lifshitz, E. M. (1941). *Teoriya Polya*. Nauka: Moscow. (English translation of a later edition, *The Classical Theory of Fields*. Addison-Wesley: Cambridge, Mass. (1951).)
Leaver, E. W. (1985). Solutions to a Generalized Spheroidal Wave Equation in Molecular Physics and General Relativity, and an Analysis of the Quasi-Normal Modes of Kerr Black Holes, unpublished PhD thesis, University of Utah: Salt Lake City.
Leaver, E. W. (1986*a*). *Proceedings of the Royal Society of London A*, **402,** 285.
Leaver, E. W. (1986*b*). *Physical Review D*, **34,** 384.
Lifshitz, E. M. (1946). *Zhurnal Eksperimentalnoi i Teoreticheskoi Fiziki*, **16,** 587.
Lindblom, L. (1986). *Astrophysical Journal*, **303,** 146.
Lindblom, L. (1987). *Astrophysical Journal* (in press).
Lindblom, L. and Detweiler, S. L. (1983). *Astrophysical Journal*, **53,** 73.
Linet, B. (1984). *General Relativity and Gravitation*, **16,** 89.
Linet, B. and Tourrenc, P. (1976). *Canadian Journal of Physics*, **54,** 1129.
Linsay, P., Saulson, P., Weiss, R. and Whitcomb, S. (1983). A Study of a Long Baseline Gravitational Wave Antenna System; report prepared for The National Science Foundation. MIT: Cambridge, Massachusetts (unpublished).
Lipunov, V. M. and Postnov, K. A. (1986). *Soviet Astronomy Letters*. In press.
Lipunov, V. M., Postnov, K. A. and Prokhorov, X. (1987). Paper in preparation.
Lyamov, V. E. and Rudenko, V. N. (1975). *Soviet Physics – JETP*, **40,** 787.
MacCallum, M. A. H., ed. (1987). *Proceedings of the Tenth International Conference on General Relativity and Gravitation*. Cambridge University Press.
MacCallum, M. A. H. and Taub, A. H. (1973). *Communications in Mathematical Physics*, **30,** 153.
Macedo, P. G. and Nelson, A. H. (1983). *Physical Review D*, **28,** 2382.
Maischberger, K., Rudiger, A., Schilling, R., Schnupp, L., Shoemaker, D. and Winkler, W. (1985). Vorschlag zum Bau eines grossen Laser-Interferometers zur Messung von Gravitationswellen. Max-Planck Institut fur Quantenoptik: Munich (unpublished).
Mashhoon, B., Carr, B. J. and Hu, B. L. (1981). *Astrophysical Journal*, **246,** 569.
Matzner, R. A., DeWitt-Morette, C., Nelson, B. and Zhang, T.-R. (1985). *Physical Review D*, **31,** 1869.
Matzner, R. A. and Tipler, F. J. (1984). *Physical Review D*, **29,** 1575.
Meers, B. (1983). Unpublished PhD thesis, University of Glasgow.
Michelson, P. F. (1983). In Deruelle and Piran (1983), p. 465.
Michelson, P. F. (1986). *Physical Review D*, **34,** 2966.
Michelson, P. F. and Taber, R. C. (1981). *Journal of Applied Physics*, **52,** 4313.
Michelson, P. F. and Taber, R. C. (1984). *Physical Review D*, **29,** 2149.
Mijic, M., Morris, M. and Suen, W.-M. (1986). *Physical Review D*, **34,** 2934.
Mironovskii, V. N. (1966). *Soviet Astronomy – AJ*, **9,** 752.
Misner, C. W., Thorne, K. S. and Wheeler, J. A. (1973). *Gravitation*. W. H. Freeman & Co.: San Francisco. Cited in text as MTW.

Moncrief, V. (1974). *Annals of Physics*, **88,** 323.
Moss, G. E., Miller, L. R. and Forward, R. L. (1971). *Applied Optics*, **10,** 2495.
Müller, E. (1982). *Astronomy and Astrophysics*, **114,** 53.
Müller, E. (1984). In *Problems of Collapse and Numerical Relativity*, eds. D. Bancel and M. Signore, p. 271. Reidel: Dordrecht.
Nakamura, T. (1983). In Deruelle and Piran (1983).
Nakamura, T. (1986). In *Gravitational Collapse and Relativity*, eds. H. Sato and T. Nakamura, p. 295. World Scientific: Singapore.
Nakamura, T. and Sasaki, M. (1981). *Physics Letters*, **106B,** 69.
Nesterikhin, Y. E., Rautian, S. G. and Smirnov, G. I. (1978). *Soviet Physics – JETP*, **48,** 1.
Newman, E. T. and Penrose, R. (1965). *Physical Review Letters*, **15,** 231.
Newman, E. T. and Tod, K. P. (1980). In *General Relativity and Gravitation, Volume 2*, ed. A. Held, p. 1. Plenum: New York.
Nutku, Y. and Halil, M. (1977). *Physical Review Letters*, **39,** 1379.
Oelfke, W. C. (1983). In *Quantum Optics, Experimental Gravitation and Measurement Theory*, eds. P. Meystre and M. O. Scully, p. 387. Plenum: New York.
Ostriker, J. P., Thompson, C. and Witten, E. (1986). *Physics Letters*, **B180,** 231.
Owa, S., Fujimoto, M.-K., Hirakawa, H., Morimoto, K., Suzuki, T. and Tsubono, K. (1986). In *Proceedings of Fourth Marcel Grossman Meeting on Recent Developments of General Relativity*, ed. R. Ruffini. North Holland: Amsterdam.
Paczynski, B. and Sienkiewicz, R. (1981). *Astrophysical Journal (Letters)*, **248,** L27.
Paik, H. J. and Wagoner, R. V. (1976). *Physical Review D*, **13,** 2694.
Pandharipande, V. R., Pines, D. and Smith, R. A. (1976). *Astrophysical Journal*, **208,** 550.
Pallotino, G. V. and Pizella, G. (1981). *Nuovo Cimento*, **4C,** 237.
Papadopoulos, D. and Esposito, F. P. (1985). *Astrophysical Journal*, **282,** 330.
Papapetrou, A. (1962). *Comptes Rendus Acad. Sci. Paris*, **255,** 1578.
Papapetrou, A. (1971). *Annales Institut Henri Poincaré*, **14,** 79.
Papapetrou, A. (1977). In deSabbata and Weber (1977), p. 83.
Pegoraro, F., Picasso, E. and Radicati, J. (1978). *Journal of Physics A*, **11,** 1949.
Penrose, R. (1963*a*). *Physical Review Letters*, **10,** 66.
Penrose, R. (1963*b*). *Relativity, Groups and Topology*, eds. C. DeWitt and B. DeWitt, p. 565. Gordon and Breach: New York.
Peres, A. (1960). *Nuovo Cimento*, **15,** 351.
Peters, P. C. (1964). *Physical Review*, **136,** B1224.
Peters, P. C. and Mathews, J. (1963). *Physical Review*, **131,** 435.
Piran, T. (1983). In Deruelle and Piran (1983), p. 203.
Piran, T. and Stark, R. F. (1986). In *Dynamical Spacetimes and Numerical Relativity*, ed. J. M. Centrella, p. 40. Cambridge University Press.
Pirani, F. A. E. (1956). *Acta Physica Polonica*, **15,** 389.
Pirani, F. A. E. (1957). *Physical Review*, **105,** 1089.
Pirani, F. A. E. (1964). In *Lectures on General Relativity*, eds. A. Trautman, F. A. E. Pirani and H. Bondi. Prentice-Hall: Englewood Cliffs, NJ.
Power, E. A. and Wheeler, J. A. (1957). *Reviews of Modern Physics*, **29,** 480.
Press, W. H. (1971). *Astrophysical Journal (Letters)*, **170,** L105.
Press, W. H. (1977). *Physical Review D*, **15,** 965.
Press, W. H. (1979). *General Relativity and Gravitation*, **11,** 105.
Price, R. H. (1972*a*). *Physical Review D*, **5,** 2419.
Price, R. H. (1972*b*). *Physical Review D*, **5,** 2439.
Price, R. H. and Thorne, K. S. (1969). *Astrophysical Journal*, **155,** 163.

Priedhorsky, W., Stella, L. and White, N. E., *International Astronomical Union Circular*, No. 4247, 28 August, 1986.
Quinlan, G. and Shapiro, S. L. (1987). *Astrophysical Journal*, in press.
Ramaty, R., Bonazzola, S., Cline, T. L., Kazanas, D. and Meszaros, P. (1980). *Nature*, **287,** 122.
Rees, M. J. (1971). *Monthly Notices of the Royal Astronomical Society*, **154,** 187.
Rees, M. J. (1983). In Deruelle and Piran (1983), p. 297.
Regge, T. and Wheeler, J. A. (1957). *Physical Review*, **108,** 1063.
Resch, G. M., Hogg, D. E. and Napier, P. G. (1984). *Radio Science*, **19,** 411.
Reynolds, S. P. and Stinebring, D. R. (1984). *Millisecond Pulsars*, National Radio Astronomy Observatory: Green Bank, West Virginia.
Romani, R. W. and Taylor, J. H. (1983). *Astrophysical Journal (Letters)*, **265,** L35.
Rosen, N. (1937). *Physikalische Zeitschift Sowjetunion*, **12,** 366.
Rubakov, V., Sazhin, M. C. and Veryaskin, A. (1982). *Physics Letters*, **115B,** 189.
Rudenko, V. N. (1975). *Soviet Astronomy – AJ*, **19,** 270.
Rudenko, V. N. and Sazhin, M. V. (1980). *Kvantovaya Elektronika*, **7,** 2344.
Ruffini, R. ed. (1986). *Proceedings of the Fourth Marcel Grossman Meeting on Recent Developments of General Relativity*. North Holland: Amsterdam.
Ruffini, R. and Wheeler, J. A. (1971). In *Proceedings of the Conference on Space Physics*. European Space Research Organization: Paris, p. 45.
Sachs, R. K. (1962). *Proceedings of the Royal Society of London A*, **270,** 103.
Sachs, R. K. (1963). In *Relativity, Groups and Topology*, eds. C. DeWitt and B. DeWitt, p. 521. Gordon and Breach: New York.
Sachs, R. K. and Wolfe, A. M. (1967). *Astrophysical Journal*, **147,** 73.
Saenz, R. A. and Shapiro, S. L. (1978). *Astrophysical Journal*, **221,** 286.
Saenz, R. A. and Shapiro, S. L. (1979). *Astrophysical Journal*, **229,** 1107.
Saenz, R. A. and Shapiro, S. L. (1981). *Astrophysical Journal*, **244,** 1033.
Sasaki, M. (1984). In *Problems of Collapse and Numerical Relativity*, eds. D. Bancel and M. Signore, p. 203. Reidel: Dordrecht.
Sasaki, M. and Nakamura, T. (1982). *Progress of Theoretical Physics*, **67,** 1788.
Sato, H. (1987). Paper in press.
Saulson, P. (1984). *Physical Review D*, **30,** 732.
Saulson, P. (1984). *Physical Review D*, **30,** 732.
Sazhin, M. V. (1978). *Soviet Astronomy – AJ*, **22,** 36.
Schmidt, B. G. (1979). In *Isolated Gravitating Systems*, ed. J. Ehlers, p. 11. North Holland: Amsterdam.
Schmidt, B. G. (1986). Gravitational Radiation Near Spatial and Null Infinity, preprint.
Schilling, R. (1986). Private communication.
Schoemaker, D., Winkler, W., Maischberger, K., Rudiger, A., Schilling, R. and Schnupp, L. (1986). In Ruffini, ed. (1986).
Schrader, R. (1984). *Physics Letters B*, **143B,** 421.
Schramm, D. N. and Arnett, W. D. (1975). *Astrophysical Journal*, **198,** 629.
Schumaker, B. L. (1986). *Physics Reports*, **135,** 318.
Schumaker, B. L. and Thorne, K. S. (1983). *Monthly Notices of the Royal Astronomical Society*, **203,** 457.
Schutz, B. F. (1972). *Astrophysical Journal Supplement Series*, **24,** 343.
Schutz, B. F. (1986*a*). In *Relativity, Supersymmetry, and Cosmology*, ed. O. Bressan, M. Castagnino and V. Hamity, p. 3. World Scientific: Singapore
Schutz, B. F. (1986*b*). *Nature*, **323,** 310.
Schutz, B. F. (1986*c*). In *Dynamical Spacetimes and Numerical Relativity*, ed. J. Centrella, p. 446. Cambridge University Press.

Schutz, B. F. (1987). In *Gravitation in Astrophysics*, eds. B. Carter and J. Hartle. Plenum: New York.
Schutz, B. F. and Tinto, M. (1987). *Monthly Notices of the Royal Astronomical Society*, **224,** 131.
Schvartzman, V. F. (1969). *Soviet Physics – JETP Letters*, **9,** 184.
Segelstein, D. J., Rawley, L. A., Stinebring, D. R., Fruchter, A. S. and Taylor, J. H. (1986). *Nature*, **322,** 714.
Shapiro, S. L. and Teukolsky, S. A. (1985). *Astrophysical Journal (Letters)*, **292,** L41.
Slusher, R. E., Hollberg, L. W., Yurke, B., Mertz, J. C. and Valley, J. F. (1985). *Physical Review Letters*, **55,** 2409.
Smarr, L. (1977*a*). *Annals of the New York Academy of Sciences*, **302,** 569.
Smarr, L. (1977*b*). *Physical Review D*, **15,** 2069.
Smarr, L., ed. (1979). *Sources of Gravitational Radiation*. Cambridge University Press.
Smarr, L. and York, J. W. (1978). *Physical Review D*, **17,** 2529.
Smarr, L., Vessot, R. F. C., Lundquist, C. A., Decher, R. and Piran, T. (1983). *General Relativity and Gravitation*, **15,** 129.
Spero, R. (1986*a*). In Ruffini, ed. (1986).
Spero, R. (1986*b*). Private communication.
Stark, R. F. and Piran, T. (1985). *Physical Review Letters*, **55,** 891 and **56,** 97.
Stark, R. F. and Piran, T. (1986). In *Proceedings of the Fourth Marcel Grossman Meeting on Recent Developments of General Relativity*, ed. R. Ruffini (in press).
Starobinsky, A. A. (1979). *Soviet Physics – JETP Letters*, **30,** 682.
Starobinsky, A. A. (1985). *Soviet Astronomy Letters*, **11,** 133.
Stewart, J. M. and Friedrich, H. (1983). *Proceedings of the Royal Society of London A*, **384,** 427.
Suzuki, T., Tsubono, K. and Hirakawa, H. (1978). *Physics Letters A*, **67,** 10.
Sullivan, W. T. (1982). *Classics in Radio Astronomy*. Reidel: Dordrecht.
Sullivan, W. T. (1984). *The Early Years of Radio Astronomy*. Cambridge University Press.
Szekeres, P. (1971). *Annals of Physics*, **64,** 599.
Tammann, G. A. (1981). In *Supernovae: A Survey of Current Research*, ed. M. J. Rees and R. J. Stoneham, p. 371. Reidel: Dordrecht.
Taylor, J. H. (1987). In *General Relativity and Gravitation*, ed. M. A. H. MacCallum. Cambridge University Press (in press).
Teissier du Cros, F. (1985). *Annales de Physique*, **10,** 263.
Taylor, J. H. (1987). In *Proceedings of Eleventh International Conference on General Relativity and Gravitation*. Cambridge University Press (in press).
Teukolsky, S. A. (1972). *Physical Review Letters*, **29,** 1114.
Teukolsky, S. A. (1973). *Astrophysical Journal*, **185,** 635.
Teukolsky, S. A. and Press, W. H. (1974). *Astrophysical Journal*, **193,** 443.
Thorne, K. S. (1969*a*). *Astrophysical Journal*, **158,** 1.
Thorne, K. S. (1969*b*). *Astrophysical Journal*, **158,** 997.
Thorne, K. S. (1972). In *Magic Without Magic: John Archibald Wheeler*. Appendix A, p. 243. W. H. Freeman: San Francisco.
Thorne, K. S. (1977). In de Sabbata and Weber (1977), p. 1.
Thorne, K. S. (1978). In *Theoretical Principles in Astrophysics and Relativity*, eds. N. R. Lebovitz, W. H. Reid and P. O. Vandervoort, p. 149. University of Chicago Press: Chicago.
Thorne, K. S. (1980*a*). *Reviews of Modern Physics*, **52,** 285.
Thorne, K. S. (1980*b*). *Reviews of Modern Physics*, **52,** 299.
Thorne, K. S. (1983). Chapter 1 of Deruelle and Piran (1983).
Thorne, K. S. (1985). In *Nonlinear Phenomena in Physics*, ed. F. Claro, p. 280. Springer-Verlag: Berlin.

Thorne, K. S. and Braginsky, V. B. (1976). *Astrophysical Journal (Letters)*, **204,** L1.
Thorne, K. S. and Campolattaro, A. (1967). *Astrophysical Journal*, **149,** 591; and **152,** 673.
Thorne, K. S., Caves, C. M., Sandberg, V. D., Zimmermann, M. and Drever, R. W. P. (1979). In Smarr (1979). p. 49.
Thorne, K. S., Drever, R. W. P., Caves, C. M., Zimmermann, M. and Sandberg, V. D. (1978). *Physical Review Letters*, **40,** 667.
Thorne, K. S. and Gürsel, T. (1983). *Monthly Notices of the Royal Astronomical Society*, **205,** 809.
Thorne, K. S. and Kovacs, S. J. (1975). *Astrophysical Journal*, **200,** 245.
Thorne, K. S., Price, R. H. and Macdonald, D. M., eds. (1986). *Black Holes: The Membrane Paradigm*. Yale University Press: New Haven, Connecticut.
Tipler, F. J. (1980). *Physical Review D*, **22,** 2929.
Tourrenc, P. (1978). *General Relativity and Gravitation*, **9,** 123 and 141.
Tourrenc, P. and Crossiord, J.-L. (1974). *Nuovo Cimento*, **19B,** 105.
Traschen, J., Turok, N. and Brandenberger, R. (1986). *Physical Review D*, **34,** 919.
Tsvetkov, V. P. (1984). *Soviet Astronomy – AJ*, **28,** 394.
Turner, M. and Wagoner, R. V. (1979). In Smarr (1979), p. 383.
Turner, M. and Will, C. M. (1978). *Astrophysical Journal*, **220,** 1107.
Turok, N. and Brandenberger, R. H. (1986). *Physical Review D*, **33,** 2175.
Tyson, J. A. and Giffard, R. P. (1978). *Annual Reviews of Astronomy and Astrophysics*, **16,** 521.
Unruh, W. G. (1982). Unpublished work reported at NATO Advanced Study Institute on Gravitational Radiation, Les Houches, France, June 1982.
Vachaspati, T. and Vilenkin, A. (1985). *Physical Review D*, **31,** 3052.
van den Heuvel, E. P. J. (1984). In Reynolds and Stinebring, ed. (1984).
Vilenkin, A. (1981*a*). *Physical Review D*, **24,** 2082.
Vilenkin, A. (1981*b*). *Physics Letters*, **107B,** 47.
Vinet, J. Y. (1985). *Annales de Physique*, **10,** 253.
Vinet, J. Y. (1986). *Journal de Physique*, **47,** 639.
Vishniac, E. T. (1982). *Astrophysical Journal*, **257,** 456.
Wagoner, R. V. (1977). In *Gravitazione Sperimentale*, ed. B. Bertotti. Accademia Nazionale dei Lincei: Rome.
Wagoner, R. V. (1984). *Astrophysical Journal*, **278,** 345.
Wagoner, R. V. and Will, C. M. (1976). *Astrophysical Journal*, **210,** 764.
Wahlquist, H. D. (1987). *General Relativity and Gravitation* (in press).
Wahlquist, H. D., Anderson, J. D., Estabrook, F. B. and Thorne, K. S. (1977). In *Gravitazione Sperimentale*, ed. B. Bertotti, p. 335. Accademia Nazionale dei Lincei: Rome.
Wainstein, L. A. and Zubakov, V. D. (1962). *Extraction of Signals from Noise*, Prentice-Hall: London.
Wald, R. M. (1973). *Journal of Mathematical Physics*, **14,** 1453.
Walgate, R. (1983). *Nature*, **305,** 665.
Walker, M. (1983). In *Relativistic Astrophysics and Cosmology*, eds. X. Fustero and E. Verdaguer. World Scientific: Singapore.
Walker, M. and Will, C. M. (1980). *Astrophysical Journal (Letters)*, **242,** L129.
Walls, D. F. (1983). *Nature*, **306,** 141.
Weber, J. (1959). *Reviews of Modern Physics*, **31,** 681 (see especially Section V).
Weber, J. (1960). *Physical Review*, **117,** 306.
Weber, J. (1967). *Physical Review Letters*, **18,** 498.
Weber, J. (1969). *Physical Review Letters*, **22,** 1302.
Weber, J. (1986). In *Proceedings of the Sir Arthur Eddington Centenary Symposium. Volume 3 – Gravitational Radiation and Relativity*, eds. J. Weber and T. M. Karade, p. 1. World Scientific: Singapore.

Weber, J. and Wheeler, J. A. (1957). *Reviews of Modern Physics*, **29,** 509.
Weyl, H. (1922). *Space–Time–Matter*. Methuen: London.
Weisberg, J. M. and Taylor, J. H. (1984). *Physical Review Letters*, **52,** 1348.
Weiss, R. (1972). *Quarterly Progress Report of the Research Laboratory of Electronics of the Massachusetts Institute of Technology*, **105,** 54.
Weiss, R. (1978). In *Sources of Gravitational Radiation*, ed. L. Smarr, p. 7. Cambridge University Press.
Weiss, R. (1979). In Smarr (1979), p. 7.
Weiss, R., Bender, P. L., Misner, C. W. and Pound, R. V. (1976). *Report of the Sub-Panel on Relativity and Gravitation, Management and Operations Working Group for Shuttle Astronomy*. NASA: Washington, DC.
Westpfahl, K. (1985). *Fortschritte der Physik*, **33,** 417.
Wheeler, J. A. (1955). *Physical Review*, **97,** 511.
Wheeler, J. A. (1962). *Geometrodynamics*. Academic Press: New York.
Wheeler, J. A. (1964). In *Relativity, Groups, and Topology*, eds. C. DeWitt and B. DeWitt. Gordon and Breach: New York.
Whitcomb, S. and Saulson, P. (1984). Unpublished research at Caltech and MIT; a few of the results are given in Fig. E.3 of Drever, Weiss *et al.* (1985).
Wiener, N. (1949). *The Extrapolation, Interpolation and Smoothing of Stationary Time Series with Engineering Applications*. Wiley: New York.
Will, C. M. (1986). *Canadian Journal of Physics*, **64,** 140.
Wilson, J. R. (1974). *Physical Review Letters*, **32,** 849.
Winkler, W. (1977). In *Gravitazione Sperimentale*, ed. B. Bertotti. Academia Nazionale dei Lincei: Rome.
Winkler, W., Maischberger, K., Rüdiger, A., Schilling, R., Schnupp, L. and Shoemaker, D. (1986). In Ruffini, ed. (1986).
Witten, E. (1984). *Physical Review D*, **30,** 272.
Wood, K. S., Michelson, P. F., Boynton, P., Yearian, M. R., Gursky, H., Friedman, H. and Dieter, J. (1986). *A Proposal to NASA for an X-Ray Large Array (XLA) for the NASA Space Station*. Stanford: Palo Alto, California.
Wood, L., Zimmermann, G., Nukolls, J. and Chapline, G. (1971). *Bulletin of the American Physical Society*, **16,** 609.
Woosley, S. E. and Weaver, T. A. (1986). *Annual Reviews of Astronomy and Astrophysics*, **24,** 205.
Wu, L.-A., Kimble, H. J., Hall, J. L. and Wu, H. (1986). *Physical Review Letters*, **57,** 2520.
Yamamoto, Y. and Haus, H. A. (1986). *Reviews of Modern Physics*, **58,** 1001.
York, J. W. (1983). In Deruelle and Piran (1983), p. 175.
Yurtsever, U. (1986). *Abstracts of Contributed Papers, 11th International Conference on General Relativity and Gravitation*, Stockholm, Sweden, 6–12 July, 1986, p. 287. University of Stockholm: Stockholm.
Yurtsever, U. (1987*a*). *Physical Review D*, submitted.
Yurtsever, U. (1987*b*). Paper in preparation.
Zel'dovich, Ya. B. (1980). *Monthly Notices of the Royal Astronomical Society*, **192,** 663.
Zel'dovich, Ya. B. and Novikov, I. D. (1983). *Relativistic Astrophysics Vol. 2. The Structure and Evolution of the Universe*. University of Chicago Press: Chicago.
Zhang, X.-H. (1986). *Physical Review D*, **34,** 991.
Zerilli, F. J. (1970). *Physical Review D*, **2,** 2141.
Zimmermann, M. (1978). *Nature*, **271,** 524.
Zimmermann, M. (1980). *Physical Review D*, **21,** 891.
Zimmermann, M. and Szedenits, E. (1979). *Physical Review D*, **20,** 351.
Zimmerman, M. and Thorne, K. S. (1980). In *Essays in General Relativity*, ed. F. J. Tipler, p. 139. Academic Press: New York.